

AI MUSCLE GAIN BRAIN

VOLUME 2

Elite AI Coaching Knowledge Base for Personalized Muscle Gain

DOCUMENT TYPE	AI Training Knowledge Base — RAG Optimized
VERSION	2.0 2025
MODULES	15 Elite Modules
COACHING RULES	500+ IF-THEN Decision Rules
FOOD DATABASE	200+ Foods with Full Macros
EXERCISE DATABASE	300+ Exercises with Coaching Cues
RECIPE LIBRARY	120+ Recipes (All Diets & Budgets)
FAQ DATABASE	170+ Expert Q&As
USE CASE	RAG · LLM Coaching · Personalized Plans · SWOT Engine
CLASSIFICATION	ZITLAS Internal — Coach System

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TABLE OF CONTENTS

1 MUSCLE GAIN FUNDAMENTALS

Hypertrophy science · Progressive overload · Beginner gains · Realistic expectations · Coaching rules

2 ASSESSMENT ENGINE

BMI · LBM · BF% · FFMI · BMR · TDEE · Scoring systems · Classification matrices · Assessment logic

3 AI SWOT ENGINE

100+ SWOT generation rules · 8 user archetypes · Strengths · Weaknesses · Opportunities · Threats

4 NUTRITION SCIENCE

Protein · Carbs · Fats · Fiber · Micronutrients · Hydration · Meal timing · MPS triggers · Gut health

5 MUSCLE GAIN FOOD DATABASE

200+ foods with full macros · Animal protein · Plant protein · Dairy · Carbs · Fats · Budget ratings

6 RECIPE LIBRARY

120+ recipes · Breakfast · Lunch · Dinner · Pre/Post workout · Snacks · Hostel · Mass gainer

7 WORKOUT DATABASE

300+ exercises · Full muscle group coverage · Equipment · Difficulty · Coaching cues · Progressions

8 WORKOUT PROGRAM LIBRARY

Beginner to Advanced · 13 user type programs · Full Body · PPL · Upper-Lower · Home · Hostel

9 RECOVERY SYSTEM

Sleep science · Cortisol · Deloads · CNS fatigue · Overtraining · Recovery scoring · Decision trees

10 SUPPLEMENTS

Evidence-based only · Recommendation engine · Budget matrix · Suitability by profile

11 SPECIAL POPULATIONS

13 dedicated coaching systems · Hostel · Vegan · Women · Teens · 40+ · Hardgainers · Travelers

12 PLATEAUS & TROUBLESHOOTING

150+ IF-THEN troubleshooting rules · Weight · Strength · Recovery · Motivation · Compliance

13 AI DECISION ENGINE

200+ coaching rules · Athlete classification · Nutrition · Training · Recovery · Special conditions

14 MASTER FAQ DATABASE

170+ expert Q&As · Nutrition · Training · Recovery · Supplements · Special populations · Beginners

15 RAG OPTIMIZATION

Retrieval tags · Decision rules per section · Related modules · AI coaching notes · Structured metadata

MODULE 1 | MUSCLE GAIN FUNDAMENTALS

Core definitions · Hypertrophy mechanisms · Progressive overload · Timelines · Coaching rules

1.1 CORE DEFINITIONS

TERM	DEFINITION
Muscle Hypertrophy	Enlargement of skeletal muscle fibers through resistance training + nutrition. Two types: myofibrillar (contractile protein growth) and sarcoplasmic (fluid/glycogen expansion).
Myofibrillar Hypertrophy	Growth of actin/myosin contractile proteins. Increases strength AND density. Targeted by heavy loads (3-6 RM range, 80-90% 1RM).
Sarcoplasmic Hypertrophy	Expansion of non-contractile fluid, glycogen, and organelles within the muscle cell. Increases size more than strength. Targeted by moderate loads (8-15 RM).
Progressive Overload	Systematically increasing the training stimulus over time. The #1 mechanism for continued adaptation. Methods: load, volume, density, ROM, tempo, frequency.
Lean Bulk	Controlled calorie surplus of +200-350 kcal/day. Maximizes muscle gain while minimizing fat accumulation. Recommended for most athletes (BF 10-18% male, 18-28% female).
Dirty Bulk	Aggressive surplus of +500-1000+ kcal/day. Causes significant fat gain. Only justified for severely underweight athletes (BMI <17) or extreme hardgainers after audit.
Body Recomposition	Simultaneous fat loss + muscle gain at TDEE. Possible for beginners, detrainees, and athletes with BF >20% (M) / >28% (F). Requires 2.0+ g/kg protein.
Beginner Gains	Rapid strength + muscle gains in first 6-12 months. Driven by neural adaptation + low training threshold. Highest rate of gain in a natural athlete's career.
Training Age	Years of consistent structured resistance training. Determines volume, intensity, and expected rate of adaptation. Beginner <1yr Intermediate 1-3yr Advanced 3+yr.
FFMI (Fat-Free Mass Index)	Lean body mass (kg) / height(m) ² . Natural athletes cap at FFMI ~25 (male) / ~22 (female). Values above this indicate likely performance-enhancing drug use.

1.2 THREE MECHANISMS OF HYPERTROPHY

MECHANISM	DESCRIPTION	HOW TO TARGET	REP RANGE
Mechanical Tension	Force generated under load. Activates mTOR / mTORC1 anabolic signalling. PRIMARY driver.	Heavy compound lifts, slow eccentrics, full ROM	3-8 reps @ 70-85% 1RM
Metabolic Stress	Lactate, H+, Pi accumulation creating the "pump". Secondary driver. Cell swelling signals anabolism.	High-rep sets, short rest, occlusion (BFR)	12-20 reps, 45-60 sec rest
Muscle Damage	Micro-tears from novel/eccentric loading. Satellite cell activation for repair and growth.	Eccentric emphasis, new exercises, stretch-loaded	8-15 reps with controlled eccentric

1.3 PROGRESSIVE OVERLOAD FRAMEWORK

METHOD	DESCRIPTION	PRIORITY / TIMING
Load (Weight)	Add weight to the bar/dumbbells when top of rep range is achieved	PRIMARY — every session (beginner), weekly (intermediate)
Volume	Add sets or reps. 4x8 → 4x10 → 5x10	PRIMARY — block by block progression
Density	Same volume in less time. Rest 90s → 60s	SECONDARY — when load stalls
Range of Motion	Increase ROM on movements. Parallel squat → full depth	SECONDARY — technique refinement
Tempo	Slower eccentrics: 2-0-1 → 3-1-1. Increases TUT.	SECONDARY — advanced technique tool
Exercise Difficulty	Progress to harder variation. Push-up → Incline Press → Bench Press	TERTIARY — beginner tool

METHOD	DESCRIPTION	PRIORITY / TIMING
Frequency	Train muscle group more often. 1x/week → 2x/week	TERTIARY — intermediate restructuring

1.4 MUSCLE FIBER TYPES

FIBER TYPE	CHARACTERISTICS	TRAINING RESPONSE	HYPERTROPHY POTENTIAL
Type I (Slow Twitch)	High endurance, low force, oxidative, fatigue-resistant	High volume, low load (15-30 reps)	Low-Moderate
Type IIa (Fast Twitch)	Mixed: moderate force + endurance	Moderate-heavy loads, moderate volume (6-12 reps)	High
Type IIx (Fast Twitch)	Highest force, lowest endurance, glycolytic, fast fatiguing	Heavy loads, low reps, explosive (1-5 reps)	Highest

Most muscles contain mixed fiber type ratios. Training across multiple rep ranges ensures full fiber type recruitment and maximises hypertrophy across both fiber types.

1.5 REALISTIC MUSCLE GAIN TIMELINES

ATHLETE PROFILE	PHASE	MONTHLY GAIN (kg)	ANNUAL GAIN (kg)	NOTES
Beginner Male	0-12 months	0.8-1.5	6-15	Neural adaptation + muscle simultaneously
Beginner Female	0-12 months	0.4-0.8	4-9	Proportionally similar % but lower absolute
Intermediate Male	1-3 years	0.2-0.5	2-5	Progressive overload consistency critical
Intermediate Female	1-3 years	0.1-0.3	1-3	Focus on strength benchmarks
Advanced Male	3-5+ years	0.05-0.15	0.5-1.5	Diminishing returns — precision required
Advanced Female	3-5+ years	0.03-0.10	0.3-1	Precise program + nutrition essential

1.6 LEAN BULK vs DIRTY BULK vs RECOMP

STRATEGY	SURPLUS	RATE OF GAIN	FAT RISK	BEST FOR
Lean Bulk	+200-350 kcal/day	0.25-0.5 kg/month	Low	Most athletes, BF 10-18% (M) / 18-28% (F)
Moderate Bulk	+300-500 kcal/day	0.5-1.0 kg/month	Moderate	Beginners, underweight athletes
Aggressive Bulk	+500-1000 kcal/day	1.0-2.0 kg/month	High	Severely underweight (BMI <18.5), hardgainers
Dirty Bulk	+1000+ kcal/day	2+ kg/month	Very High	NOT recommended. Short-term only.
Body Recomp	TDEE ±0	Muscle+ / Fat-	None	Beginners, returning athletes, BF >20% (M) / >28% (F)

1.7 MODULE 1 COACHING RULES

- [R001] Set muscle gain expectations at onboarding. Show realistic monthly and annual gain targets by training age.
- [R002] Classify athlete as Beginner / Intermediate / Advanced before generating any training or nutrition plan.
- [R003] If training history < 1 year, classify as Beginner. Apply beginner-specific protocols and volume guidelines.
- [R004] Never recommend dirty bulk unless athlete is severely underweight (BMI <17) AND has documented high caloric expenditure.
- [R005] Recommend lean bulk (+200-350 kcal) as default for athletes with BF 10-20% (M) or 18-28% (F).
- [R006] If athlete BF > 20% (M) or > 28% (F): do NOT recommend calorie surplus. Use recomp or mild deficit first.

[R007] Progressive overload must be the core driving principle of every training plan. No overload = no growth.

[R008] If beginner, prioritize load progression over volume manipulation. Add weight before adding sets.

[R009] Body recomposition is a valid and effective strategy for beginners, returning athletes, and obese individuals.

[R010] Muscle adaptation requires all three pillars simultaneously: adequate stimulus + adequate nutrition + adequate recovery.

MODULE 2 | ASSESSMENT ENGINE

Intake formulas · Body composition · Readiness scoring · Classification matrices · AI assessment logic

2.1 USER INTAKE PARAMETERS

INPUT PARAMETER	FORMAT / UNIT	USE IN COACHING
Age	Years	Activity multiplier, hormone profile, recovery capacity
Gender	M / F / Other	BMR formula, body fat thresholds, hormonal context
Height	cm	BMI, FFMI, LBM estimation
Weight	kg	BMR, TDEE, protein target, calorie target
Goal Weight	kg	Rate of gain or loss to target. Surplus size.
Activity Level	Sedentary → Extra Active	TDEE multiplier selection
Occupation	Type (desk, physical, mixed)	NEAT estimation, energy expenditure
Sleep	Hours per night	Recovery score, hormone context, volume adjustment
Stress Level	1-10 self-rated	Cortisol risk, volume modification, supplement need
Training Experience	Years of consistent training	Training age classification, program complexity
Diet Preference	Veg / Non-Veg / Vegan / Egg-only	Protein source selection, supplementation needs
Budget	INR per day for food	Meal plan construction, supplement priority
Available Time	Hours/week for training	Training split selection, session length
Medical Conditions	List (diabetes, hypertension, etc.)	Contraindications, modification rules
Equipment Access	Full gym / Basic / Home / None	Exercise selection and program type
Previous Injuries	Region + severity	Exercise modification, recovery protocol
Gym Access	Yes / No + type	Program type and exercise library access

2.2 BODY COMPOSITION FORMULAS

BMR Formulas (AI Coach Uses Mifflin-St Jeor by Default)

FORMULA	MALE	FEMALE	ACCURACY
Mifflin-St Jeor (DEFAULT)	$(10 \times W) + (6.25 \times H) - (5 \times A) + 5$	$(10 \times W) + (6.25 \times H) - (5 \times A) - 161$	Best accuracy ±5%
Harris-Benedict (Revised)	$(13.397 \times W) + (4.799 \times H) - (5.677 \times A) + 88.362$	$(9.247 \times W) + (3.098 \times H) - (4.330 \times A) + 447.593$	Slightly overestimates
Katch-McArdle	$370 + (21.6 \times LBM)$	$370 + (21.6 \times LBM)$	Best with known LBM
Quick Estimate	$W(kg) \times 22-24$	$W(kg) \times 21-23$	Rough estimation only

W = Weight (kg), H = Height (cm), A = Age (years), LBM = Lean Body Mass (kg)

TDEE = BMR x Activity Multiplier

ACTIVITY LEVEL	MULTIPLIER	DESCRIPTION
Sedentary	1.20	No exercise, desk job — office worker, no gym

ACTIVITY LEVEL	MULTIPLIER	DESCRIPTION
Lightly Active	1.375	1-3 days light exercise/week — walks 30 min, gym 2x/week
Moderately Active	1.55	3-5 days moderate exercise/week — gym 4x/week
Very Active	1.725	6-7 days hard exercise/week — daily training, physical job
Extra Active	1.90	2x daily training or very physical job — athletes, construction
NEAT Adjustment	+100-300 kcal	Add for fidgeting, standing, housework, incidental steps

Key Body Composition Calculations

METRIC	FORMULA / METHOD	NOTES / THRESHOLDS
BMI	Weight(kg) / Height(m)^2	<18.5 Under 18.5-24.9 Normal 25-29.9 Over 30+ Obese
LBM (Lean Body Mass)	Weight x (1 - BF%/100)	Used in Katch-McArdle BMR and protein targeting
BF% (Boer Formula M)	(0.407xW) + (0.267xH) - 19.2	Rough estimate. DEXA or skinfold calipers for accuracy.
BF% (Boer Formula F)	(0.252xW) + (0.473xH) - 48.3	Female Boer formula for BF% estimation
FFMI	LBM(kg) / Height(m)^2	Natural ceiling: M=25, F=22. Above = likely enhanced.
Maintenance Calories	TDEE x 1.0	No weight change target. Baseline for all calculations.
Lean Bulk Target	TDEE + 200-350 kcal	Default muscle gain protocol for most athletes.
Protein Target	BW(kg) x 1.8-2.0g (standard)	Vegetarians: +10-20%. Deficit phase: 2.2-2.6 g/kg.
Fat Target	0.8-1.0 g/kg (min 20% kcal)	Never below 20% kcal — testosterone protection.
Carb Target	Remaining kcal after P + F	Typically 45-55% of total calories in muscle gain phase.
Hydration Target	BW(kg) x 35-40 ml/day	Add 500-750ml per training hour. More in Indian summer.
Step Goal	8,000-10,000 steps/day	NEAT proxy. Track with phone/band. Adjust TDEE if >12,000.

2.3 CLASSIFICATION MATRIX

CLASSIFICATION	CRITERIA	AI COACHING ACTION
Underweight	BMI < 18.5	Aggressive lean bulk (+400-600 kcal). Liquid calories. No cardio.
Normal Weight	BMI 18.5-24.9	Standard lean bulk (+200-350 kcal). Full program.
Overweight	BMI 25-29.9	Recomp (TDEE ±0) or mild deficit. High protein 2.0-2.2 g/kg.
Obese	BMI 30+	Mild deficit (-300 kcal). High protein. Resistance training primary.
Skinny Fat	Normal BMI + high BF%	Recomp: TDEE ±0. 2.0 g/kg protein. Build muscle while losing fat.
Hardgainer	Gaining <0.1 kg/week despite surplus	Audit intake first. Liquid calories. Reduce NEAT. Increase by +150 kcal/2wks.
Beginner	Training history < 1 year	Full body 3x/week. Compound only. Linear progression.
Intermediate	Training 1-3 years	Upper-Lower or PPL. Double progression. 2x/week/muscle.
Advanced	3+ years structured training	Block periodization. High frequency. 2-4x/week/muscle.
High Recovery	Sleep 8+hrs, low stress, good nutrition	Higher training volume tolerated. Can run aggressive surplus.
Poor Recovery	Sleep <7hrs OR high stress OR poor diet	Reduce training volume 20%. Address recovery before increasing load.
Vegan	No animal products	+20% protein target. B12, D3, zinc, omega-3 supplement mandatory.
Vegetarian	No meat, may have dairy/eggs	+10-15% protein target. Focus on soy, dairy, eggs.

2.4 READINESS & SCORING SYSTEMS

Muscle Gain Readiness Score (0-100)

COMPONENT	INPUT	SCORING
Sleep Score	Avg hrs/night	< 6hrs = 0 6-7hrs = 10 7-8hrs = 20 8-9hrs = 30 9+ = 25
Nutrition Score	Protein + calorie tracking	No tracking = 0 Occasional = 5 Consistent = 20 Precise = 25
Training Score	Sessions/week + progressive overload	0-1 sess = 0 2-3 = 15 4-5 = 25 5+ with log = 30
Recovery Score	Stress level + DOMS frequency	Chronic high stress = 0 Moderate = 7 Low stress = 15
TOTAL	0-100 points	0-40: Optimize foundations first 41-70: Good base 71-100: Optimised

2.5 ASSESSMENT WORKED EXAMPLES

PROFILE	BMR	TDEE	LEAN BULK TARGET	MACRO TARGETS
Male, 70kg, 175cm, 22yr, Mod Active	1,695 kcal	2,627 kcal	2,877 kcal	140g prot, 338g carbs, 67g fat
Male, 60kg, 170cm, 20yr, Lightly Active (Hostel)	1,544 kcal	2,123 kcal	2,373 kcal	120g prot, 296g carbs, 53g fat
Female, 55kg, 162cm, 21yr, Mod Active	1,356 kcal	2,102 kcal	2,302 kcal	110g prot, 288g carbs, 51g fat
Male, 85kg, 180cm, 28yr, Very Active	1,985 kcal	3,423 kcal	3,673 kcal	170g prot, 459g carbs, 82g fat
Female, 65kg, 168cm, 25yr, Lightly Active	1,468 kcal	2,019 kcal	2,219 kcal	130g prot, 278g carbs, 49g fat

2.6 MODULE 2 COACHING RULES

- [R011] Always calculate TDEE before generating a calorie target. Never use generic calorie numbers.
- [R012] Default surplus recommendation: +250 kcal/day for lean bulk starting point.
- [R013] Adjust calorie targets every 2 weeks based on actual weight trend, not estimates alone.
- [R014] If athlete gains >0.6 kg/week, reduce calories by 200 kcal to minimize excess fat gain.
- [R015] If athlete not gaining weight for 2 consecutive weeks on surplus, increase calories by 150-200 kcal.
- [R016] Minimum protein must be calculated and set FIRST before distributing remaining calories.
- [R017] For hostel students: calculate minimum affordable calorie floor (BW x 33-35 kcal/kg).
- [R018] Never prescribe same calorie target for different body weights. Always personalise to individual.
- [R019] Account for NEAT: active students and workers need higher TDEE multipliers than desk workers.
- [R020] Hardgainers: increase calories by 150 kcal increments every 2 weeks until gaining 0.25 kg/week.

MODULE 3 | AI SWOT ENGINE

100+ SWOT generation rules · 8 user archetypes · Strengths · Weaknesses · Opportunities · Threats

3.1 SWOT ENGINE OVERVIEW

The ZITLAS SWOT Engine generates personalised Strengths, Weaknesses, Opportunities, and Threats for each user based on their assessment profile. Every SWOT is dynamically assembled from the ruleset below based on the user's classification across 9 key dimensions. SWOT output drives onboarding messaging, program priority-setting, and risk flagging.

SWOT: HOSTEL STUDENT

S	Consistent training environment available (hostel gym or body weight training in room)
S	Young biological age = peak anabolic hormone production (testosterone, IGF-1)
S	Flexible schedule allows multiple training sessions and recovery time
W	Very limited food budget (typically INR 80-150/day) constrains protein choices
W	No personal kitchen → dependency on mess/canteen food with poor macro control
W	No refrigeration → shelf-stable proteins only (soy chunks, peanuts, canned fish)
W	High academic stress elevates cortisol, impairs recovery and MPS
O	Soy chunks: 52g protein per 100g dry weight at INR 20-25 — highest protein/INR ratio
O	Eggs + milk + soy chunks = complete protein strategy at INR 80-100/day
O	Bodyweight training eliminates gym fee barrier — progressive overload via harder variations
T	Mess food often high in carbs, low in protein — nutritional gaps accumulate
T	Peer social culture (late nights, alcohol) disrupts sleep and hormonal environment
T	Exam seasons cause training disruption — progress loss if not managed proactively

SWOT: VEGETARIAN ATHLETE

S	Typically higher fiber and micronutrient intake supports gut health and recovery
S	Lower saturated fat intake from absence of red meat reduces inflammation
S	Soy, dal, paneer, eggs, dairy = complete protein strategy within vegetarian framework
W	Lower leucine density per gram of plant protein requires higher total intake
W	Higher risk of Vitamin B12, zinc, iron (females), and omega-3 EPA/DHA deficiency
W	No heme iron = lower iron bioavailability (affects oxygenation and performance)
W	Phytates in legumes reduce mineral absorption unless soaked/cooked properly
O	Soy protein isolate (90% protein, complete EAA profile) = best plant protein for MPS
O	Paneer + eggs + Greek yogurt covers all essential amino acids efficiently
O	Algae oil provides direct EPA+DHA without fish — ethical and effective supplement
T	Risk of insufficient leucine if protein target not met — blunted MPS response
T	Social situations: vegetarian options at restaurants often low protein (paneer only option)

SWOT: BUSY PROFESSIONAL

S	Higher income allows better food quality, whey protein, and supplement access
S	Structured routine enables batch cooking and pre-planned nutrition
W	Time scarcity — typically only 45-60 min training windows available
W	High chronic stress from work elevates cortisol, suppresses testosterone

W	Frequent business travel disrupts training consistency and nutrition control
W	Sedentary desk job reduces NEAT — TDEE often overestimated without step tracking
O	Meal prep services or tiffin delivery can achieve protein targets without cooking time
O	Whey protein + pre-packed nuts = portable nutrition for travel and office
O	Early morning training (6-7 AM) shows highest long-term adherence in professionals
T	Client dinners and social events = alcohol exposure, poor food choices, late nights
T	Work project deadlines can eliminate training for weeks — detraining risk

SWOT: TEEN ATHLETE (14-18)

S	Highest natural anabolic hormone levels (testosterone/IGF-1 peaks in teens)
S	Rapid beginner adaptations — fastest strength gains possible at this age
S	Most resilient recovery capacity of any age group
W	Epiphyseal plates not fused — heavy spinal loading carries growth plate risk
W	Irregular eating patterns, fast food consumption impairs protein and calorie targets
W	Sleep often disrupted by social media, late-night study, circadian shift
W	Body image concerns can lead to unhealthy extremes (undereating or supplement abuse)
O	Compound bodyweight and dumbbell training safe and highly effective at this age
O	Learning correct technique now builds foundation for decades of safe training
T	Supplement marketing targets teens aggressively — creatine should be minimum age 16 with parental consent
T	Social pressure to achieve rapid results leads to program-hopping and poor adherence

SWOT: UNDERWEIGHT USER (BMI < 18.5)

S	Low body fat = excellent muscle-to-fat ratio gains during bulk
S	Can sustain aggressive calorie surplus without rapid fat accumulation
W	Low appetite makes hitting calorie surplus consistently difficult
W	High NEAT (unconscious movement) burns additional calories reducing surplus
W	Often misreports "eating a lot" — tracking reveals actual intake is far below target
O	Liquid calories (milk shakes, mass gainers) bypass low appetite — 500ml whole milk + PB = +600 kcal
O	Beginner status means fastest possible muscle gain rate alongside caloric surplus
T	Risk of dirty bulk → rapid fat accumulation if surplus is set too aggressively
T	Medical causes of underweight should be ruled out before starting aggressive protocol

SWOT: SKINNY FAT USER

S	Normal or higher BMI but body composition improvement is achievable without bulk
S	Insulin sensitivity often high initially — recomp protocol effective
W	Simultaneous goals (lose fat, gain muscle) create conflicting macro requirements
W	Scale weight may not change even when body is recomposing — risk of "plateau" perception
W	Often incorrectly attempts bulk → accelerates fat gain, worsens appearance
O	Recomp strategy (TDEE ±0, 2.0 g/kg protein) simultaneously improves both metrics
O	Resistance training alone drives significant improvements in first 3-6 months
T	High body fat with normal BMI = elevated metabolic disease risk if untreated
T	Giving up early when scale doesn't move — must track measurements and progress photos

SWOT: FEMALE ATHLETE

S	Estrogen is anabolic and anti-inflammatory — supports recovery, especially in follicular phase
S	Generally better pain tolerance and higher training volume capacity than males
W	Lower absolute testosterone limits absolute muscle gain rate vs males
W	Monthly menstrual cycle affects energy, strength, and recovery unpredictably
W	Higher risk of iron deficiency anemia (monthly blood loss + often low meat intake)
W	Society/media pressure creates fear of "getting bulky" — limits heavy lifting potential
O	Programme heaviest sessions in follicular phase (days 1-14) when estrogen + strength peak
O	Resistance training provides bone density benefits — reduces osteoporosis risk long-term
T	Red flags: menstrual irregularity during high training + caloric deficit = hormonal disruption
T	Essential BF% floor is 15% for females — below this causes hormonal and health risks

SWOT: HARDGAINER

S	Typically lean with low body fat — can sustain larger surplus without excessive fat gain
S	High NEAT means cardiovascular health is excellent
W	Chronically under-eating despite believing otherwise — intake audit always reveals gap
W	High NEAT burns hundreds of extra calories — TDEE significantly higher than calculated
W	Low appetite and high satiety from protein makes hitting calorie target daily difficult
O	Liquid calorie strategy (1L whole milk + 2 tbsp PB + banana + whey = ~750 kcal) bypasses satiety
O	Calorie-dense foods (nuts, ghee, avocado, PB, dry fruits) maximise calorie per bite
T	Excessive cardio kills surplus — must quantify and reduce NEAT during building phase
T	Weekend caloric drop: many hardgainers eat less on weekends, erasing weekday surplus

3.2 SWOT GENERATION RULES

- [SR001] Always generate full SWOT (all 4 quadrants) during onboarding assessment.
- [SR002] SWOT must be personalised per user archetype — never use generic template output.
- [SR003] Strengths section: highlight assets the user may not recognise (age, diet type, training age).
- [SR004] Weaknesses: frame constructively as "areas to improve" not deficiencies.
- [SR005] Opportunities: always include at least 2 budget/accessible-level opportunities.
- [SR006] Threats: flag health risks first (hormonal, injury, medical) before lifestyle threats.
- [SR007] IF user is female AND in luteal phase → THEN flag recovery priority as active SWOT threat.
- [SR008] IF hostel student AND no kitchen → THEN soy chunks cold soak method listed as key Opportunity.
- [SR009] IF FFMI > 25 (male) or > 22 (female) → THEN flag PED suspicion as SWOT threat.
- [SR010] SWOT updates should be triggered every 90 days or after major lifestyle change.

MODULE 4 | NUTRITION SCIENCE

Protein · Carbohydrates · Fats · Fiber · Micronutrients · Hydration · MPS triggers · Meal timing

4.1 PROTEIN SCIENCE

Daily Targets by Athlete Type

Athlete Type	Minimum (g/kg)	Optimal (g/kg)	Max Useful (g/kg)	Notes
Beginner (any)	1.4	1.6-1.8	2.0	Newer trainees benefit from lower end
Intermediate	1.6	1.8-2.0	2.2	Higher end during caloric deficit
Advanced	1.8	2.0-2.2	2.4	High training volume demands more
Lean Bulk Phase	1.6	1.8-2.0	2.2	Surplus provides anabolic environment
Cutting Phase	2.0	2.2-2.6	3.0	High protein preserves muscle in deficit
Recomp	1.8	2.0-2.4	2.8	Critical for simultaneous goals
Vegetarian	+10%	2.0-2.4 g/kg	2.8	Lower bioavailability needs compensation
Female Athlete	1.4	1.6-2.0	2.2	Similar to male relative to LBM
Older Adult (40+)	1.8	2.0-2.4	2.8	Anabolic resistance increases with age

Leucine Threshold & MPS Activation

Protein Source	Leucine %	Needed for 3g LEU	Notes
Whey Protein	10-11%	~27g	Fast-absorbing. Best leucine source per gram.
Eggs (whole)	8.5%	~35g (6 eggs)	Complete EAA profile. Versatile.
Chicken Breast	8%	~38g	Lean, complete, high quality.
Fish (Tuna/Salmon)	8-9%	~33-38g	Salmon adds omega-3 bonus.
Paneer	7%	~43g	Key lacto-vegetarian source.
Soy Protein Isolate	7.5%	~40g	Best plant protein for MPS.
Soy Chunks	6.5%	~46g dry	Budget-friendly. Complete protein.
Lentils / Dal	6-7%	~43-50g	Incomplete — combine with rice for full EAA.
Pea Protein	6-8%	~38-50g	Good plant alternative to whey.
Peanuts / PB	4%	~75g	Low leucine density. Supplement with dairy.

Minimum leucine to maximally stimulate MPS: 2-3g per meal. A 30-40g dose of high-quality protein typically delivers this threshold. Plant proteins require higher doses due to lower leucine density.

Meal Timing Protocol

Timing Window	Protein	Carbs	Fat	Examples
Pre-Workout (2-3 hrs)	25-40g	60-80g complex	Low fat (<10g)	Rice + chicken, Oats + milk + eggs
Pre-Workout (30-60 min)	15-20g	30-50g fast	Minimal	Banana + curd, Bread + egg whites
Post-Workout (0-60 min)	30-50g fast	60-120g high GI	Low (<10g)	Whey + milk + banana, Eggs + rice
Between Meals (every 3-5 hrs)	30-40g	Mixed	Normal	Maintain MPS stimulation throughout day
Pre-Sleep (30-40g casein)	30-40g slow	50-100g moderate	Low	Paneer (200g), Greek yogurt, Casein shake
Morning (within 60 min)	30-40g	Moderate	Normal	Break overnight fast, initiate MPS

4.2 CARBOHYDRATE SCIENCE

Daily Carbohydrate Targets

GOAL	TARGET (g/kg BW)	DAILY EXAMPLE (70kg)	NOTES
Lean Bulk (moderate activity)	4-5 g/kg	280-350g (70kg)	Fill remaining calories after protein + fats
Lean Bulk (high activity)	5-7 g/kg	350-490g (70kg)	High-volume athletes need significantly more
Cutting (preserve muscle)	2-3 g/kg	140-210g (70kg)	Maintain training performance despite deficit
Recomp	3-4 g/kg	210-280g (70kg)	Cycle carbs: higher on training days
Rest Day	2-3 g/kg	140-210g (70kg)	Lower carbs acceptable on non-training days
Heavy Training Day	5-7 g/kg	350-490g (70kg)	Load up for performance and glycogen resynthesis

Glycogen System

PARAMETER	CAPACITY / RATE	FUNCTION
Muscle Glycogen	300-600g (1,200-2,400 kcal)	Fuel for working muscle during training
Liver Glycogen	70-100g (280-400 kcal)	Blood glucose maintenance between meals
Depletion Rate (intense)	~1-2g/min during high-intensity training	Full depletion in 45-90 min of intense work
Replenishment Rate	~5-8g carbs/hr/kg body weight	Very fast with adequate post-workout carb intake

4.3 DIETARY FAT SCIENCE

FAT TYPE	SOURCES	MUSCLE GAIN ROLE	RECOMMENDATION
Saturated Fat	Ghee, butter, eggs, whole milk, red meat	Testosterone precursor, cell membrane structure	15-20% of fat intake. Include, don't eliminate.
Monounsaturated (MUFA)	Olive oil, peanuts, almonds, avocado	Anti-inflammatory, hormone support, heart health	Primary fat source. 40-50% of fat intake.
Omega-6 PUFA	Sunflower oil, safflower, corn oil	Cell membrane, inflammatory signalling	Limit to 5-10%. Most Indians already over-consume.
Omega-3 PUFA	Fatty fish, flaxseed, walnuts, fish oil	Anti-inflammatory, MPS enhancement, insulin sensitivity	PRIORITIZE. 2-4g EPA+DHA per day target.
Trans Fats (artificial)	Vanaspati, margarine, ultra-processed	Reduces testosterone, promotes inflammation	AVOID entirely. Zero tolerance.

4.4 MICRONUTRIENTS, FIBER & HYDRATION

MICRONUTRIENT	DAILY TARGET	IMPORTANCE FOR ATHLETES
Vitamin D	1,000-2,000 IU/day	Testosterone production, MPS, bone health. Majority of Indians deficient.
Magnesium	400-420 mg/day (M), 310-360 (F)	ATP production, protein synthesis, muscle relaxation. Lost in sweat.
Zinc	11 mg/day (M), 8 mg/day (F)	Testosterone cofactor, immune function. Athletes at high deficiency risk.
Iron	8 mg/day (M), 18 mg/day (F)	Oxygen transport. Female athletes and vegetarians at high risk.
Vitamin B12	2.4 mcg/day	Critical for vegans/vegetarians. Nerve function, RBC production.
Calcium	1,000-1,200 mg/day	Muscle contraction, bone health, nerve function.
Fiber	25-35g/day	Gut health, satiety, blood sugar control. Avoid >35g/day to prevent GI distress.
Hydration	35-40 ml/kg BW/day	Add 500-750ml per training hour. Electrolytes for heavy sweaters.

4.5 MODULE 4 COACHING RULES

- [R021] Default protein target for muscle gain: 1.8-2.0 g/kg bodyweight. Vegetarians: add 10-20%.
- [R022] Each meal should contain 30-50g protein to maximally stimulate MPS via leucine threshold.
- [R023] Distribute protein evenly across 4-5 meals. Never front-load or back-load all protein in one meal.
- [R024] Post-workout: 30-50g fast protein (whey preferred) + 60-120g high-GI carbs within 60 minutes.
- [R025] Pre-sleep protein: 30-40g casein/paneer/Greek yogurt. Extends overnight MPS by 7-8 hours.
- [R026] Default carb recommendation for bulking: 4-6 g/kg BW/day. Never below 2g/kg during muscle gain.
- [R027] Minimum fat intake: 20% of total calories. Never prescribe below this — testosterone risk.
- [R028] Target 2-4g EPA+DHA omega-3 per day. Supplement with fish oil if fatty fish intake is <2x/week.
- [R029] Vitamin D3: supplement routinely for Indian athletes who spend most time indoors.
- [R030] Hydration target: 35-40ml/kg BW/day. Add 500-750ml per training hour in Indian summer.

MODULE 5 | MUSCLE GAIN FOOD DATABASE

200+ foods with full macros · Animal protein · Plant protein · Dairy · Carbs · Fats · Budget ratings

5.1 HOW TO READ THIS DATABASE

Each entry shows: **Name** | **Cal** (per 100g) | **Prot** (g) | **Carbs** (g) | **Fat** (g) | **Fiber** (g) | **Serving** | **Cost (INR)** | **Tags**. Tags: NV=Non-Veg | V=Vegetarian | VG=Vegan | H=Hostel-Friendly | B=Budget | A=Athlete-focus. All values per 100g unless serving size specified differently. Use serving column for real-world portions.

5.x ANIMAL PROTEIN SOURCES

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Chicken Breast (cooked)	165	31	0	3.6	0	100g	60-90	NV A
Chicken Thigh (cooked)	209	26	0	11	0	100g	50-70	NV B
Chicken Leg (cooked)	185	27	0	8	0	100g	40-60	NV B
Chicken Liver	167	25	0.7	6.5	0	100g	30-50	NV B
Chicken Wings	203	30	0	9	0	100g	60-80	NV
Turkey Breast (cooked)	135	30	0	1	0	100g	100-150	NV A
Beef (lean 95%)	152	26	0	4.7	0	100g	150-200	NV A
Beef (regular 80%)	215	26	0	12	0	100g	100-150	NV
Lamb (lean)	258	25	0	17	0	100g	200-300	NV
Mutton (cooked)	294	25	0	21	0	100g	150-250	NV
Pork Tenderloin	143	26	0	3.5	0	100g	150-200	NV A
Fish - Rohu	97	17	0	2.7	0	100g	60-100	NV B
Fish - Catla	111	19	0	3.4	0	100g	70-110	NV B
Fish - Pomfret	87	17	0	2	0	100g	150-250	NV
Fish - Tilapia	128	26	0	2.7	0	100g	80-130	NV A
Fish - Salmon (imported)	208	20	0	13	0	100g	400-600	NV A
Fish - Sardines (canned)	208	24	0	11	0	100g	40-80	NV H B
Fish - Tuna (canned)	128	29	0	0.5	0	100g	50-80	NV H B A
Fish - Mackerel (Bangda)	205	19	0	14	0	100g	80-120	NV B
Prawns (cooked)	99	24	0	0.5	0	100g	200-400	NV A
Squid (Calamari)	92	16	3.1	1.4	0	100g	150-250	NV
Duck Breast	201	27	0	10	0	100g	200-300	NV

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Fish Oil Capsule	900	0	0	100	0	1 capsule (1g)	2-5	NV A
Chicken Breast (boiled, packaged)	165	31	0	3.6	0	100g	80-120	NV H A
Canned Tuna in Water	116	26	0	0.5	0	1 can (185g)	50-80	NV H B

5.x PLANT PROTEIN SOURCES

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Soy Chunks / Nutri Nuggets (dry)	345	52	33	0.5	4.5	100g dry	20-30	V H B VG
Soy Chunks (cooked)	149	22	14	0.2	2	100g cooked	20-30	V H B VG
Tofu (firm)	144	17	3	9	0.3	150g	40-70	V VG B
Tempeh	195	19	9	11	0	100g	80-150	V VG A
Edamame (soybeans)	121	11	9	5	5	100g	60-100	V VG B
Moong Dal (raw)	347	24	63	1.2	7	100g dry	30-50	V H B VG
Masoor Dal (raw)	352	26	60	1	8	100g dry	25-40	V H B VG
Chana Dal (raw)	364	25	61	6	9	100g dry	30-45	V H B VG
Toor Dal / Arhar (raw)	335	22	63	1.5	7	100g dry	60-90	V H B VG
Urad Dal (raw)	347	26	59	1.5	7	100g dry	50-80	V H B VG
Rajma / Kidney Beans	337	22	61	1.5	7	100g dry	40-70	V H B VG
Chickpeas / Chole (raw)	364	19	61	6	10	100g dry	40-70	V H B VG
Roasted Chana (Bhuna Chana)	384	22	63	6	10	100g	20-30	V H B VG
Black-Eyed Peas (Lobia)	336	24	60	1.4	10.5	100g dry	30-50	V H B VG
Moth Beans (Matki)	343	23	62	1.5	9	100g dry	30-50	V H B VG

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Peanuts (raw)	567	26	16	49	8.5	100g	20-35	V H B VG
Peanut Butter (natural)	588	25	20	50	6	32g (2 tbsp)	25-40	V H B VG
Almonds	579	21	22	50	12.5	30g (20 nuts)	50-90	V VG
Cashews	553	18	30	44	3.3	30g	50-90	V VG
Walnuts	654	15	14	65	6.7	30g	60-100	V VG A
Hemp Seeds	553	31	9	49	4	30g	100-200	V VG A
Pumpkin Seeds	559	30	11	49	6	30g	30-60	V VG A
Sunflower Seeds	584	21	20	51	8.6	30g	15-30	V VG
Spirulina (powder)	290	57	24	8	3.6	10g	100-200	V VG A
Chickpea Flour (Besan)	387	22	58	6	10	100g	20-35	V H B VG
Pea Protein Powder	360	80	4	3	1	30g scoop	100-200	V VG A
Soy Protein Isolate	370	90	1	0.5	0	30g scoop	150-250	V VG A
Oats (rolled, dry)	389	17	66	7	10.6	100g dry	8-15	V H B VG
Wheat Roti / Chapati	297	11	62	1.7	2.7	2 medium (60g)	5-10	V H B VG
Multigrain Roti	279	12	55	3.5	5	2 medium (60g)	8-15	V B VG
Whole Wheat Bread	247	13	44	3.4	7	2 slices (60g)	10-20	V H B VG
White Bread	265	9	51	3.2	2.7	2 slices (60g)	8-15	V H B VG
Maize Flour / Makki Atta	358	9	74	3.9	7.3	100g	10-20	V B VG
Jowar / Sorghum Flour	349	10	74	1.7	3.9	100g	12-22	V B VG
Bajra / Pearl Millet	361	11	68	5	1.2	100g	10-18	V B VG
Quinoa (dry)	368	14	64	6	7	100g	150-300	V VG A

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Oat Bran	246	17	66	7	15	100g	15-30	V VG
Popcorn (air-popped)	387	13	78	4.5	15	100g popped	10-20	V VG
Almond Butter	614	21	18	56	10	32g (2 tbsp)	60-120	V VG
Tahini (Sesame Paste)	595	17	21	54	9.3	30g (2 tbsp)	50-100	V VG
Chia Seeds	486	17	42	31	34	30g (2 tbsp)	20-40	V VG A
Flaxseeds / Alsi	534	18	29	42	27	30g (2 tbsp)	10-20	V VG B
Dark Chocolate (85%+)	598	12	22	57	10	30g	30-60	V VG
Mixed Nuts (unsalted)	610	16	21	54	7	30g	50-100	V VG
Peanut Butter (Sundrop/Pintola)	588	25	20	50	6	32g	25-40	V H B VG
Sattu (Roasted Gram Flour)	406	25	65	6.5	7	50g (2-3 tbsp)	10-20	V H B VG

5.x DAIRY & DAIRY PROTEIN

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Whole Eggs	155	13	1.1	11	0	2 eggs (100g)	15-20	V H B
Boiled Eggs	155	13	1.1	11	0	2 eggs (100g)	15-20	V H B
Tofu (silken)	61	5.5	2	3.6	0	150g	40-70	V VG
Paneer (full fat)	296	18	3.1	23	0	100g	60-100	V
Paneer (low fat)	193	20	3	12	0	100g	70-110	V A
Greek Yogurt (plain)	97	9	3.6	5	0	200g	60-100	V A
Hung Curd (Chakka)	140	7	5	10	0	100g	20-30	V
Whey Protein (concentrate)	380	75	6	5	0	30g scoop	50-100	V A
Cottage Cheese (Chena)	99	12	3.4	4.3	0	100g	40-70	V
Hard Cheese (Cheddar)	402	25	1.3	33	0	30g	60-100	V
Pesto	370	8	7	35	2.5	30g	40-80	V VG
Whey Protein (MuscleBlaze)	390	78	3.5	5	0	33g scoop (1 serving)	60-100	V A

5.x CARBOHYDRATE SOURCES

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Condensed Milk	321	7.9	54	8.7	0	100g	30-50	V

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Poha / Flattened Rice	367	7	80	1.5	0.8	100g dry	8-15	V H B VG
Ragi / Finger Millet	328	7	73	1.5	11	100g	15-25	V B VG
Dates (Khajoor)	282	2.5	75	0.4	8	5 dates (50g)	30-60	V H B VG
Cornflakes (fortified)	375	7	84	0.4	3	100g	25-40	V H
Rice Cakes	387	7	81	1.9	1.9	100g	30-60	V H
Sabudana / Tapioca	352	0.3	88	0.2	0.9	100g dry	15-25	V H
Puffed Rice / Murmura	402	7	90	0.5	0	100g	8-15	V H B VG
Garlic (raw)	149	6.4	33	0.5	2.1	10g (3 cloves)	2-5	V VG B
Dates (dried Khajoor)	282	2.5	75	0.4	8	5 dates (50g)	30-60	V VG H B
Raisins / Kishmish	299	3.1	79	0.5	3.7	30g	20-40	V VG H
Figs / Anjeer (dried)	249	3.3	64	0.9	9.8	4 figs (60g)	30-70	V VG
Honey	304	0.3	82	0	0.2	1 tbsp (21g)	15-30	V
Jaggery / Gud	383	0.4	98	0.1	0	1 tbsp (20g)	5-10	V VG H B
Dark Chocolate (>70%)	598	7.8	45	42	8	30g	30-60	V VG

5.x FAT SOURCES

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Butter	717	0.9	0.1	81	0	10g	10-20	V
Ghee	900	0	0	100	0	1 tsp (5g)	10-20	V
Groundnut Oil	884	0	0	100	0	1 tbsp (14g)	5-10	V VG
Mustard Oil	884	0	0	100	0	1 tbsp (14g)	5-10	V VG
Coconut Oil	862	0	0	100	0	1 tbsp (14g)	10-20	V VG
Olive Oil	884	0	0	100	0	1 tbsp (14g)	20-40	V VG A
Sunflower Oil	884	0	0	100	0	1 tbsp (14g)	5-10	V VG
Sesame Oil	884	0	0	100	0	1 tbsp (14g)	15-30	V VG
Ghee (clarified butter)	900	0	0	100	0	1 tsp (5g)	10-20	V
Butter (salted)	717	0.9	0.1	81	0	10g	10-20	V

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Coconut Cream	330	3.5	6	34	2.2	50g	20-40	V VG
Coconut Milk (canned)	197	2	3	21	0.5	100ml	20-40	V VG
Mayonnaise	680	1	0.6	75	0	15g (1 tbsp)	15-30	V
Coconut (fresh, grated)	354	3.3	15	33	9	50g	10-20	V VG B

5.x VEGETABLES

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Skimmed Milk	36	3.5	5.1	0.1	0	250ml	12-22	V H B A
Chaas / Buttermilk	40	1.4	4.8	1.3	0	250ml	5-15	V H B
Spinach / Palak (raw)	23	2.9	3.6	0.4	2.2	100g	5-15	V VG B
Spinach (cooked)	41	3	6.7	0.5	2.4	100g	5-15	V VG B
Broccoli (cooked)	35	2.4	7.2	0.4	3.3	100g	25-50	V VG
Broccoli (raw)	34	2.8	7	0.4	2.6	100g	25-50	V VG
Cauliflower (cooked)	29	2.2	5.3	0.4	2.9	100g	15-30	V VG B
Cabbage (raw)	25	1.3	5.8	0.1	2.5	100g	8-15	V VG B
Kale / Lauki Saag	35	2.9	4.4	1.5	4.1	100g	10-20	V VG
Bottle Gourd / Lauki	14	0.6	3.4	0.1	0.5	100g	8-15	V VG B
Ridge Gourd / Turai	18	1.2	3.6	0.2	0.7	100g	8-15	V VG B
Bitter Gourd / Karela	17	1	3.5	0.2	2.8	100g	10-20	V VG B
Capsicum / Bell Pepper (red)	31	1	6	0.3	2.1	100g	20-40	V VG
Capsicum (green)	20	0.9	4.6	0.2	1.7	100g	15-30	V VG B
Tomato (raw)	18	0.9	3.9	0.2	1.2	100g	5-15	V VG B
Onion (raw)	40	1.1	9.3	0.1	1.7	100g	5-15	V VG B

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Mushroom / Khumb	22	3.1	3.3	0.3	1	100g	30-60	V VG
Beans (French/Green)	31	1.8	7	0.1	3.4	100g	15-30	V VG B
Drumstick / Moringa	37	2.1	8.5	0.2	4.5	100g	10-25	V VG B
Cucumber / Kheera	15	0.7	3.6	0.1	0.5	100g	5-15	V VG B
Zucchini / Courgette	17	1.2	3.1	0.3	1	100g	20-40	V VG
Pumpkin (cooked)	26	1	6.5	0.1	0.5	100g	8-15	V VG B
Ash Gourd / Petha	13	0.4	3	0.2	0.6	100g	8-15	V VG B
Lady Finger / Okra (Bhindi)	33	1.9	7.5	0.2	3.2	100g	10-20	V VG B
Cluster Beans / Guar Phali	16	3.2	1.8	0.4	5.7	100g	8-15	V VG B
Raw Papaya	32	0.6	8	0.1	0.9	100g	8-15	V VG B
Turnip (Shalgam)	28	0.9	6.4	0.1	1.8	100g	8-15	V VG B
Radish / Mooli (raw)	16	0.7	3.4	0.1	1.6	100g	5-12	V VG B
Watermelon	30	0.6	7.6	0.2	0.4	200g slice	5-15	V VG H B
Strawberry	32	0.7	7.7	0.3	2	100g	40-80	V VG
Creatine Monohydrate	0	0	0	0	0	5g	15-25/month	V VG A
Electrolyte Powder	30	0	7.5	0	0	1 sachet	10-30	V VG H
Green Tea (brewed)	1	0.2	0.2	0	0	250ml	2-5	V VG H B
Coffee (black)	2	0.3	0	0	0	250ml	5-10	V VG H B
Coconut Water	19	0.7	3.7	0.2	1.1	250ml	10-20	V VG H B
Sports Drink (Gatorade etc.)	26	0	6.8	0	0	500ml	20-40	V VG

5.x FRUITS

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Green Peas (fresh)	81	5.4	14	0.4	5.1	100g	15-30	V VG B
Oats (cooked)	71	2.5	12	1.5	1.7	200g cooked	8-15	V H B VG
Sweet Potato (cooked)	86	1.6	20	0.1	3	200g	15-25	V H B VG
Regular Potato (boiled)	77	2	17	0.1	2	200g	8-15	V H B VG
Idli (steamed)	58	2	11	0.4	0.5	2 pieces (100g)	5-15	V H B VG
Ginger (raw)	80	1.8	18	0.8	2	10g	2-5	V VG B
Peas (fresh/frozen)	81	5.4	14	0.4	5.1	100g	15-30	V VG B
Lotus Stem / Kamal Kakdi	74	1.6	17	0.1	3.1	100g	20-40	V VG
Banana (medium)	89	1.1	23	0.3	2.6	1 medium (118g)	5-15	V VG H B
Apple	52	0.3	14	0.2	2.4	1 medium (182g)	20-40	V VG
Mango (Alphonso)	65	0.8	17	0.4	1.8	100g	15-40	V VG H B
Papaya	43	0.5	11	0.3	1.7	100g	5-15	V VG H B
Orange	47	0.9	12	0.1	2.4	1 medium (131g)	10-25	V VG B
Guava	68	2.6	14	1	5.4	100g	5-15	V VG B
Grapes (black/green)	69	0.7	18	0.2	0.9	100g	15-30	V VG B
Pineapple	50	0.5	13	0.1	1.4	100g	10-20	V VG B
Pomegranate (Anar)	83	1.7	19	1.2	4	100g arils	20-50	V VG
Chickoo / Sapodilla	83	0.4	20	1.1	5.3	100g	10-20	V VG B
Kiwi	61	1.1	15	0.5	3	1 fruit (69g)	30-60	V VG
Blueberries (fresh/frozen)	57	0.7	14	0.3	2.4	100g	80-200	V VG

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Lychee	66	0.8	17	0.4	1.3	100g	20-50	V VG B
Pear	57	0.4	15	0.1	3.1	1 medium (166g)	20-40	V VG

5.x MISCELLANEOUS / CONDIMENTS

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tags
Egg Whites	52	11	0.7	0.2	0	3 whites (90g)	10-15	V H B A
Moong Dal (cooked)	105	7	19	0.4	2	100g cooked	30-50	V H B VG
Quinoa (cooked)	120	4.4	21	1.9	2.8	100g cooked	80-150	V VG A
Soy Milk (fortified)	54	3.3	6	1.8	0.5	250ml	30-50	V VG
Whole Milk	61	3.2	4.8	3.3	0	250ml	15-25	V H B
Toned Milk (2%)	50	3.5	5	1.5	0	250ml	12-22	V H B
Curd / Dahi (full fat)	98	3.5	4.7	6.4	0	200g	15-25	V H B
Curd (low fat)	63	4	5.5	2	0	200g	15-25	V H B A
Whey Protein (isolate)	375	90	2	1.5	0	30g scoop	80-150	V A
Casein Protein	360	80	4	2	0	30g scoop	80-150	V A
Lassi (homemade, salted)	72	2.5	5	3.4	0	250ml	10-20	V H B
Ice Cream (vanilla)	207	3.5	24	11	0	100g	20-40	V
White Rice (cooked)	130	2.7	28	0.3	0.4	200g cooked	8-15	V H B VG
Brown Rice (cooked)	112	2.6	24	0.9	1.8	200g cooked	12-20	V H B VG
Basmati Rice (cooked)	121	3.5	25	0.4	0.5	200g cooked	12-22	V H B VG
Pasta (cooked)	131	5	25	1.1	1.8	200g cooked	15-25	V H
Whole Wheat Pasta (cooked)	124	5.5	24	0.8	3.5	200g cooked	20-35	V
Dosa (plain)	168	3.5	30	3.7	0.5	1 medium (100g)	10-20	V H B VG
Upma	173	3	28	5	1.5	200g	10-20	V H B VG
Corn / Maize (boiled)	99	3.3	21	1.5	2.4	200g	10-20	V H B VG
Muesli (no sugar)	350	9	65	5	7	100g	30-60	V

FOOD	Cal	Prot(g)	Carbs(g)	Fat(g)	Fiber(g)	Serving	Cost(INR)	Tag s
Bread Sticks / Rusks	390	11	76	4.5	2	100g	20-40	V H
Suji / Semolina	360	12	73	1	3.9	100g	8-15	V H B
Maida / Refined Flour	364	10	76	0.8	2.7	100g	5-12	V H B
Vermicelli (Sevai, cooked)	148	3.2	30	1.2	1	100g cooked	8-15	V H B
Avocado	160	2	9	15	7	100g (half)	80-150	V VG A
Carrot (raw)	41	0.9	10	0.2	2.8	100g	8-15	V VG B
Beetroot (cooked)	44	1.7	10	0.2	2	100g	10-20	V VG B
Corn on Cob (boiled)	96	3.4	21	1.5	2.4	100g kernels	10-20	V VG B
Colocasia / Arbi	112	1.5	26	0.1	4.1	100g cooked	10-20	V VG B
Yam / Suran	118	1.5	28	0.2	4.1	100g cooked	15-25	V VG B
Custard Apple / Sitaphal	94	2.1	23	0.6	4.4	100g	20-50	V VG
Amla / Indian Gooseberry	44	0.9	10	0.6	4.3	100g	5-15	V VG B
Mass Gainer (commercial)	370	30	65	3	1	75g serving	50-90	V
DIY Mass Gainer (oats+PB+whey+milk)	900	70	95	25	8	1 serving	60-100	V B
Protein Bar (Quest/RiteBite)	200	21	22	6	14	1 bar (60g)	50-100	V H
Protein Muffin (homemade)	250	20	25	7	3	1 muffin (80g)	30-50	V
Protein Shake (homemade-milk+whey+banana)	450	48	52	8	2	1 glass	50-80	V
Egg White Powder	350	83	3	0.3	0	30g	80-150	V A

5.8 PROTEIN COST-EFFICIENCY RANKING

Ranked by grams of protein per INR 10 spent. Best value for budget athletes.

FOOD	PROTEIN CONTENT	COST RANGE	EFFICIENCY	NOTES
Soy Chunks (dry)	52g/100g	INR 20-25	20-26g protein per INR 10	VG, H, B — Best overall
Eggs (whole)	13g/2 eggs	INR 15-20	8-10g protein per INR 10	V, H, B — Most versatile
Moong Dal (dry)	24g/100g	INR 30-50	5-8g protein per INR 10	VG, H, B — Good amino profile
Roasted Chana	22g/100g	INR 20-30	8-10g protein per INR 10	VG, H, B — Snack-friendly
Peanuts / PB	26g/100g	INR 20-35	7-13g protein per INR 10	VG, B — Low leucine

FOOD	PROTEIN CONTENT	COST RANGE	EFFICIENCY	NOTES
Tuna (canned)	29g/100g	INR 50-80	4-6g protein per INR 10	NV, H, B — Complete protein
Chicken Breast	31g/100g	INR 60-90	3-5g protein per INR 10	NV, A — Best animal source
Paneer (full fat)	18g/100g	INR 60-100	2-3g protein per INR 10	V — Slow-digesting casein
Whey Protein	75g/100g	INR 50-100/scoop	7-15g protein per INR 10	V, A — Supplement only
Greek Yogurt	9g/200g	INR 60-100	1-2g protein per INR 10	V, A — Casein protein

MODULE 6 | RECIPE LIBRARY

120+ recipes · All diet types · All budgets · Hostel-friendly · Pre/Post workout · Mass gainer

6.1 HOW TO USE THIS LIBRARY

Each recipe shows: **Name** | **Category** | **Kcal** | **Protein (g)** | **Carbs (g)** | **Fat (g)** | **Prep Time** | **Key Ingredients** | **Notes**. Recipes are sorted by category. Use filters: BK=Breakfast | LN=Lunch | DN=Dinner | PRE=Pre-Workout | PST=Post-Workout | SN=Snack | HS=Hostel | MG=Mass Gainer. All macros are approximate per serving.

6.x BREAKFAST RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Egg & Oats Power Bowl	520	38	55	14	10	4 eggs, 80g oats, 300ml milk, salt, pepper	Cook oats in milk. Scramble eggs. Combine. Add seasoning.
Masala Omelette	340	28	8	22	8	4 eggs, onion, tomato, green chilli, coriander, oil	Beat eggs with veggies. Cook in non-stick pan. Serve with bread.
Protein Poha	420	32	55	8	12	100g poha, 100g soy chunks, veggies, mustard seeds, turmeric	Soak poha, cook soy chunks, temper spices, mix all together.
Peanut Butter Banana Smoothie	480	22	62	16	5	2 bananas, 2 tbsp PB, 300ml milk, 1 scoop whey, ice	Blend all ingredients until smooth. Serve chilled.
Paneer Bhurji	380	28	8	26	10	200g paneer, onion, tomato, spices, oil	Crumble paneer. Saute onion+tomato+spices. Mix paneer, cook 3 min.
Boiled Egg Plate	290	30	28	10	8	5 boiled eggs, 2 slices bread, tomato, salt, pepper	Boil eggs 8-10 min. Serve with bread and raw veggies.
Sattu Protein Drink	320	22	52	5	3	60g sattu, 300ml water/milk, lemon, jaggery or salt	Mix sattu in liquid. Add lemon and sweetener. Stir well.
Soy Chunk Upma	390	28	52	8	15	100g semolina, 60g soy chunks, veggies, mustard, ghee	Roast semolina. Cook soy chunks. Temper spices, add all together.
Greek Yogurt Parfait	350	24	40	8	5	200g Greek yogurt, 50g oats, 1 banana, honey	Layer yogurt, oats, sliced banana. Drizzle honey.
Protein Dosa	420	30	48	10	20	100g urad dal, 50g chana dal, 50g soy flour, ferment overnight	Grind mixed dal batter. Ferment. Make thin dosas on hot griddle.
Scrambled Eggs & Multigrain Toast	380	30	35	15	8	4 eggs, 3 tbsp milk, 2 multigrain slices, butter	Whisk eggs with milk. Soft scramble in butter. Serve on toast.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Besan Chilla (Protein Pancake)	360	24	42	10	12	100g besan, 50g paneer, veggies, spices, oil	Mix besan with water, add crumbled paneer+veggies. Pan cook.
Dal Paratha Plate	480	22	68	12	20	3 parathas stuffed with spiced dal, curd 100g	Stuff dal in paratha dough. Cook on tawa with ghee. Serve with curd.
Milk + Banana + PB Stack	450	20	58	16	2	500ml whole milk, 1 banana, 2 tbsp peanut butter	Drink milk, eat banana, eat PB directly or blend together.
Curd & Chana Breakfast Bowl	330	22	42	7	3	200g curd, 100g roasted chana, salt, cumin, cucumber	Mix curd with chana and cucumber. Season with cumin salt.
Sprouts Salad Plate	260	18	38	4	10	100g mixed sprouts, tomato, cucumber, lemon, salt	Soak overnight. Rinse and boil if needed. Toss with veggies.
Moong Dal Chilla	340	22	48	6	15	100g split moong dal (soaked), veggies, spices	Grind soaked moong. Add veggies+spices. Pan cook like pancakes.
Oats Protein Pancakes	480	34	52	12	15	80g oat flour, 3 eggs, 1 scoop whey, milk, baking powder	Mix all ingredients. Pour batter into pan. Cook 2-3 min each side.
Egg Bhurji + 3 Rotis	520	34	62	16	12	4 eggs, onion, tomato, spices, oil, 3 rotis	Make bhurji masala. Scramble eggs in. Serve hot with rotis.
Milk Oats with Almonds	430	18	58	14	5	80g oats, 300ml milk, 10 almonds, honey, cinnamon	Cook oats in milk. Top with crushed almonds. Add honey+cinnamon.

6.x LUNCH RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Chicken Breast + Rice + Dal	720	58	85	14	30	200g chicken breast, 250g cooked rice, 200g cooked dal, veggies	Boil/grill chicken with spices. Serve with rice and dal.
Tuna Brown Rice Bowl	520	45	55	8	10	1 can tuna (185g), 200g brown rice, cucumber, lemon, olive oil	Drain tuna. Mix with brown rice. Add raw veggies and lemon dressing.
Rajma Chawal	650	28	95	8	40	200g rajma (cooked), 300g rice, onion-tomato masala, ghee	Cook rajma in pressure cooker. Make masala. Serve with steamed rice.
Paneer Tikka Wrap	560	36	52	20	25	200g paneer, 2 rotis, hung curd marinade, veggies, chutney	Marinate paneer in spiced yogurt. Grill. Wrap in roti with veggies.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Soy Chunk Curry + 3 Rotis	580	40	72	12	25	100g soy chunks dry, onion-tomato gravy, spices, 3 rotis	Boil soy chunks, squeeze water. Cook in spiced gravy. Serve with rotis.
Grilled Fish + Sweet Potato Mash	490	42	42	10	25	200g fish (rohu/tilapia), 300g sweet potato, olive oil, herbs	Marinate fish. Grill 4-5 min each side. Mash sweet potato with oil.
Egg Curry + Brown Rice	560	32	58	18	30	4 eggs (boiled), onion-tomato-spice curry, 200g brown rice	Hard boil eggs. Make curry base. Add eggs, simmer 10 min.
Chole Bhature Muscle Version	680	24	95	18	40	200g chickpeas, 2 medium bhature, masala, onion, ginger	Pressure cook chickpeas. Make spiced masala. Serve with bhature.
Dal Makhani + 2 Parathas	640	26	82	20	60	200g black urad dal, butter, cream, spices, 2 whole wheat parathas	Pressure cook dal. Slow cook in butter+cream gravy 30 min.
Mutton Stew + Rice	720	48	60	24	60	200g mutton, 250g rice, mixed veggies, coconut milk optional	Pressure cook mutton with spices. Serve over rice with gravy.
Chicken Dal Rice Bowl	680	55	80	12	30	200g chicken, 200g cooked dal, 200g rice, turmeric, ghee	Mix rice-dal-chicken in bowl. Season with turmeric, top with ghee.
Palak Paneer + Roti	540	28	58	22	30	200g paneer, 200g spinach puree, spices, 3 rotis	Blanch and puree spinach. Cook paneer in spinach gravy.
Kadhi Chawal	480	18	72	12	30	200g curd, besan 50g, tempering, 300g cooked rice	Whisk curd with besan. Cook kadhi 20 min. Temper. Serve with rice.
Fish Curry + Rice	560	40	60	14	30	200g fish, coconut-tomato curry base, rice 250g	Marinate fish. Cook in curry base 10-15 min. Serve with rice.
Protein Buddha Bowl	580	38	62	14	20	150g grilled chicken/paneer, 200g brown rice, chickpeas, veggies, hummus	Assemble rice base. Top with protein, chickpeas, veggies, hummus.
Egg Fried Rice (high protein)	550	30	70	14	20	4 eggs, 300g cooked rice, soy sauce, veggies, sesame oil	Scramble eggs. Fry rice with veggies. Add eggs + soy sauce.
Sarson Ka Saag + Makki Roti	460	16	62	16	40	200g mustard leaves, 50g spinach, makki flour 100g, ghee	Cook greens in pressure cooker. Blend. Serve with makki roti.
Tofu Stir Fry + Noodles	520	28	62	16	20	200g firm tofu, 200g noodles, veggies, soy sauce, ginger-garlic	Press and fry tofu. Stir fry veggies. Toss with noodles.
Chickpea Salad Power Bowl	460	22	60	12	10	200g cooked chickpeas, cucumber, tomato, onion, lemon, olive oil	Toss chickpeas with chopped veggies and dressing.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Millet Khichdi + Curd	490	18	72	10	25	100g bajra/jowar, 50g moong dal, veggies, ghee, curd 150g	Pressure cook millet+dal together. Season. Serve with curd.

6.x DINNER RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Grilled Chicken + Dal + Sabzi	620	52	60	14	30	200g chicken breast, 200g dal, seasonal sabzi, roti	Grill seasoned chicken. Serve with dal and vegetable dish.
Paneer Bhurji + 3 Rotis	560	32	60	22	15	200g paneer, onion, tomato, spices, 3 rotis	Crumble and saute paneer with masala. Serve with rotis.
Fish Tikka + Rice	530	40	55	12	35	200g fish, tikka marinade, 250g rice, salad	Marinate fish 30 min. Grill or bake. Serve with steamed rice.
Mixed Dal (Panchmel) + Chapati	500	24	72	10	35	Mixed dal 200g (moong, masoor, chana), 3 chapatis, ghee	Pressure cook mixed dal. Make tadka. Serve with whole wheat chapati.
Egg Masala Curry + 2 Rotis	480	30	54	18	25	4 boiled eggs, tomato-onion masala, 2 rotis	Make rich masala. Add boiled eggs. Simmer. Serve with rotis.
Soy Keema + Bhatura	620	38	78	16	30	100g soy chunks (ground), onion-tomato, spices, 2 bhaturas	Pulse soy chunks. Cook keema masala style. Serve with bhatura.
Chicken Biryani (light)	680	45	75	18	60	200g chicken, 200g basmati, whole spices, yogurt, saffron	Marinate chicken. Layer with par-boiled rice. Dum cook 25 min.
Tuna Pasta	540	38	62	10	20	1 can tuna, 200g pasta, tomato sauce, olive oil, garlic	Cook pasta. Make tomato sauce. Mix in tuna. Toss together.
Lentil Soup + Bread	400	22	58	8	25	200g masoor/moong dal, tomato, onion, cumin, 2 slices bread	Boil dal until soft. Make tadka with tomatoes. Serve with bread.
Rajma + Brown Rice (night)	580	24	88	6	40	200g cooked rajma, 250g brown rice, tomato masala	Pressure cook rajma. Make curry. Serve with brown rice.
Stuffed Capsicum (Protein)	480	28	50	16	35	4 capsicums, stuffed with soy/paneer/chicken+rice+spices	Hollow capsicums. Fill with spiced protein+rice mix. Bake 20 min.
Chicken Stew (light)	420	40	28	12	40	200g chicken, potato, carrot, onion, pepper, light coconut milk	Brown chicken. Add veggies and coconut milk. Simmer 30 min.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Moong Dal Khichdi + Ghee	460	20	68	10	20	100g rice, 100g moong dal, ghee 1 tbsp, turmeric, cumin	Pressure cook rice+dal. Mash lightly. Season with ghee+spices.
Prawn Masala + Rice	520	36	58	14	30	200g prawns, onion-tomato masala, coconut, 250g rice	Cook prawn in spiced masala. Serve hot with steamed rice.
Paneer + Soy Protein Bowl	560	40	48	22	20	150g paneer, 60g soy chunks, curry base, 2 rotis	Cook both proteins in same curry base. High protein dinner.
Turkey/Chicken Meatballs + Pasta	580	45	55	15	35	200g minced chicken, 200g pasta, tomato sauce, herbs	Mix minced chicken+egg+herbs. Form balls. Bake 20 min. Serve on pasta.
Vegetable Korma + Roti	430	14	58	16	30	Mixed veggies, coconut cream, cashew paste, 3 rotis	Blend onion-cashew. Cook veggies in creamy korma sauce.
Egg White Omelette + Salad	280	28	12	10	10	6 egg whites, mushroom, spinach, tomato, olive oil	Whisk whites. Cook veggie omelette. Serve with raw salad.
Sarso Chicken + 2 Parathas	600	45	58	20	30	200g chicken, mustard paste, curd, 2 parathas	Marinate in mustard-curd. Grill/cook. Serve with hot parathas.
Chana Masala + Jeera Rice	530	20	80	10	35	200g chickpeas, tomato-onion masala, 250g jeera rice	Pressure cook chickpeas. Make rich masala. Serve with jeera rice.

6.x PRE-WORKOUT RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Banana + PB Toast	380	14	52	14	3	2 slices wheat toast, 2 tbsp PB, 1 large banana	Toast bread. Spread PB. Top with banana slices. Eat 60-90 min before.
Oats + Dates Shake	350	12	62	6	5	50g oats, 5 dates, 300ml milk, pinch cinnamon	Blend oats+dates+milk. Drink 45-60 min before training.
Chicken Rice Bowl (light)	460	38	52	8	30	150g chicken breast, 200g white rice, minimal oil, salt	Boil plain chicken. Steam rice. Simple pre-training fuel 2 hrs before.
Banana + Boiled Eggs	290	22	30	10	8	2 boiled eggs, 2 medium bananas	Easy, portable, complete pre-workout fuel 60 min before gym.
Sweet Potato + Curd Bowl	320	10	55	5	20	200g sweet potato (boiled), 150g curd, honey, cinnamon	Mash sweet potato. Top with curd and honey. Eat 90 min before.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Fruit Smoothie + Whey	360	28	52	4	5	1 banana, 100g mango, 1 scoop whey, 250ml water	Blend fruits with water. Mix in whey. Drink 30-45 min before.
Date Balls (Energy)	280	8	52	8	15	100g dates, 30g oats, 30g PB, 15g cocoa powder	Blend dates. Mix in oats+PB+cocoa. Roll into balls. Refrigerate.
Roti + Dal (simple)	380	18	60	8	20	2 rotis, 200g dal, minimal oil, no heavy spices	Light roti-dal combo. Avoid high fat versions before training.
Rice Cakes + Honey + Whey Shake	320	26	48	2	5	3 rice cakes, 1 tbsp honey, 1 scoop whey in water	Eat rice cakes with honey. Drink whey shake 30 min before.
Semolina (Upma) Light	340	10	58	8	15	80g semolina, veggies, minimal oil, no heavy tempering	Light upma with minimal oil. Good 90 min pre-training fuel.
Banana + Milk Combo	280	10	45	6	2	1-2 bananas, 300ml milk	Drink milk, eat bananas. Simple, effective, 45-60 min pre-workout.
Poha with Peanuts	380	12	60	12	12	100g poha, 30g peanuts, mustard, turmeric, lemon	Classic light meal. Complex carbs + protein. 90 min before training.

6.x POST-WORKOUT RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Whey + Banana + Milk Shake	450	48	52	8	3	1.5 scoops whey, 1 banana, 300ml milk	Blend or shake together. Drink within 30 min of training.
4 Eggs + 200g White Rice	480	36	65	18	15	4 whole eggs (scrambled), 200g cooked white rice	Fast protein + fast carbs. Optimal glycogen + MPS combo.
Chicken + White Rice + Banana	580	45	72	10	30	150g grilled chicken, 200g white rice, 1 banana	Complete post-workout meal. Ideal 45-60 min after training.
Soy Chunks + Rice (Veg Post-Workout)	490	40	65	6	15	100g soy chunks dry, 200g white rice, minimal spices	Boil soy chunks. Mix with white rice. Simple veg post-workout.
Greek Yogurt + Oats + Berries	380	28	48	8	5	200g Greek yogurt, 80g oats, 100g fruit	Layer yogurt over oats. Top with fruit. Casein + fast carbs.
Tuna + Bread Sandwich	420	38	48	8	5	1 can tuna, 3 slices white bread, mustard, tomato	Drain tuna. Mix with mustard. Sandwich in bread. Eat immediately.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Paneer + White Rice Bowl	520	28	68	16	10	150g paneer, 250g white rice, salt, cumin powder	Simple paneer rice bowl. Good veg post-workout option.
Egg White Omelette + Potato	350	30	40	6	15	5 egg whites, 200g boiled potato, salt, pepper, oil spray	Protein from whites + carbs from potato. Light and effective.
Homemade Protein Shake (no powder)	460	36	55	12	5	500ml whole milk, 4 egg whites, 1 banana, honey	Blend all. Natural post-workout shake without supplements.
Canned Sardines + Rice Cakes	380	30	40	12	3	1 can sardines, 5 rice cakes, lemon juice	Hostel-friendly. Drain sardines. Eat with rice cakes immediately.
Dal + White Rice (quick)	420	20	68	6	10	200g cooked moong dal, 250g white rice, salt, ghee 0.5 tsp	Simplest veg post-workout. Mix dal into rice. Add small amount ghee.
Whey + Water + Dates	280	25	38	2	2	1.5 scoops whey, 250ml water, 5-6 dates	Shake whey in water. Eat dates immediately after training.

6.x SNACK RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Roasted Chana + Peanuts Mix	290	16	32	14	2	75g roasted chana, 25g peanuts, lemon, salt, spices	Mix chana and peanuts. Season. Shelf-stable hostel protein snack.
Peanut Butter + Apple	280	10	32	16	2	1 apple, 2 tbsp peanut butter	Slice apple. Dip in PB. Protein + fiber + natural sugars.
Boiled Egg Snack Box	250	26	2	16	8	4 boiled eggs, salt, pepper, paprika	Pre-boil eggs. Refrigerate. Eat as quick protein snack anytime.
Makhana (Fox Nut) Protein Trail	240	10	40	5	5	100g makhana (roasted), 30g peanuts, herbs	Roast makhana with oil+spices. Mix with peanuts.
Greek Yogurt + Nuts	300	18	16	18	2	200g Greek yogurt, 30g mixed nuts, honey	Top yogurt with nuts and honey. High protein afternoon snack.
Protein Smoothie (quick)	380	32	42	8	3	1 scoop whey, 1 banana, 250ml milk	Blend and drink. 5-minute high-protein snack.
Paneer Cubes (raw)	240	16	3	18	2	120g paneer, rock salt, pepper, chat masala	Cut paneer into cubes. Season. Simple no-cook protein snack.
Sprouts Chaat	200	14	30	3	10	150g mixed sprouts, tomato, onion, lemon, spices	Boil or use raw sprouts. Add chopped veggies and spices.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Curd + Roasted Cumin	180	10	14	8	2	250g full-fat curd, roasted cumin powder, salt	Mix curd with seasoning. Add honey if needed. Easy gut-friendly snack.
Soy Milk + Oats Cup	280	14	40	6	5	200ml soy milk, 50g quick oats, banana	Heat soy milk. Add oats. Microwave 2 min. Top with banana.
Cheese Toast (protein)	300	18	28	12	5	2 bread slices, 40g cheddar/cheese, tomato	Top toast with cheese. Grill or toast. Quick protein snack.
Banana + Milk (hostel staple)	280	10	44	6	2	2 medium bananas, 300ml milk	Most accessible snack. Drink milk, eat bananas. Protein+carbs.
Trail Mix (budget version)	320	12	32	18	2	30g peanuts, 30g roasted chana, 20g raisins	Mix and store in box. Cheap, shelf-stable, portable snack.
Sattu Lemonade + Egg	260	20	30	7	5	50g sattu, water, lemon, 2 boiled eggs	Mix sattu drink. Eat alongside boiled eggs. Rural performance combo.
Tofu Scramble (quick)	260	18	8	16	10	150g firm tofu, onion, peppers, soy sauce, oil	Crumble tofu. Stir fry with veggies. Season. Vegan protein snack.
Protein Bar (homemade)	280	22	28	10	20	50g oats, 30g whey, 30g PB, 2 tbsp honey, 30g nuts	Mix all. Press in tray. Refrigerate. Cut into bars.
Boiled Corn + Butter	200	5	38	5	5	2 corn cobs, butter, lemon, salt, pepper	Boil/roast corn. Rub butter+lemon +spices. Good carb snack.
Milk + Bournvita / Health Drink	280	8	42	8	2	300ml milk, 3 tsp health drink powder	Budget recovery drink. Milk provides protein, drink provides carbs.
Soy Chunks in Room (cold soak)	200	32	14	1	5	100g soy chunks dry, cold water 500ml, salt, spices	Soak soy chunks in cold water 45 min. Drain. Season. Eat directly.
Egg + Bread Hostel Stack	340	28	36	12	8	4 eggs (boiled/fried), 3 bread slices, salt, ketchup	Boil eggs in kettle or immersion rod. Layer with bread.
Curd Rice	380	12	60	8	10	200g cooked rice (mess), 150g curd, salt, mustard optional	Mix mess rice with curd. Season. Easy recovery meal.
Sattu + Milk Protein Shake	360	24	48	8	3	60g sattu, 300ml milk, jaggery or honey, cardamom	Mix sattu in warm milk. Sweeten. Protein shake without supplements.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Dal + Rice (mess optimization)	420	18	70	6	0	Mess dal 200g, extra rice 250g, extra roti if available	Take double dal at mess. Add extra rice. Core muscle fuel.
Peanut Butter Roti Roll	340	14	48	14	3	2 rotis, 3 tbsp PB, 1 banana (optional)	Spread PB on roti. Roll with banana. Budget snack+mini meal.
Chana Chaat (instant)	260	14	38	5	3	200g canned/boiled chana, lemon, chaat masala, onion	Mix chana with seasoning. Can add mess sabzi on top.
Milk + Banana + Eggs Breakfast	470	32	52	16	5	3 boiled eggs, 2 bananas, 400ml milk	Complete hostel breakfast combo. ■50-60. 32g protein.
Oats Cup (kettle method)	380	18	58	8	5	80g oats, 300ml milk (heat in kettle), 2 tbsp PB	Heat milk in kettle. Pour over oats in cup. Wait 3 min. Add PB.
Bread + Egg + Milk Plate	400	30	45	14	5	4 boiled eggs, 3 bread slices, 300ml milk	Most hostel-friendly meal. Boil eggs in kettle. Budget: ■40-50.

6.x MASS GAINER RECIPES

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Super Mass Shake	1200	68	145	30	10	100g oats, 50g PB, 2 scoops whey, 500ml whole milk, 2 bananas, honey	Blend all. Drink in 2 portions if too much at once.
GOMAD (Gallon of Milk Approach - Modified)	730	40	52	36	5	1 liter whole milk throughout day + normal meals	Add 1L milk across the day as liquid calorie supplement.
Weight Gain Bowl	900	50	110	25	20	250g rice, 200g chicken, 200g dal, 1 tbsp ghee, 2 eggs	Combine rice+chicken+dal. Add ghee+eggs on top. Massive bowl.
Peanut Butter Mass Toast	680	30	75	30	8	4 thick bread slices, 4 tbsp PB, 2 bananas, honey, milk 300ml	Layer toast with PB+banana. Drink milk alongside.
Calorie Dense Smoothie	850	42	110	22	8	2 scoops whey, 500ml milk, 100g dates, 3 tbsp PB, 100g oats	Blend everything. Split across 2 servings if needed.
Paneer + Potato Sabzi Combo	720	32	72	30	30	200g paneer, 300g potato, onion-tomato gravy, 3 rotis, ghee	Rich calorie-dense vegetarian bulking meal.
Ghee Rice + Dal Combo (heavy)	750	22	112	22	15	300g white rice, 200g dal, 2 tbsp ghee, pickle	Add extra ghee to rice for calorie density. Traditional bulk meal.
High Cal Dal Makhani	680	22	80	28	60	200g black urad, 2 tbsp butter, 100ml cream, 3 parathas	Slow cook dal with butter cream. Serve with butter parathas.

RECIPE	Kcal	Prot(g)	Carbs(g)	Fat(g)	Prep	Key Ingredients	Notes
Banana Oat Pancakes (stack)	620	30	82	16	20	3 bananas, 100g oats, 2 eggs, 1 scoop whey, honey	Mash bananas, blend oats. Mix all. Cook 6-8 large pancakes.
Night Mass Bowl (pre-sleep)	520	38	40	20	5	250g paneer, 100ml milk, 30g casein if available, honey	Slow-digesting casein from paneer supports overnight MPS.

6.9 HIGH-PROTEIN RECIPE COMBINATIONS

RECIPE	MACROS	TAGS	METHOD / INGREDIENTS
Hostel Mass Shake	~700 kcal / 55g prot	MG, HS	500ml full-fat milk + 2 scoops whey + 2 tbsp PB + 1 banana + 50g oats. Blend. Drink.
Soy Power Bowl	~480 kcal / 42g prot	V, VG, HS, B	100g soy chunks cooked + 200g rice + dal + jeera tadka. Zero cook skill needed.
Egg Feast	~520 kcal / 44g prot	V, B	5 whole eggs scrambled + 300ml milk + 3 slices bread. Classic budget protein bomb.
Paneer Bhurji Plate	~580 kcal / 38g prot	V	200g paneer crumbled + onion + tomato + 3 rotis. High casein night meal.
Tuna Rice Bowl	~450 kcal / 42g prot	NV, H, B	1 can tuna (150g) + 200g rice + soy sauce + lemon. 5-min no-cook protein.
Chicken Rice Bowl	~600 kcal / 52g prot	NV, A	200g chicken breast + 250g cooked rice + vegetables + 1 tsp ghee.
Greek Yogurt Parfait	~380 kcal / 30g prot	V	300g Greek yogurt + 50g oats + 1 banana + honey. High casein pre-sleep.
Moong Dal Chilla	~340 kcal / 24g prot	V, VG, B	Soaked moong dal batter pancakes. 4 chillas with mint chutney.

MODULE 7 | WORKOUT DATABASE

300+ exercises · Full muscle group coverage · Equipment · Coaching cues · Progressions

7.1 HOW TO READ THIS DATABASE

Each entry shows: **Exercise** | **Primary Muscles** | **Equipment** | **Difficulty** (1=Beginner, 2=Intermediate, 3=Advanced) | **Key Coaching Cue** | **Regression** (easier) | **Progression** (harder). Use these to build customized programs matching athlete equipment and skill level.

7.x CHEST

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Explosive Push-Up	Chest, Triceps, Power	Bodyweight	Intermediate	Explode to leave floor; clap optional	Push-Up	Clapping Push-Up
Clapping Push-Up	Chest, Triceps, Explosiveness	Bodyweight	Advanced	Explosive push; clap in air; land soft	Explosive Push-Up	Between Bench Clap
Nordic Push-Up	Chest, Triceps, Core	Bodyweight +Anchor	Advanced	Feet anchored; lean forward; catch on hands	Diamond Push-Up	Full Nordic
Assisted One-Arm Push-Up	Chest, Core (unilateral)	Bodyweight	Advanced	One arm on floor; one on light support	Archer Push-Up	Full One-Arm

7.x BACK

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Archer Push-Up	Pec Major (unilateral)	Bodyweight	Advanced	One arm wide; shift weight to primary side	Wide Push-Up	Assisted Archer
Barbell Bent-Over Row	Lats, Rhomboids, Traps, Biceps	Barbell	Intermediate	45deg torso; pull to lower chest; retract scapulae	DB Row	Pendlay Row
Pendlay Row (strict)	Lats, Rhomboids, Traps	Barbell	Intermediate	Dead stop at floor each rep; explosive pull	Bent-Over Row	Weighted Pendlay
T-Bar Row	Lats, Rhomboids, Mid Back	T-Bar Station	Intermediate	Chest on pad; full stretch at bottom	Cable Row	Weighted T-Bar
Dumbbell Row (single arm)	Lats, Rhomboids	Dumbbell	Beginner	Knee on bench; pull elbow to hip; full ROM	Cable Row	Kroc Row
Kroc Row	Lats, Traps (heavy)	Heavy Dumbbell	Advanced	Strict for first 5 reps; controlled swing for last 5	DB Row	Weighted Kroc
Pull-Up (overhand)	Lats, Biceps, Rear Delts	Pull-Up Bar	Intermediate	Full hang start; pull chest to bar	Assisted Pull-Up	Weighted Pull-Up
Chin-Up (underhand)	Lats, Biceps	Pull-Up Bar	Beginner	Supinated grip; more bicep involvement	Band-Assisted	Weighted Chin-Up
Neutral Grip Pull-Up	Lats, Brachialis	Pull-Up Bar	Intermediate	Palms facing each other; shoulder friendly	Assisted	Weighted
Wide Grip Pull-Up	Upper Lats	Pull-Up Bar	Intermediate	Wide grip emphasizes lat width	Regular Pull-Up	Weighted Wide

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Close Grip Pull-Up	Lower Lats, Biceps	Pull-Up Bar	Intermediate	Hands close together; elbows tight	Regular Pull-Up	Weighted
Band-Assisted Pull-Up	Lats, Biceps	Band+Bar	Beginner	Band removes body weight; learn pull-up pattern	Lat Pulldown	Unassisted
Lat Pulldown (wide)	Lats, Teres Major	Cable Machine	Beginner	Pull to upper chest; control up	Band Pulldown	Behind-Neck (caution)
Close Grip Lat Pulldown	Lower Lats, Biceps	Cable Machine	Beginner	Elbows close to sides; full stretch at top	Wide Pulldown	Weighted
Straight Arm Pulldown	Lats (isolation)	Cable Machine	Intermediate	Straight arms; pull bar to hips; feel lat stretch	Band Pulldown	Weighted
Seated Cable Row	Mid Back, Rhomboids	Cable Machine	Beginner	Straight torso; pull handle to abs; retract scap	Band Row	Single Arm Cable
Single Arm Cable Row	Lat (unilateral)	Cable Machine	Intermediate	Greater ROM; rotate slightly for stretch	DB Row	Heavier Weight
Rack Pull (partial DL)	Traps, Erectors, Glutes	Barbell+Rack	Advanced	Start from knee height; loads upper back	Full DL	Straps for heavy
Meadows Row	Lat (unilateral, stretch)	Barbell Landmine	Advanced	Unilateral; torso twist allows deep stretch	DB Row	Heavier
Renegade Row	Lats, Core, Shoulders	Dumbbells	Intermediate	Plank position; row one arm; maintain hips level	DB Row	Weighted
Inverted Row (bodyweight)	Rhomboids, Lats, Biceps	Barbell on Rack	Beginner	Body at angle; pull chest to bar; squeeze back	Band Row	Weighted Vest
Towel / Sheet Row (hostel)	Lats, Biceps	Door + Towel	Beginner	Tie towel in door; sit/lean back; row yourself up	Inverted Row	Single Arm
Archer Row	Lats (unilateral)	Bodyweight+Bar	Advanced	Inverted row; one arm extended; pull to one side	Inverted Row	Weighted
Chest Supported Row (DB)	Rhomboids, Mid Back	Incline Bench+DB	Intermediate	Chest on bench removes cheating; strict form	DB Row	T-Bar Row
Lat Prayer (pullover)	Lats, Serratus	Cable Machine	Intermediate	Arms straight; pull from overhead to hips	Straight Arm Pulldown	Weighted
Barbell Pullover	Lats, Serratus, Long Head Tri	Barbell+Bench	Advanced	Across bench; deep stretch overhead	DB Pullover	Weighted
Dumbbell Pullover	Lats, Serratus	Dumbbell+Bench	Intermediate	Deep stretch overhead; squeeze lat at bottom	Cable Pullover	Heavier DB
Dumbbell Shrugs	Upper Traps	Dumbbells	Beginner	Hold up 2 seconds; no rolling; straight up	Barbell Shrug	Heavy Shrug
Barbell Shrugs	Upper Traps	Barbell	Beginner	Heavy weight; hold 2 sec at top; don't roll	DB Shrug	Behind-Back Shrug
Cable Upright Row	Traps, Medial Deltoid	Cable Machine	Intermediate	Pull cable up; elbows lead; stop at chin	Barbell Upright	DB Upright
Overhead Carry (farmer's)	Traps, Deltoid, Core	Dumbbells/KB	Advanced	Arm locked overhead; walk for distance	Shoulder Rack Walk	Heavier

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Barbell High Pull	Traps, Deltoid, Hamstrings	Barbell	Advanced	Power movement; pull bar explosively to chin	Upright Row	Power Clean
Cable Tricep Pushdown (rope)	Triceps Lateral+Medial Head	Cable+Rope	Beginner	Flare rope at bottom; full lockout; elbows pinned	Straight Bar Down	Weighted Pushdown
Cable Pushdown (straight bar)	Triceps Lateral Head	Cable Machine	Beginner	Full lockout; elbows at sides; control eccentric	Rope Pushdown	Reverse Grip Down
Tricep Kickback (DB)	Triceps Lateral Head	Dumbbell	Beginner	Torso parallel; extend arm fully back; lock elbow	Cable Kickback	Heavier
Single Arm Cable Kickback	Triceps Lateral Head	Cable Machine	Intermediate	One arm; full lockout; feel squeeze	DB Kickback	Heavier
Leg Extension	Quads (isolation)	Machine	Beginner	Slow eccentric; squeeze at top; adjust seat	Bodyweight Terminal	Weighted Extension
Sissy Squat	Quads (extreme isolation)	Bodyweight/ Machine	Advanced	Lean back; knees forward; controlled descent	Leg Extension	Weighted
Single-Leg Standing Calf Raise	Gastrocnemius (unilateral)	Bodyweight/ DB	Beginner	One leg; full ROM; hold top; fix imbalances	Standing Calf	Weighted Single
Suitcase Carry	Core (anti-lateral flexion)	Dumbbell	Intermediate	Walk with weight in one hand; do NOT lean	Farmer Carry	Heavier
Muscle-Up (bar)	Back, Chest, Shoulders (transition)	Pull-Up Bar	Elite	Pull-up + dip transition; strict form	Weighted Pull-Up+Dip	Ring Muscle-Up
Archer Pull-Up	Lat (unilateral)	Pull-Up Bar	Advanced	One arm close; one arm wide; pull to close side	Wide Pull-Up	One-Arm Negatives
One-Arm Pull-Up (negative)	Lats, Biceps	Pull-Up Bar	Elite	Start at top; lower on one arm; jump up	Weighted Pull-Up	Full One-Arm
Band Row (anchored)	Mid Back, Rhomboids, Biceps	Band+Door	Beginner	Anchor band; sit/stand; row handles to torso	Seated Cable Row	Heavier Band
Band Lat Pulldown	Lats, Biceps	Band+Anchor	Beginner	Anchor overhead; pull down to chest; feel lats	Bodyweight Row	Heavier Band
Lat Pulldown Machine	Lats, Biceps	Machine	Beginner	Wide or close grip; pull to upper chest	Band Pulldown	Weighted
Row Machine (chest supported)	Rhomboids, Lats, Biceps	Machine	Beginner	Chest supported; removes cheating; strict form	DB Row	Cable Row

7.x SHOULDERS

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Face Pull	Rear Delts, External Rotators, Traps	Cable Machine	Beginner	Pull to face; elbows high; flare out at end	Band Pull-Apart	Weighted Face Pull
Barbell Overhead Press (standing)	Ant+Med Deltoid, Triceps, Core	Barbell	Intermediate	Brace abs+glutes; press straight up; lockout	DB OHP	Push Press

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Seated Barbell OHP	Ant+Med Deltoid, Triceps	Barbell	Intermediate	Supported; allows heavier loading	Standing OHP	BTN OHP (caution)
Dumbbell OHP (seated)	Deltoid (all heads)	Dumbbells	Beginner	Each arm moves independently; fix imbalances	Machine OHP	Barbell OHP
Dumbbell OHP (standing)	Deltoid, Core	Dumbbells	Intermediate	More core demand than seated	Seated DB OHP	Arnold Press
Arnold Press	All 3 Deltoid Heads	Dumbbells	Intermediate	Rotate from palms-in to palms-out while pressing	DB OHP	Standing Arnold
Lateral Raise (dumbbell)	Medial Deltoid	Dumbbells	Beginner	Lead with elbow; slight lean forward; stop at ear	Cable Lateral	Weighted Lateral
Cable Lateral Raise	Medial Deltoid	Cable Machine	Beginner	Constant tension; cross cable or single side	DB Lateral	Unilateral Cable
Leaning Away Cable Lateral	Medial Deltoid	Cable Machine	Intermediate	Lean away from stack; better ROM	Cable Lateral	Weighted
Bent-Over Dumbbell Lateral	Rear Deltoid, Rhomboids	Dumbbells	Beginner	Torso parallel; slight elbow bend; pinch shoulder blades	Face Pull	Cable Bent Lateral
Reverse Pec Deck	Rear Deltoid	Machine	Beginner	Arms parallel; pull back; squeeze rear delt	Bent-Over Lateral	Cable Reverse
Cable Rear Delt Fly	Rear Deltoid	Cable Machine	Beginner	Set cables high; pull to sides; feel rear delt	Face Pull	Weighted
Face Pull (rope)	Rear Delt, External Rotators	Cable+Rope	Beginner	Pull to face; elbows high; external rotate at end	Band Pull-Apart	Heavy Face Pull
Band Pull-Apart	Rear Delt, External Rotators	Resistance Band	Beginner	Arms straight; pull band apart; squeeze rear delt	Face Pull	Weighted Band
Front Raise (dumbbell)	Anterior Deltoid	Dumbbells	Beginner	Raise to eye level; control lowering	Cable Front Raise	Barbell Front Raise
Barbell Front Raise	Anterior Deltoid	Barbell	Intermediate	Overhand grip; straight arms; raise to eye level	DB Front Raise	Plate Raise
Plate Front Raise	Anterior Deltoid, Core	Weight Plate	Beginner	Arms straight; raise plate forward to eye level	DB Front Raise	Heavier Plate
Push Press	Deltoid, Triceps, Legs	Barbell	Intermediate	Slight dip; drive bar overhead with leg momentum	OHP	Jerk
Handstand Push-Up (assisted)	Ant Deltoid, Triceps	Wall	Advanced	Feet on wall; lower head to floor	Pike Push-Up	Full Handstand
Pike Push-Up	Anterior Deltoid, Triceps	Bodyweight	Intermediate	Hips high; head aims for floor between hands	Regular Push-Up	Handstand PU
Machine Shoulder Press	Deltoid, Triceps	Machine	Beginner	Adjust seat; press overhead; full ROM	DB OHP	Barbell OHP
Landmine Lateral Raise	Medial Deltoid	Landmine+Plate	Intermediate	Hold end of bar; arc laterally; feel squeeze	DB Lateral	Heavier
Y-W-T Raises	Rear Delt, Lower Trap, Rhomboids	Light DB/Cable	Beginner	Lie prone; form Y, W, T positions	Band Pull-Apart	Prone on Incline

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Resistance Band OHP	Deltoid, Triceps	Resistance Band	Beginner	Stand on band; press overhead; great for home	DB OHP	Heavier Band
Single Arm DB Press (standing)	Deltoid, Core	Dumbbell	Intermediate	One arm; forces core stability and antirotation	Seated DB OHP	KB Press
Band Pull-Apart (core tension)	Rear Delt, Core Tension	Band	Beginner	Maintain hollow body while pulling band apart	Plank	Heavier
Crab Walk	Shoulders, Triceps, Glutes, Core	Bodyweight	Beginner	Reverse table position; walk laterally	Reverse Plank	Distance Increase
Wall Handstand Hold	Shoulders, Core	Bodyweight +Wall	Advanced	Kick up to wall; maintain alignment; hold	Pike Push-Up	Free Handstand
Band Pull-Apart	Rear Delt, Rhomboids	Band	Beginner	Arms straight; pull band apart at chest level	Face Pull	Heavier Band
Band Overhead Press	Deltoid, Triceps	Band	Beginner	Stand on band; press overhead; core braced	DB OHP	Barbell OHP
Band Lateral Raise	Medial Deltoid	Band	Beginner	Stand on band; lateral raise; constant tension	DB Lateral	Heavier Band
Band Face Pull	Rear Delt, External Rotators	Band+Anchor	Beginner	Anchor at face height; pull to face; elbows high	Band Pull-Apart	Heavier
Shoulder Press Machine	Deltoid, Triceps	Machine	Beginner	Fixed path; good starting point for beginners	DB OHP	Barbell OHP
Lateral Raise Machine	Medial Deltoid	Machine	Beginner	Pads on forearms; raise to ear level; control	DB Lateral	Cable Lateral

7.x BICEPS

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Barbell Curl	Biceps Brachii, Brachialis	Barbell	Beginner	Elbows pinned; full ROM; control lowering	DB Curl	Weighted Curl
EZ Bar Curl	Biceps, Brachialis	EZ Bar	Beginner	Neutral-ish grip; easier on wrists than straight bar	DB Curl	Barbell Curl
Dumbbell Bicep Curl (alternating)	Biceps Brachii	Dumbbells	Beginner	Supinate at top; control; squeeze hard	Cable Curl	Concentration Curl
Concentration Curl	Biceps Brachii (peak)	Dumbbell	Intermediate	Elbow on inner knee; full ROM; peak contraction	DB Curl	Cable Conc. Curl
Preacher Curl	Biceps (lower portion)	EZ Bar+Preacher	Intermediate	Arm supported on pad; full ROM	Machine Preacher	Weight Increase
Incline DB Curl	Biceps (long head stretch)	Dumbbells+Incline	Intermediate	Arms hanging back; full stretch at bottom	DB Curl	Cable Incline
Cable Curl (standing)	Biceps	Cable Machine	Beginner	Constant tension; squeeze at top	DB Curl	Rope Cable Curl

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Rope Cable Curl	Biceps, Brachialis	Cable+Rope	Intermediate	At top flare rope handles out; peak squeeze	Cable Curl	Heavy Rope
Machine Curl	Biceps Brachii	Machine	Beginner	Adjust seat; full ROM; consistent resistance	DB Curl	Supinated Machine
Zottman Curl	Biceps + Brachioradialis	Dumbbells	Intermediate	Supinate up; pronate at top; control lowering reverse	DB Curl	Weighted Zottman
21s (Curl Variation)	Biceps (all ranges)	Barbell/DB	Intermediate	7 reps bottom half + 7 top half + 7 full = 21	Standard Curl	Weighted 21s
Spider Curl	Biceps (stretch + contraction)	EZ Bar/Barbell	Intermediate	Chest on incline bench; arms hang; curl up	Preacher Curl	Weighted
Cable Flex Curl (peak)	Biceps Peak	Cable	Intermediate	Flex and hold at top; really feel the contraction	Machine Curl	Weighted
Chin-Up (supinated)	Biceps, Lats	Pull-Up Bar	Intermediate	Underhand grip; biceps heavily involved	Assisted Chin	Weighted Chin-Up
Band Bicep Curl	Biceps Brachii	Resistance Band	Beginner	Stand on band; curl up; home workout alternative	DB Curl	Heavier Band
Barbell Drag Curl	Biceps (long head)	Barbell	Intermediate	Drag bar up torso (not arc); elbows move back	Barbell Curl	Heavy Drag
Dumbbell Scott Curl	Biceps Lower	Dumbbell	Intermediate	Use incline bench as preacher support; full ROM	Preacher Curl	Heavier
Lying Leg Curl	Biceps Femoris, Semimembranosus	Machine	Beginner	Toes pointed; curl to glutes; hold 1 sec	Seated Leg Curl	Weighted
Nordic Hamstring Curl	Biceps Femoris (eccentric)	Partner/Machine	Advanced	Knees fixed; lean forward eccentrically; catch if needed	Band Curl	Full Nordic
Cable Hamstring Curl (standing)	Biceps Femoris	Cable Machine	Beginner	Ankle attachment; curl to glutes; support body	Machine Curl	Heavier
Outward Rotation Curl	Outer Hamstring (Biceps Femoris)	Machine	Intermediate	Externally rotate foot slightly during curl	Standard Leg Curl	Heavier
Band Bicep Curl	Biceps Brachii	Band	Beginner	Stand on band; curl handles to shoulders	DB Curl	Heavier Band
Preacher Curl Machine	Biceps Lower	Machine	Beginner	Arm fully supported; full ROM; strict	DB Curl	EZ Preacher

7.x TRICEPS

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Diamond Push-Up	Triceps, Inner Pec	Bodyweight	Intermediate	Hands form diamond; elbows close to body	Push-Up	Weighted Diamond
Close-Grip Bench Press	Triceps, Inner Pec	Barbell	Intermediate	Grip inside shoulder width; elbows close	DB Narrow	Weighted Close-Grip
Close-Grip Bench Press	Triceps, Inner Pec	Barbell	Intermediate	Grip inside shoulder width; elbows tuck in	DB Close Press	Weighted CGBP
Skull Crusher (EZ Bar)	Triceps Long Head	EZ Bar+Bench	Intermediate	Bar to forehead; elbows stationary; flare slightly	DB Extension	Barbell Skull Crusher

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Skull Crusher (DB)	Triceps All Heads	Dumbbells	Intermediate	Arms slightly past vertical; lower DBs to ears	Cable Extension	Heavier
Overhead Tricep Extension (DB)	Long Head Tricep	Dumbbell	Beginner	Both hands on one DB; lower behind head	Cable Overhead	EZ Bar Overhead
Overhead Tricep Extension (cable)	Long Head Tricep	Cable	Beginner	Face away from stack; rope or bar; extend overhead	DB Overhead	Rope OH Extension
Reverse Grip Pushdown	Triceps Medial Head	Cable+Bar	Intermediate	Underhand grip; great for medial head isolation	Standard Pushdown	Heavier
Dips (tricep focus)	Triceps, Pecs	Parallel Bars	Intermediate	Stay more upright to target triceps more	Bench Dip	Weighted Dip
Bench Dip	Triceps	Bench/Chair	Beginner	Hands on bench; dip down; elbows back	Incline Push-Up	Feet-Elevated Dip
Diamond Push-Up	Triceps, Inner Pec	Bodyweight	Beginner	Hands in diamond shape; elbows close	Push-Up	Weighted Diamond
Lying DB Tricep Extension	Long Head Tricep	Dumbbells+Bench	Intermediate	Arms slightly past vertical; lower to ears	Skull Crusher	Heavier
Machine Dip (weighted)	Triceps, Pecs	Dip Machine	Beginner	Machine-assisted for beginners; add weight progressively	Bench Dip	Bodyweight Dip
Tate Press	Triceps Medial Head	Dumbbells	Advanced	Pinkies point up; elbows flare; lower DBs to chest	Close Grip Press	Heavier Tate
JM Press	Triceps	Barbell	Advanced	Hybrid of skull crusher and close grip; elbows flare	CGBP	Heavy JM
Band Tricep Pushdown	Triceps	Resistance Band	Beginner	Anchor band overhead; push down; home workout	DB Extension	Heavier Band
Pike Push-Up (tricep emphasis)	Triceps, Ant Deltoid	Bodyweight	Intermediate	Elbows tuck in vs flare for more tri	Push-Up	Handstand PU
Rope Overhead Extension	Long Head Tricep	Cable+Rope	Intermediate	Face away; rope behind head; extend fully	DB Overhead Ext	Heavy Rope
Dip (bodyweight)	Triceps, Chest, Shoulders	Parallel Bars/Chairs	Beginner	Full ROM; chest slight forward lean	Bench Dip	Weighted Dip
Band Tricep Pushdown	Triceps	Band+Overhead	Beginner	Anchor overhead; push down; full lockout	Cable Pushdown	Heavier Band
Tricep Extension Machine	Triceps	Machine	Beginner	Fixed axis; full extension; slow return	Cable Pushdown	Dips

7.x QUADS & LEGS

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Trap Bar Deadlift	Quads, Glutes, Hamstrings, Back	Trap Bar	Beginner	Stand inside bar; more squat-like pattern	Conv DL	Paused Trap DL
Back Squat	Quads, Glutes, Hams, Core	Barbell	Intermediate	Chest up; knees track toes; hip below knee depth	Goblet Squat	Front Squat

EXERCISE	PRIMARY MUSCLES	EQUIPMEN T	LVL	KEY CUE	REGRESSION	PROGRE SSION
Front Squat	Quads (emphasis), Upper Back	Barbell	Adv anc ed	Elbows high; upright torso; clean or cross grip	Goblet Squat	Paused Front Squat
Goblet Squat	Quads, Glutes, Core	Dumbbell/K B	Beg inne r	Hold weight at chest; sit between knees; upright	Box Squat	Back Squat
Box Squat	Quads, Glutes, Hams	Barbell+Box	Inte r	Sit to box; pause 1 sec; drive up explosively	Goblet Squat	Back Squat
Bulgarian Split Squat	Quads, Glutes (unilateral)	DB or Barbell	Inte r	Rear foot on bench; front knee over toe	Lunge	Barbell BSS
Lunges (forward)	Quads, Glutes	Bodyweight/ DB	Beg inne r	Long step; knee to floor; push back up	Reverse Lunge	Walking Lu nge+Weig ht
Reverse Lunge	Quads, Glutes	Bodyweight/ DB	Beg inne r	Step back; more knee-friendly than forward lunge	Lunge	Barbell Reverse
Walking Lunges	Quads, Glutes, Balance	Bodyweight/ DB	Beg inne r	Continuous forward lunge steps; works balance	Static Lunge	Weighted Walking
Leg Press	Quads, Glutes (depends on stance)	Machine	Beg inne r	Don't lock knees; full ROM; vary stance for emphasis	Goblet Squat	Heavier Leg Press
Hack Squat (machine)	Quads	Machine	Beg inne r	Feet low on plate; upright torso	Leg Press	Barbell Hack Squat
Barbell Hack Squat	Quads	Barbell	Adv anc ed	Bar behind legs; squat down; quad dominant	Machine Hack	Heavier
Wall Sit	Quads (isometric)	Bodyweight	Beg inne r	90deg knee angle; hold for time	Squat	Weighted Wall Sit
Step-Up (weighted)	Quads, Glutes	Box+DB	Beg inne r	Drive through heel; step up on box; alternate	Lunge	Single Leg Press
Jump Squat	Quads, Glutes, Calves (power)	Bodyweight	Inte r	Squat then explode up; land softly	Squat	Weighted Jump Squat
Pistol Squat (assisted)	Quads, Glutes (unilateral)	Bodyweight +TRX	Adv anc ed	Single leg; use TRX/band for assistance; build to full	BSS	Full Pistol
Sumo Squat	Quads, Inner Thigh, Glutes	DB or Barbell	Beg inne r	Wide stance; toes out; hold weight between legs	Goblet Squat	Barbell Sumo
Box Step-Down (eccentric)	Quads (eccentric control)	Box	Inte r	Stand on box; lower slowly on one leg; build knee strength	Leg Press	Add Resistance
Sled Push/Pull	Quads, Glutes, Conditioning	Sled	Inte r	Low push or pull; power from legs	Leg Press	Heavier Sled
Cycle/Stationary Bike (resistance)	Quads (conditioning)	Bike	Beg inne r	High resistance; sprint intervals for quad development	Light Cycling	HIIT Cycling
Bulgarian Split Squat	Quads, Glutes	DB/Barbell+ Bench	Inte r	Rear elevated; torso upright; deep lunge	Lunge	Barbell BSS
Box Jump	Quads, Glutes, Explosiveness	Box	Inte r	Swing arms; explode; land soft; step down	Jump Squat	Depth Jump
Broad Jump	Quads, Glutes, Power	Bodyweight	Inte r	Horizontal jump; triple extension; stick landing	Jump Squat	Weighted Vest

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Tuck Jump	Quads, Core, Explosiveness	Bodyweight	Intermediate	Jump and tuck knees to chest; land soft	Jump Squat	Weighted Vest
Plyometric Lunge	Quads, Glutes, Power	Bodyweight	Intermediate	Jump between lunge positions; alternate legs	Static Lunge	Weighted Vest
Squat Jump	Quads, Glutes, Conditioning	Bodyweight	Beginner	Squat to parallel; jump; land soft; repeat	Squat	Tuck Jump
Pistol Squat	Quads, Glutes, Balance	Bodyweight	Advanced	Single leg squat to full depth; heel flat	Assisted Pistol	Weighted Pistol
Shrimp Squat	Quads, Glutes, Balance	Bodyweight	Advanced	Single leg; rear leg bent behind; squat down	BSS	Pistol Squat
Band Squat	Quads, Glutes	Band	Beginner	Band overhead or under feet; adds resistance to squat	Bodyweight Squat	Barbell Squat
Band Walking Lunge	Quads, Glutes	Band	Intermediate	Band provides resistance to forward lunge	Walking Lunge	Weighted Vest
Leg Press	Quads, Glutes, Hamstrings	Machine	Beginner	Full ROM; do not lock knees; vary foot position	Goblet Squat	Single Leg Press
Leg Extension	Quadriceps	Machine	Beginner	Full lockout; squeeze VMO; control lowering	TKE Band	Single Leg Ext

7.x HAMSTRINGS & GLUTES

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Conventional Deadlift	Glutes, Hams, Erectors, Lats, Traps	Barbell	Intermediate	Bar over midfoot; push floor away; hips fwd lock	Romanian DL	Deficit DL
Romanian Deadlift (RDL)	Hamstrings, Glutes, Erectors	Barbell	Intermediate	Soft knee; hinge at hip; bar close to legs	Stiff Leg DL	Single-Leg RDL
Sumo Deadlift	Glutes, Inner Thigh, Back	Barbell	Intermediate	Wide stance; toes out; more upright torso	Conv DL	Banded Sumo
Romanian Deadlift	Hamstrings, Glutes, Erectors	Barbell/DB	Intermediate	Hinge at hips; soft knees; bar close to legs; feel stretch	Good Morning	Single-Leg RDL
Single-Leg RDL	Hamstrings, Glutes (unilateral)	DB/Barbell	Intermediate	Balance on one leg; hinge back hip; DB reaches floor	RDL	Barbell SL-RDL
Seated Leg Curl	Hamstrings	Machine	Beginner	Seated; knee angle allows full ROM	Lying Curl	Weighted Seated
Good Morning	Hamstrings, Erectors	Barbell	Advanced	Bar on shoulders; hip hinge; slight knee bend	RDL	Seated Good Morning
Glute Bridge (hamstring bias)	Hamstrings, Glutes	Bodyweight/Barbell	Beginner	Feet far from body; contract ham at top	Hip Thrust	Weighted Bridge
Stiff-Leg Deadlift	Hamstrings, Glutes, Erectors	Barbell	Intermediate	Minimal knee bend; more stretch on hams than RDL	RDL	Barbell SLDL
Swiss Ball Hamstring Curl	Hamstrings, Glutes, Core	Swiss Ball	Intermediate	Feet on ball; bridge up; curl ball to glutes	Machine Curl	Single Leg Ball

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Band Hamstring Curl (standing)	Hamstrings	Resistance Band	Beginner	Anchor band; stand; curl heel to glutes	Machine Curl	Heavier Band
Hip Hinge Drill	Hamstrings, Glutes, Hinge Pattern	Bodyweight	Beginner	Hands on hips; hinge back; feel hamstring stretch	RDL	Barbell RDL
Prone Hamstring Stretch Hold (active)	Hamstring Flexibility	Bodyweight	Beginner	Lie prone; lift leg toward ceiling; active stretch	Static Stretch	PNF Stretch
Deadlift (conventional) - ham bias	Hamstrings (conventional stance)	Barbell	Intermediate	Conventional stance loads hams more than sumo	Trap Bar DL	Deficit DL
Barbell Hip Hinge (hex bar)	Hamstrings, Glutes	Hex Bar	Beginner	Trap bar forces more upright; accessible for beginners	Bodyweight Hinge	Full DL
Leg Press (low foot position)	Hamstrings, Glutes	Leg Press Machine	Beginner	Low foot placement = more ham emphasis	Standard Leg Press	Single-Leg Press
Natural Glute Ham Raise (GHR)	Hamstrings, Glutes, Calves	GHD Machine	Advanced	Curl body up from prone; hardest ham exercise	Nordic	Weighted GHR
Inward Rotation Curl	Inner Hamstring (Semitendinosus)	Machine	Intermediate	Internally rotate foot slightly during curl	Standard Leg Curl	Heavier
Wall Hamstring Stretch (dynamic)	Hamstring Flexibility	Bodyweight	Beginner	Dynamic swinging leg stretch; prep for training	Static Stretch	PNF Advanced
Barbell Hip Thrust	Glutes, Hamstrings	Barbell+Bench	Intermediate	Chin tucked; drive hips fully up; squeeze at top	Glute Bridge	Banded Hip Thrust
Glute Bridge	Glutes, Hamstrings	Bodyweight/Barbell	Beginner	Feet flat; drive hips up; squeeze hard at top	Hip Thrust	Weighted Bridge
Single-Leg Glute Bridge	Glutes, Hamstrings (unilateral)	Bodyweight	Beginner	One foot elevated; drive hip up with one leg	Glute Bridge	Hip Thrust Unilateral
Cable Pull-Through	Glutes, Hamstrings, Erectors	Cable Machine	Beginner	Hip hinge with cable; squeeze glutes at top	RDL	Banded Pull-Through
Banded Monster Walk	Glutes (med), Hip Abductors	Resistance Band	Beginner	Band around ankles/knees; shuffle laterally	Hip Abduction	Heavier Band
Clamshell	Glute Med, Hip Abductor	Resistance Band	Beginner	Side-lie; feet together; open top knee like clam	Hip Abduction	Heavier Band
Sumo Deadlift	Glutes, Inner Thigh, Back	Barbell	Intermediate	Wide stance targets glutes more than conventional	Romanian DL	Banded Sumo
Hip Abduction Machine	Glute Medius, TFL	Machine	Beginner	Seated; push pads apart; slow return	Banded Walk	Heavier Machine
Kickback (cable)	Gluteus Maximus	Cable Ankle	Beginner	On all fours; extend hip back; squeeze at end	Bodyweight Kickback	Weighted
Fire Hydrant	Glute Med, Hip Abductors	Bodyweight/Band	Beginner	On all fours; lift knee out sideways	Clamshell	Banded Hydrant
Donkey Kick	Gluteus Maximus	Bodyweight/Band	Beginner	All fours; kick heel to ceiling; squeeze glute	Cable Kickback	Ankle Weight

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Step-Up (glute bias)	Glutes, Quads	Box+DB	Beginner	High box; lean forward slightly; drive heel down	Lunge	Weighted Step-Up
Romanian Deadlift (glute bias)	Glutes, Hamstrings	Barbell	Intermediate	Full hip extension at top; squeeze glutes hard	Glute Bridge	Banded RDL
Lateral Lunge (Cossack Squat)	Glutes, Inner Thigh, Quads	Bodyweight/DB	Intermediate	Wide step lateral; shift weight; go deep	Sumo Squat	Weighted
Reverse Hyper	Glutes, Hamstrings, Erectors	Reverse Hyper Machine	Advanced	Legs hang; swing up to parallel; squeeze glutes	Superman	Weighted
Banded Squat	Glutes, Quads	Band+Bodyweight	Beginner	Band above knees; push knees out; activate glutes	Squat	Back Squat
Seated Hip Abduction (banded)	Glute Med	Band	Beginner	Seated; band above knees; push knees apart	Clamshell	Machine
Barbell Glute Bridge	Glutes (heavy)	Barbell+Mat	Intermediate	Weight on hips; drive up; hold 1-2 sec	Hip Thrust	Heavier Barbell
Pendulum Quadruped Extension	Glute Max, Hamstrings	Cable/Ankle	Intermediate	Quad position; extend hip back in controlled arc	Cable Kickback	Heavier
Band Good Morning	Hamstrings, Erectors	Band	Beginner	Band behind neck or on shoulders; hinge at hips	Bodyweight Hinge	RDL
Band Hip Thrust	Glutes, Hamstrings	Band+Bench	Beginner	Band over hips; anchor or sit on; thrust up	Glute Bridge	Barbell Thrust
Band Glute Kickback	Glutes	Band+Anchor	Beginner	Attach band to ankle; kick back; hip extension	Bodyweight Kickback	Cable Kickback
Band Monster Walk	Glute Med, Hip Abductors	Band	Beginner	Band around ankles; squat slightly; shuffle	Clamshell	Heavier Band
Band Hamstring Curl	Hamstrings	Band+Anchor	Beginner	Anchor band to ankle; lie prone; curl heel to glute	Lying Leg Curl	Machine Curl
Band Deadlift	Glutes, Hams, Back	Band+Floor	Beginner	Stand on band; hinge; straighten up; deadlift pattern	Bodyweight Hinge	Barbell DL
Leg Curl (lying)	Hamstrings	Machine	Beginner	Toes pointed; squeeze at top; slow eccentric	Band Curl	Seated Leg Curl
Leg Curl (seated)	Hamstrings, Soleus	Machine	Beginner	90deg knee at start; full ROM; control down	Lying Curl	Single Leg Curl
Hip Abduction Machine	Glute Med, TFL	Machine	Beginner	Push pads out; contract glute med; hold briefly	Band Walk	Heavier
Hip Adduction Machine	Hip Adductors, Inner Thigh	Machine	Beginner	Squeeze pads together; control opening	Sumo Squat	Heavier
Glute Kickback Machine	Gluteus Maximus	Machine	Beginner	Drive heel backward; full hip extension; squeeze	Cable Kickback	Heavier

7.x CALVES

EXERCISE	PRIMARY MUSCLES	EQUIPMEN T	LVL	KEY CUE	REGRESSION	PROGRE SSION
Standing Calf Raise	Gastrocnemius (upper calf)	Machine/Ste p	Beg inner	Full ROM; pause at top; hold 1-2 sec squeeze	Seated Calf	Weighted Single
Seated Calf Raise	Soleus (lower calf)	Machine	Beg inner	Knee bent emphasizes soleus; full ROM	Standing Calf	Heavier Machine
Donkey Calf Raise	Gastrocnemius	Machine/Par tnerweight	Inter	Bent-over position; optimal muscle angle	Standing Calf	Heavier
Leg Press Calf Raise	Gastrocnemius, Soleus	Leg Press Machine	Beg inner	Toes on plate edge; full ROM; safe for heavy loads	Standing Calf	Heavier
Calf Jumps (explosive)	Gastrocnemius, Soleus, Power	Bodyweight	Inter	Rapid jumps; minimal knee bend; calf power	Jump Squat	Weighted Jump
Hack Machine Calf Raise	Gastrocnemius	Machine	Beg inner	Hack squat machine; toes on edge; full ROM	Standing Calf	Heavier
Smith Machine Calf Raise	Gastrocnemius	Smith Machine	Beg inner	Bar on shoulders; stand on step; full ROM	Standing Calf	Heavier
Box Calf Raise (step)	Gastrocnemius	Step/Box	Beg inner	Stand on step edge; full ROM; no equipment needed	Floor Calf Raise	Weighted
Bodyweight Calf Raise	Gastrocnemius, Soleus	Bodyweight	Beg inner	Basic starting point; progress to single-leg	Box Calf	Weighted
Jumping Rope (calves)	Gastrocnemius, Coordination	Jump Rope	Beg inner	Stay on balls of feet; great calf conditioning	Calf Raise	Double Under
Hill Walking/Running	Gastrocnemius, Soleus	Outdoors/Tr eadmill	Beg inner	Uphill activates calves more than flat	Walking	Sprint Hills
Band Calf Raise (hostel)	Gastrocnemius	Band+Door/ Step	Beg inner	Band for resistance; perform on step	Bodyweight	Heavier Band
Barbell Calf Raise	Gastrocnemius	Barbell+Ste p	Inter	Heavy loading; bar on back; full ROM on step	Standing Calf	Heavier
Band Calf Raise	Gastrocnemius	Band	Beg inner	Stand on band; hold handles; raise on toes	Bodyweight Calf	Machine Calf
Calf Raise Machine	Gastrocnemius	Machine	Beg inner	Full ROM; pause at top; slow eccentric	Standing Calf	Single Leg

7.x CORE & ABS

EXERCISE	PRIMARY MUSCLES	EQUIPMEN T	LVL	KEY CUE	REGRESSION	PROGRE SSION
Plank (standard)	Core, Shoulders, Glutes	Bodyweight	Beg inner	Straight line head to heel; 360 deg brace; breathe	Incline Plank	Weighted Plank
Side Plank	Obliques, Core	Bodyweight	Beg inner	Hips up; straight body; hold for time	Knee Side Plank	Weighted Side Plank
Dead Bug	Deep Core, Transverse Abdominis	Bodyweight	Beg inner	Lower limbs opposite to each other; low back flat	Bear Crawl	Weighted Dead Bug

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Hollow Body Hold	Core (anterior)	Bodyweight	Intermediate	Round low back to floor; arms+legs extended; hold	Dead Bug	Hollow Rock
Hollow Rock	Core (anterior), Hip Flexors	Bodyweight	Advanced	Rock forward and back in hollow position	Hollow Hold	Weighted
Hanging Knee Raise	Lower Abs, Hip Flexors	Pull-Up Bar	Beginner	Hang; bring knees to chest; control lowering	Lying Leg Raise	Hanging Leg Raise
Hanging Leg Raise	Lower Abs, Hip Flexors	Pull-Up Bar	Intermediate	Straight legs; raise to parallel or beyond	Hanging Knee Raise	Weighted
Ab Wheel Rollout	Core, Lats, Shoulders	Ab Wheel	Intermediate	Knees on floor; roll out; fully extend; roll back	Dead Bug	Standing Rollout
Standing Ab Wheel Rollout	Core (extreme)	Ab Wheel	Advanced	Standing rollout; maintain neutral spine	Knee Rollout	Add Resistance
Dragon Flag	Core (entire)	Bench/Bar	Advanced	Lay on bench; extend fully; lower controlled	Hanging Leg Raise	Weighted
Cable Crunch (kneeling)	Rectus Abdominis	Cable Machine	Beginner	Rope at top; kneel; crunch elbows to knees	Crunch	Weighted
Crunch (basic)	Rectus Abdominis	Bodyweight	Beginner	Hands at temples; curl shoulders to hips	Dead Bug	Weighted Crunch
Reverse Crunch	Lower Abs	Bodyweight	Beginner	Lie flat; bring knees to chest; control down	Lying Knee Raise	Weighted
Bicycle Crunch	Obliques, Rectus Abs	Bodyweight	Beginner	Opposite elbow to knee; rotate fully; slow	Crunch	Weighted Bicycle
Russian Twist	Obliques	Bodyweight/Medicine Ball	Beginner	Lean back 45deg; twist torso; control each side	Seated Twist	Weighted Russian
Pallov Press (anti-rotation)	Deep Core, Anti-Rotation	Cable/Band	Intermediate	Press cable out; resist rotation; hold; return	Band Pallov	Heavier
Farmer's Carry	Core, Traps, Grip	Dumbbells/KB	Beginner	Heavy; walk distance; upright posture	Suitcase Carry	Heavier
Turkish Get-Up	Core, Shoulders, Hip (complex)	Kettlebell/DB	Advanced	Complex movement; arm locked overhead throughout	Half Get-Up	Heavier
L-Sit Hold	Core, Hip Flexors, Triceps	Parallel Bars	Advanced	Support on hands; legs parallel to floor; hold	Hanging Knee Hold	Heavier
Barbell Rollout	Core, Lats	Barbell on Floor	Advanced	Like ab wheel but with barbell; larger ROM	Ab Wheel Rollout	Wider Grip
Mountain Climbers	Core, Shoulders, Hip Flexors	Bodyweight	Beginner	Plank position; alternate knee drives to chest	High Plank	Speed MCS
Bear Crawl	Core, Shoulders, Hips	Bodyweight	Beginner	All fours; alternate hand+foot; low hips	Plank	Forward-B backward Combo
Windmill	Obliques, Hip, Shoulder	Kettlebell/DB	Advanced	Arm overhead; hinge sideways; look up at arm	Side Bend	Heavier

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
L-Hang (pull-up bar)	Core, Hip Flexors, Grip	Pull-Up Bar	Advanced	Hang; raise straight legs to L position; hold	Hanging Knee Raise	Weighted
Wood Chop (cable)	Obliques, Core, Shoulders	Cable Machine	Intermediate	Rotate from high to low; diagonal pattern	Russian Twist	Heavier
Plank with Hip Dips	Obliques, Core	Bodyweight	Intermediate	Side plank position; dip hip to floor and return	Side Plank	Weighted
Inchworm	Core, Hamstrings, Shoulders	Bodyweight	Beginner	Walk hands out to plank; return; warm-up	Dead Bug	Weighted Inchworm
High Knees	Core, Hip Flexors, Cardio	Bodyweight	Beginner	Alternate knee drives to waist height; fast	March	Resisted High Knees
Inchworm Walk-Out	Core, Shoulders, Hamstrings	Bodyweight	Beginner	Walk hands to plank; hold; walk feet to hands	Plank	Add Push-Up
L-Sit on Floor	Core, Triceps, Hip Flexors	Bodyweight+Blocks	Advanced	Seated; press up; legs parallel; hold	Hanging L-Sit	Parallel Bar L-Sit
Human Flag (progression)	Obliques, Lats, Full Body	Pole	Elite	Horizontal flag hold on vertical pole	Dragon Flag	Full Flag
Band Pallof Press	Deep Core, Anti-Rotation	Band+Anchor	Beginner	Anchor sideways; press band out; resist twist	Plank	Heavier Band
Abdominal Crunch Machine	Rectus Abdominis	Machine	Beginner	Rotate against pad; full ROM; squeeze abs	Cable Crunch	Weighted
Rotary Torso Machine	Obliques	Machine	Beginner	Rotate torso; hold briefly; control return	Russian Twist	Cable Wood Chop

7.x FULL BODY / BODYWEIGHT

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Dumbbell Deadlift	Full Posterior Chain	Dumbbells	Beginner	Same cues as barbell DL; good for learning	Hip Hinge Drill	Barbell DL
Burpee	Full Body, Conditioning	Bodyweight	Beginner	Squat-plank-push-up-jump; continuous flow	Modified Burpee	Weighted Burpee
Jumping Jack	Full Body, Cardio	Bodyweight	Beginner	Arms and legs simultaneous; high intensity	Step Jack	Weighted Jack
Animal Flow (bear crawl + crab)	Full Body	Bodyweight	Advanced	Complex combination of ground movements	Bear Crawl	Full Animal Flow

7.x ACCESSORIES & BANDS

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Barbell Bench Press	Pec Major, Ant Deltoid, Triceps	Barbell+Bench	Intermediate	Scapulae retracted; chest up; 45deg elbow flare	DB Press	Weighted+Pause
Incline Barbell Bench Press	Upper Pec, Ant Deltoid	Barbell+Incline	Intermediate	30-45deg incline; same form as flat	Incline DB	Close-Grip Incline

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Decline Barbell Bench Press	Lower Pec, Triceps	Barbell+Decline	Intermediate	Control descent; feet hooked in bench	DB Decline	Weighted Decline
Dumbbell Bench Press (flat)	Pec Major, Triceps	Dumbbells+Bench	Beginner	Full ROM; controlled eccentric; slight arc at top	Push-Up	Barbell Bench
Incline DB Press	Upper Pec, Ant Deltoid	Dumbbells+Incline	Beginner	30deg incline for max upper chest	Incline Push-Up	Barbell Incline
Decline DB Press	Lower Pec	Dumbbells+Decline	Intermediate	Control descent; squeeze at top	Dip	Barbell Decline
Dumbbell Flyes (flat)	Pec Major (stretch)	Dumbbells+Bench	Intermediate	Slight elbow bend; feel stretch at bottom	Cable Fly	Weighted Fly
Incline DB Fly	Upper Pec	Dumbbells+Incline	Intermediate	Stretch at bottom; squeeze at top	Cable Incline	Full ROM Incline Fly
Cable Chest Fly (mid)	Pec Major	Cable Machine	Intermediate	Elbows slightly bent; cross body	DB Fly	High-to-Low Cable
High-to-Low Cable Fly	Lower Pec	Cable Machine	Intermediate	Set cables high; pull down and across	Decline Fly	Weighted Low Cable
Low-to-High Cable Fly	Upper Pec	Cable Machine	Intermediate	Set cables low; pull up and across	Incline Fly	Pulley Cross
Chest Dips	Lower Pec, Triceps	Parallel Bars	Intermediate	Lean forward slightly; feet behind you	Bench Dip	Weighted Dip
Push-Up (standard)	Pec Major, Triceps, Core	Bodyweight	Beginner	Straight body; full ROM; elbows at 45deg	Incline Push-Up	Decline Push-Up
Wide-Grip Push-Up	Pec Major	Bodyweight	Beginner	Wider than shoulder width; more chest activation	Wall Push-Up	Weighted Push-Up
Decline Push-Up	Upper Pec, Shoulders	Bodyweight+Chair	Intermediate	Feet elevated; hands on floor	Push-Up	Archer Push-Up
Incline Push-Up	Lower Pec	Bodyweight+Elevated	Beginner	Hands elevated; good for beginners	Wall Push-Up	Push-Up
Single Arm Push-Up (assisted)	Pec Major	Bodyweight	Advanced	Wide feet; lower single arm; torso stays level	Archer Push-Up	Full Single Arm
Pec Deck / Chest Fly Machine	Pec Major	Machine	Beginner	Elbows slightly bent; pads at mid-chest	DB Fly	Cable Fly
Smith Machine Bench	Pec Major, Triceps	Smith Machine	Beginner	Fix bar path; good for beginners to learn form	DB Press	Barbell Bench
Hex Press (close grip)	Inner Pec, Triceps	Dumbbells	Intermediate	Press DBs together throughout movement	DB Bench	Paused Hex
Plate Press (overhead circle)	Pec Minor, Serratus	Weight Plate	Intermediate	Circle plate over head; squeeze throughout	Cable Press	Heavy Plate
Resistance Band Push-Up	Pec Major	Resistance Band	Intermediate	Band across upper back for added resistance	Push-Up	Weighted Push-Up
Svend Press	Inner Pec	Weight Plates	Intermediate	Squeeze plate between palms while pressing out	Cable Press	Add Weight
Guillotine Press	Upper Pec	Barbell	Advanced	Bar to neck level; wider grip; advanced only	Incline Press	Paused Guillotine
Push-Up Plus (serratus)	Serratus Ant, Pec	Bodyweight	Beginner	At top of push-up, push upper back to ceiling	Push-Up	Weighted Push-Up Plus

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Landmine Press	Pec Minor, Upper Chest	Barbell+Landmine	Intermediate	Angled press; good for shoulder issues	Cable Cross	Double Landmine
Cable Crossover	Pec Major	Dual Cable	Intermediate	Cross cables at end range; big squeeze	DB Fly	Weighted Crossover
Good Morning	Erectors, Hamstrings, Glutes	Barbell	Advanced	Bar on shoulders; hinge like RDL	RDL	Banded Good Morning
Superman Hold	Erectors, Glutes	Bodyweight	Beginner	Lie prone; lift arms+legs simultaneously	Back Extension	Weighted Superman
Back Extension	Erectors, Glutes	GHD/Hyperext Bench	Beginner	Control up and down; don't hyperextend at top	Superman	Weighted Hold
Cuban Press	External Rotators, Rear Delt	Dumbbells	Intermediate	Upright row to 90deg; rotate to OHP position	Face Pull	Barbell Cuban
Hammer Curl	Brachialis, Brachioradialis	Dumbbells	Beginner	Neutral grip; builds arm thickness and forearm	DB Curl	Cable Hammer
Cross Body Hammer Curl	Brachialis, Brachioradialis	Dumbbell	Intermediate	Curl across body to opposite shoulder	Hammer Curl	Heavier
Reverse Curl	Brachioradialis, Extensor Digitorum	Barbell/EZ	Intermediate	Overhand grip; builds forearm and brachialis	Hammer Curl	Weighted Reverse
Terminal Knee Extension (TKE)	VMO, Knee Stabilizer	Band	Beginner	Band behind knee; straighten fully; squeeze VMO	Leg Extension	Weighted TKE
Tibialis Raise (anterior lower leg)	Tibialis Anterior	Bodyweight/Machine	Beginner	Heels down; pull toes up; often neglected	Standing Calf	Weighted
Superman (floor)	Erectors, Glutes	Bodyweight	Beginner	Prone; lift opposite arm+leg; hold 2 sec	Back Extension	Weighted
Jump Rope (single)	Calves, Cardio, Coordination	Jump Rope	Beginner	Balls of feet; wrists; light jumps	Step Overs	Double Under
Band Chest Press	Pec Major, Triceps	Band+Anchor	Beginner	Band behind back; press forward at chest level	Push-Up	Heavier Band
Band Chest Fly	Pec Major	Band+Anchor	Beginner	Anchor behind; fly motion; slight elbow bend	Cable Fly	Heavier Band
Seated Calf Machine	Soleus	Machine	Beginner	Pads on knees; full ROM; soleus emphasis	Standing Calf	Heavier
Chest Press Machine	Pec Major, Triceps	Machine	Beginner	Fixed path; safe for beginners; full ROM	DB Press	Barbell Bench
Incline Chest Press Machine	Upper Pec	Machine	Beginner	Adjust incline; full ROM; squeeze at top	Incline DB	Barbell Incline
Pec Deck / Fly Machine	Pec Major	Machine	Beginner	Elbows level with shoulder; squeeze at front	DB Fly	Cable Fly
Back Extension Machine	Erectors, Glutes	Machine	Beginner	Pad at hips; extend back; control lowering	Superman	Weighted

EXERCISE	PRIMARY MUSCLES	EQUIPMENT	LVL	KEY CUE	REGRESSION	PROGRESSION
Smith Machine (general)	Multiple Groups	Smith Machine	Beginner	Fixed bar path; good for learning; less stabilizer	Free Weights	Free Bar

7.12 KEY COMPOUND LIFT COACHING GUIDES

BACK SQUAT

Primary Muscles	Quads, Glutes, Hamstrings, Spinal Erectors, Core
Setup	Bar on traps (high bar) or rear delts (low bar). Feet shoulder-width+, toes 15-30° out.
Key Coaching Cues	Brace core 360° (Valsalva). Initiate: hip hinge + knee push simultaneously. Depth: hip crease below knee. Knees track toes. Drive through heels.
Regression (Easier)	Goblet Squat → Box Squat → Front Squat
Progression (Harder)	Bulgarian Split Squat → Pause Squat → Safety Bar Squat

CONVENTIONAL DEADLIFT

Primary Muscles	Glutes + Hamstrings (primary), Erectors, Lats, Traps, Core
Setup	Bar over midfoot, hip-width stance. Hip hinge, grab bar just outside legs. Neutral spine.
Key Coaching Cues	"Protect armpits" for lat engagement. Brace before lift. Drive: leg press the floor → hips forward at lockout. Bar drags up body through thighs.
Regression (Easier)	Romanian DL → Trap Bar DL
Progression (Harder)	Sumo DL → Deficit DL → Snatch-grip DL

FLAT BENCH PRESS

Primary Muscles	Pectorals (primary), Anterior Deltoid, Triceps
Setup	Eyes under bar. 5-point contact: head, upper back, glutes, both feet. Grip 1.5-2x shoulder width.
Key Coaching Cues	Upper back arched, scapulae retracted+depressed. Touch lower chest, no bounce. Elbows 45° in during press. Leg drive through feet.
Regression (Easier)	DB Press → Push-up
Progression (Harder)	Incline Press → Close-grip Press → Board Press

PULL-UP / CHIN-UP

Primary Muscles	Latissimus Dorsi (primary), Biceps, Rear Delts, Rhomboids
Setup	Overhand (pull-up) or underhand (chin-up). Shoulder-width or wider. Full hang start.
Key Coaching Cues	Initiate with scapular depression (pack shoulders). Pull chest to bar, not chin. Control lowering — eccentric drives hypertrophy. Feet planted.
Regression (Easier)	Band-assisted → Lat Pulldown
Progression (Harder)	Weighted Pull-up → L-sit Pull-up → Muscle-up

OVERHEAD PRESS (OHP)

Primary Muscles	Anterior + Medial Deltoid (primary), Triceps, Upper Traps
Setup	Bar at shoulder height, elbows slightly forward. Tight grip, core braced, glutes squeezed.
Key Coaching Cues	Press straight up — head moves back to clear path, forward at lockout. Squeeze glutes + brace to prevent lumbar extension. Feet planted.
Regression (Easier)	DB OHP → Arnold Press
Progression (Harder)	Behind-neck Press → Push Press → Jerk

BARBELL ROW (BENT-OVER)

Primary Muscles	Lats, Rhomboids, Traps, Rear Delts, Biceps
Setup	Hip hinge, torso 30-45° from floor. Overhand grip, shoulder-width. Bar free-hanging.
Key Coaching Cues	Lead with elbows, not hands. Pull to lower chest/upper abdomen. Retract scapulae at top — squeeze between shoulder blades.
Regression (Easier)	DB Row → Chest-Supported Row
Progression (Harder)	Pendlay Row → T-bar Row → Cable Row

7.13 REP RANGE CONTINUUM

REP RANGE	LOAD (%1RM)	PRIMARY ADAPTATION	HYPERTROPHY POTENTIAL	REST PERIOD
1-3 reps	95-100%	Max strength, neural efficiency	Low (insufficient TUT)	3-5 min
4-6 reps	85-95%	Strength + myofibrillar hypertrophy	Moderate-High	2-4 min
6-12 reps	67-85%	Myofibrillar + sarcoplasmic hypertrophy	HIGH (sweet spot)	1-3 min
12-20 reps	50-67%	Sarcoplasmic + metabolic stress	Moderate-High	60-90 sec
20-30 reps	40-50%	Metabolic stress + muscular endurance	Low-Moderate	30-60 sec
30+ reps	<40%	Muscular endurance, occlusion	Low (except BFR)	30 sec

All rep ranges produce hypertrophy when taken near failure (RIR 1-3). 6-12 is optimal for most. Include variety for complete fiber type development.

MODULE 8 | WORKOUT PROGRAM LIBRARY

Beginner to Advanced · 13 user archetypes · Full Body · PPL · Upper-Lower · Home · Hostel

8.1 PROGRAM SELECTION MATRIX

USER TYPE	PROGRAM	FREQUENCY	EQUIPMENT	KEY NOTES
Absolute Beginner (0-3 months)	Full Body A/B	3 days/wk	Full gym or home	Compound only. Linear progression. Master form before load.
Beginner (3-12 months)	Full Body 3x	3 days/wk	Full gym	Add isolation. Progressive overload every session.
Intermediate (1-3 years)	Upper-Lower or PPL	4-6 days/wk	Full gym	Double progression. 2x/week per muscle. Volume block periodization.
Advanced (3+ years)	PPL or Bro Split	5-6 days/wk	Full gym	Block periodization. Specialisation phases. Auto-regulation.
Hostel Student	Bodyweight + Bands	4-5 days/wk	No gym	Progressive via harder variations. Pull-up bar + bands ideal.
Home Workout User	Home Dumbbell or BW	4 days/wk	Home	Unilateral work, tempo, isometrics for progressive overload.
Busy Professional	Upper-Lower or Full Body	3-4 days/wk	Full gym	45-60 min sessions. Supersets for efficiency. Minimal cardio.
Women (Intermediate)	Upper-Lower	4 days/wk	Full gym	Program heavy weeks in follicular phase. Glute/hamstring focus.
Teen Athlete (14-18)	Full Body	3-4 days/wk	Any	No heavy spinal loading. Dumbbell-first approach. Technique priority.
40+ Adult	Full Body or Upper-Lower	3-4 days/wk	Full gym	Extended warm-up. Moderate loads. Joint health emphasis.
Hardgainer	PPL or Bro Split	5-6 days/wk	Full gym	High volume. Compound-first. Minimal cardio. Focus on eating.
Skinny Fat	Full Body	3-4 days/wk	Full gym	Recomp protocol. Compound + isolation balance. Cardio 2x/week.
Strength + Hypertrophy	Conjugate or PHUL	4-5 days/wk	Full gym	Heavy days (3-5 RM) + volume days (8-15 RM) per week.

8.2 FULL BODY BEGINNER PROGRAM (3 DAYS/WEEK)

Each session trains all major muscle groups. 48+ hours between sessions. Linear progression: add weight every session for compound lifts.

EXERCISE	SETS	REPS	INTENSITY	REST	COACHING NOTE
Squat (Barbell/Goblet)	4	5-8	75-85%	3 min	Focus on depth and knee tracking
Bench Press (DB or BB)	3	8-10	70-80%	2 min	Scapulae retracted. Full ROM.
Barbell Row / DB Row	3	8-10	70-80%	2 min	Pull to lower chest. Squeeze.
Romanian Deadlift	3	10-12	65-75%	2 min	Hip hinge. Feel hamstring stretch.
OHP (DB or BB)	3	8-10	70-80%	2 min	Press vertical. Core tight.
Pull-up / Lat Pulldown	3	Max / 10-12	BW / 70%	2 min	Full extension. Lead with elbows.
Plank / Dead Bug	3	30-45 sec	BW	60 sec	Core stability foundation.

8.3 UPPER-LOWER INTERMEDIATE PROGRAM (4 DAYS/WEEK)

UPPER A (Mon) — STRENGTH FOCUS

EXERCISE	SETS	REPS	INTENSITY	REST
Bench Press	4	5-6	82-87%	3 min
Barbell Row	4	5-6	82-87%	3 min
OHP (DB)	3	8-10	70-80%	2 min
Lat Pulldown	3	10-12	70%	90 sec
Barbell Curl	3	10-12	70%	90 sec
Tricep Pressdown	3	12-15	65%	60 sec

LOWER A (Tue) — STRENGTH FOCUS

EXERCISE	SETS	REPS	INTENSITY	REST
Back Squat	4	5-6	82-87%	3 min
Romanian DL	3	8-10	75%	2 min
Leg Press	3	10-12	70%	90 sec
Leg Curl	3	12-15	65%	90 sec
Calf Raise	4	15-20	60%	60 sec

UPPER B (Thu) — HYPERTROPHY FOCUS

EXERCISE	SETS	REPS	INTENSITY	REST
Incline DB Press	4	10-12	68-72%	90 sec
Cable Row	4	12-15	65%	90 sec
Lateral Raise	4	15-20	60%	60 sec
Face Pull	3	15-20	Band	60 sec
Hammer Curl	3	12-15	65%	60 sec
Skull Crusher	3	12-15	65%	60 sec

LOWER B (Fri) — HYPERTROPHY FOCUS

EXERCISE	SETS	REPS	INTENSITY	REST
Deadlift (Conv)	4	4-5	85-90%	4 min
Bulgarian Split Squat	3	10/leg	70%	2 min
Hip Thrust	4	12-15	70%	90 sec
Leg Extension	3	15-20	60%	60 sec
Lying Leg Curl	3	12-15	65%	60 sec
Ab Wheel / Plank	3	10 / 30s	BW	60 sec

8.4 HOSTEL/HOME BODYWEIGHT PROGRAM (5 DAYS/WEEK)

SESSION	WORKOUT
MON — PUSH	Push-ups (4xMax) + Diamond Push-ups (3x10) + Pike Push-ups (3x10) + Chair Dips (3xMax)
TUE — PULL	Pull-ups / Door Rows (4xMax) + Table Rows (3x15) + Band Rows (3x20) + Dead Hangs (3x30s)
WED — LEGS	Air Squats (4x20) + Bulgarian Split Squat (3x12/leg) + Glute Bridges (4x20) + Wall Sit (3x40s)
THU — ACTIVE REST	Walk 30-45 min + Stretching + Foam Roll. No structured training.
FRI — FULL BODY CIRCUIT	Squat + Push-up + Table Row + Plank x 5 rounds (30s work / 15s rest each)
SAT — CORE + ARMS	Chin-ups (3xMax) + Band Curls (3x20) + Planks (3x40s) + Hollow Body (3x30s) + Leg Raises (3x15)
SUN — REST	Complete rest or 30-min gentle walk.

8.5 PUSH-PULL-LEGS (PPL) ADVANCED 6-DAY

SESSION	EXERCISES (SETS x REPS)
PUSH (Mon/Thu)	Bench 4x4-6 Incline DB 4x8-12 OHP 3x8-10 Lateral Raise 4x15 Tricep EXT 3x12 Skull Crusher 3x12
PULL (Tue/Fri)	Deadlift 4x4-5 Pull-up 4xMax Barbell Row 4x8-10 Cable Row 3x12 Face Pull 3x15 Barbell Curl 3x12
LEGS (Wed/Sat)	Squat 4x5-8 Romanian DL 3x8-10 Leg Press 3x12 Hip Thrust 4x12 Leg Curl 3x12 Calf Raise 4x20
SUN	Rest — active recovery: 30-min walk, stretching, foam rolling

8.6 MODULE 8 COACHING RULES

- [R071] Beginners: 3-day full body program with compound movements only for first 3-6 months.
- [R072] Progressive overload is mandatory in every session. Track all lifts — record weight, sets, reps.
- [R073] Rep range 6-12 is the primary hypertrophy zone. Include 4-6 heavy and 12-20 rep ranges for variety.
- [R074] Compound movements always first in every session, when the athlete is freshest.
- [R075] Beginner: add weight every session. Intermediate: every week. Advanced: every mesocycle.
- [R076] Minimum 2x/week frequency per muscle group for intermediate-advanced athletes.
- [R077] Volume: start at 10 sets/muscle/week. Increase 2 sets per 4-week block.
- [R078] Rest: 2-3 min for heavy compound work. 60-90 sec for isolation and pump work.
- [R079] Bodyweight athletes: progression through harder variations. No equipment = no excuse.
- [R080] Form before weight, always. Flag form breakdown as injury risk. Reduce load and fix first.

MODULE 9 | RECOVERY SYSTEM

Sleep science · Cortisol management · Deloads · CNS fatigue · Overtraining · Recovery scoring

9.1 SLEEP SCIENCE FOR MUSCLE GAIN

SLEEP PARAMETER	RECOMMENDATION	CONSEQUENCE OF DEFICIENCY
Duration	7-9 hours minimum (8-9 optimal for athletes)	1 week at 5hr = 15% testosterone drop, impaired MPS
Timing	Same sleep/wake time daily (±30 min)	Irregular schedule disrupts cortisol rhythm and anabolic hormones
Sleep Quality	Uninterrupted REM + deep sleep cycles	Fragmented sleep reduces GH pulse amplitude by 50-70%
Environment	Dark, cool (18-20°C), quiet	Bright light and heat impair melatonin and deep sleep onset
Pre-sleep Protein	30-40g casein 30-60 min before bed	Enhances overnight MPS by ~22% vs no protein
Alcohol	Avoid completely during bulk	Even 1 drink impairs GH release by 70% in first sleep cycle
Caffeine Cut-off	No caffeine after 2-3 PM	Caffeine half-life 5-7 hrs. Impairs sleep latency and quality.
Naps	20-30 min naps acceptable	90+ min naps can reduce nighttime sleep quality

9.2 RECOVERY METHODS EVIDENCE TABLE

RECOVERY METHOD	EVIDENCE	MECHANISM	PRACTICAL APPLICATION
Sleep (7-9 hrs)	STRONG	GH release, protein synthesis, cortisol reduction	Non-negotiable. First priority.
Post-workout Nutrition	STRONG	Glycogen resynthesis, MPS stimulation	30-50g protein + 60-120g carbs within 2 hrs
Active Recovery	MODERATE	Blood flow, metabolite clearance, DOMS reduction	Walk, swim, yoga, light cycling on rest days
Cold Water Immersion	MODERATE	Reduces acute inflammation and DOMS perception	10-15 min cold shower or ice bath post-training
Foam Rolling	MODERATE	Increases blood flow, reduces soreness perception	10-15 min of target muscle groups post-workout
Contrast Therapy	MODERATE	Vasodilation/vasoconstriction cycling improves circulation	3 min hot, 1 min cold — repeat 3-4 cycles
Sauna (dry/infrared)	MODERATE	Heat shock proteins, modest GH boost, relaxation	15-20 min session post-workout or on rest days
Compression Garments	LOW-MOD	Reduces swelling, improves venous return	Wear post-exercise for 24-48 hrs
Static Stretching	LOW	Increases ROM, may marginally reduce DOMS	5-10 min after training for trained muscle groups
Deload Week	STRONG	Reduces accumulated fatigue, allows supercompensation	1 week every 4-6 weeks of hard training

9.3 DELOAD PROTOCOLS

DELOAD TYPE	VOLUME	INTENSITY	FREQUENCY	BEST FOR
Volume Deload (PREFERRED)	Reduce 50%	Maintain (70-85% 1RM)	Maintain	Most athletes. Maintains neural drive.
Intensity Deload	Maintain	Reduce to 50-60% 1RM	Maintain	Joint pain, technique refinement focus.
Full Deload (Rest Week)	0 — rest or walk	0	0	Severe fatigue, illness, overtraining syndrome.

DELOAD TYPE	VOLUME	INTENSITY	FREQUENC Y	BEST FOR
Frequency Deload	Reduce 30%	Maintain	Reduce 50%	Busy schedule, moderate fatigue levels.

When to Deload — Trigger Criteria

- Every 4-6 weeks of hard training (scheduled, proactive)
- Joints are chronically sore or achy beyond normal training soreness
- Strength has declined for 2+ consecutive weeks without lifestyle explanation
- Sleep is disrupted and recovery feels consistently incomplete
- Mood, motivation, and energy are persistently and significantly low
- Training sessions are consistently poor quality despite adequate nutrition + sleep

9.4 OVERTRAINING SYNDROME (OTS)

CATEGORY	SEVERITY	SYMPTOMS
Performance	HIGH	Strength decline >10% over 2+ weeks. Slower times, reduced power output.
Physical	HIGH	Persistent fatigue, DOMS >72 hrs, frequent illness, elevated resting HR.
Hormonal	HIGH	Testosterone/cortisol ratio drops. Libido reduction. Menstrual irregularity (F).
Psychological	MODERATE	Reduced motivation, mood swings, depression, irritability, poor concentration.
Sleep	MODERATE	Insomnia despite fatigue, restless sleep, or excessive sleeping.
Appetite	MODERATE	Loss of appetite, weight loss despite adequate calorie tracking.

OTS Management: Phase 1 (1-2 wks): Complete rest. Sleep 9+ hrs. Maintenance calories. Phase 2 (1-2 wks): Light activity only (walking, yoga). Phase 3 (1-2 wks): Resume at 50% volume. Phase 4: Gradually return over 4-6 weeks.

9.5 MODULE 9 COACHING RULES

- [R081] Sleep is the #1 recovery priority. Below 7 hours consistently = flag as high-risk for muscle loss.
- [R082] Schedule deloads every 4-6 weeks proactively. Volume deload (50% reduction) is preferred method.
- [R083] If strength declines 2 consecutive weeks without explanation, prescribe immediate deload.
- [R084] Chronic life/work stress must reduce training volume. Do not add volume during high-stress periods.
- [R085] Pre-sleep protein: 30-40g casein/paneer/Greek yogurt recommended for all athletes in bulk phase.
- [R086] OTS: prescribe complete rest, not deload. Differentiate overtraining from normal overreaching.
- [R087] Alcohol: impairs GH, testosterone, and sleep quality simultaneously. Zero tolerance during bulk.
- [R088] Active recovery on rest days improves blood flow and reduces DOMS without taxing CNS.
- [R089] Foam rolling + static stretching post-workout: 10-15 min for injury prevention and recovery.
- [R090] Recovery formula: Adequate Sleep + Calorie Surplus + Low Stress + Deload = Optimal Recovery.

MODULE 10 | SUPPLEMENTS

Evidence-based only · Recommendation engine · Budget matrix · Suitability by profile

10.1 SUPPLEMENT EVIDENCE HIERARCHY

TIER	DEFINITION	SUPPLEMENTS
Tier 1 — STRONG	Consistent replicated human evidence. Recommend freely.	Creatine Monohydrate, Whey Protein, Caffeine
Tier 2 — MODERATE	Good evidence, some conflicting studies. Contextual use.	Fish Oil (Omega-3), Vitamin D, Magnesium, Zinc
Tier 3 — EMERGING	Early or limited human evidence. Use with caution.	Ashwagandha, HMB (older adults), Citrulline Malate
Tier 4 — WEAK/NO EVIDENCE	Mostly marketing. Do NOT recommend.	BCAAs (if protein met), Glutamine, Testosterone boosters, Fat burners
BANNED / ILLEGAL	Never recommend. Flag immediately if athlete mentions.	Steroids, SARMs, HGH, EPO, Clenbuterol, Prohormones

10.2 CREATINE MONOHYDRATE

PARAMETER	DETAIL
Mechanism	Replenishes PCr stores → more ATP for high-intensity work → higher training volume → more muscle
Muscle Gain Effect	Meta-analyses: +1.2-2.5 kg lean mass over 4-12 weeks vs placebo
Strength Effect	+5-15% strength gains greater than training alone
Standard Dose	3-5g per day continuously. No loading phase required.
Loading Protocol (optional)	20g/day in 4 doses x 5 days, then 3-5g/day maintenance
Timing	Post-workout is marginally better. Consistency of daily use matters most.
Form	CREATINE MONOHYDRATE ONLY. Kre-alkalyn, buffered, ethyl ester = marketing gimmicks.
Side Effects	Intramuscular water retention (normal, desirable). Safe for kidneys in healthy individuals.
Contraindications	Pre-existing kidney disease: medical clearance required before use.
Cost (India)	INR 400-800/month generic. INR 800-1500/month branded.
Evidence Level	TIER 1. 1,000+ studies. Strongest evidence of any sports supplement.

10.3 SUPPLEMENT PRIORITY STACK BY PROFILE

PROFILE	PRIORITY 1	PRIORITY 2	PRIORITY 3	OPTIONAL
Budget Athlete (any)	Food first	Creatine (INR 400-500/mo)	—	Multivitamin if diet is poor
Vegetarian + Low Protein	Whey Protein	Creatine	Vitamin B12 + D3	Algae Oil + Zinc
Non-Veg, Gym 4x/week	Creatine	Whey (if needed)	Fish Oil	Multivitamin + Caffeine
Hostel Student	Creatine (if >INR 500 free)	Multivitamin	—	—
Working Professional	Creatine	Whey (convenience)	Fish Oil	Vitamin D + Caffeine
Female Athlete	Whey (if protein low)	Creatine	Iron + Vitamin D	Fish Oil + Magnesium
Advanced Bulking Athlete	Creatine + Whey	Fish Oil	Vitamin D + Magnesium	Ashwagandha (if high stress)
Vegan Athlete	Vegan Protein (soy/pea blend)	Creatine	B12 + D3 + Zinc	Algae Oil (EPA+DHA)

10.4 MODULE 10 COACHING RULES

- [R091] Supplements are optional. Food is mandatory. Never recommend supplement before dietary foundation is solid.
- [R092] Creatine monohydrate is the #1 supplement recommendation for all resistance training athletes.
- [R093] Whey protein: recommend only when athlete cannot meet protein targets from whole food sources.
- [R094] Never recommend BCAAs if athlete is meeting total daily protein target — it is a waste of money.
- [R095] Fish oil: recommend if athlete is not eating fatty fish (sardines/mackerel/salmon) 2-3x/week.
- [R096] Vitamin D3: supplement routinely for Indian athletes who spend most time indoors or avoid sun.
- [R097] NEVER recommend steroids, SARMS, HGH, or testosterone boosters. Flag and educate athlete.
- [R098] Caffeine: cycle off 1-2 weeks per month to maintain sensitivity. Avoid for sleep-compromised athletes.
- [R099] For athletes under 18: food only. No supplements required or recommended beyond a multivitamin.
- [R100] Supplement budget priority: Creatine > Whey Protein > Fish Oil > Vitamin D > Multivitamin > Others.

MODULE 11 | SPECIAL POPULATIONS

13 dedicated coaching systems · Hostel · Vegan · Women · Teens · 40+ · Hardgainers · Travelers

11.1 HOSTEL STUDENTS

Calorie Target	TDEE + 200-400 kcal. Lean bulk. Account for mess food quality.
Budget Protein Stack	Eggs (6/day=INR 35) + Soy chunks 100g dry (INR 20) + Dal from mess + 500ml milk (INR 30) = ~130g protein at INR 85
Training	Bodyweight + resistance bands. Push-ups, pull-ups, squats, rows. 4-5 days/week.
Key Opportunity	Cold-soak soy chunks for 30 min, drain, season with salt. 5-min zero-cook 52g protein.
Supplement	Creatine if budget allows (INR 400-500/month). Multivitamin for food variety gaps.
Biggest Risk	Mess food carb-heavy, protein-poor. Must supplement protein via eggs + soy chunks daily.

11.2 COLLEGE STUDENTS

Calorie Target	TDEE + 200-350 kcal. Full lean bulk possible with modest budget.
Budget	INR 150-250/day food. Can include chicken 3x/week + eggs + milk + soy chunks.
Training	College gym (if available) or nearby gym. 4-5 days/week. PPL or Upper-Lower.
Time Management	Morning training before class. Evening if flexible schedule. Avoid exam week training peaks.
Biggest Risk	Exam season: reduce training 30% and sleep becomes priority.

11.3 VEGETARIANS

Protein Target	2.0-2.4 g/kg BW. 10-20% higher than non-vegetarian baseline.
Best Protein Sources	Soy chunks > Paneer > Eggs (if lacto-ovo) > Greek Yogurt > Milk > Roasted Chana
Critical Combinations	Rice + Dal = complete EAA profile. Corn + Beans = complete. Oats + Milk = complete.
Key Deficiency Risks	Vitamin B12 (supplement always), Zinc (pumpkin seeds, chickpeas), Iron (especially females), Omega-3 (algae oil)
Daily Template (75kg)	Eggs x4 + 300ml milk + 150g paneer + soy chunks 100g dry + Greek yogurt = 160g+ protein

11.4 VEGANS

Protein Target	2.0-2.6 g/kg BW. Lower bioavailability requires higher total intake.
Best Sources	Soy Protein Isolate (90g/100g) + Pea Protein (80g/100g blend) + Soy chunks + Tofu + Tempeh
Critical Supplements	B12 (500mcg/day), D3 (2000 IU/day), Zinc (15mg/day), Algae Oil EPA+DHA (2g/day), Iodine (if no iodized salt)
Meal Strategy	Soy/pea protein blend at every meal. Rice + lentil combinations. Add flaxseed and hemp seed.
Leucine Gap	Plant proteins have lower leucine. Add leucine supplement or increase protein by 20-30% per meal.

11.5 WOMEN

Protein Target	1.6-2.0 g/kg BW. Same principle relative to LBM as males.
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Calorie Surplus	+150-250 kcal/day typically sufficient due to lower TDEE.
Menstrual Cycle	Follicular (day 1-14): strength peaks — program heavy sessions here. Luteal (day 15-28): fatigue higher — reduce volume 15-20%.
Iron Needs	18 mg/day vs 8 mg/day for males. Vegetarian females at very high risk. Monitor ferritin.
Body Fat Floor	Never below 15% BF. Below this causes hormonal disruption and health risks.
Myth Correction	Heavy lifting DOES NOT make women "bulky." Testosterone levels prevent excessive muscle mass. Heavy lifting is safe and beneficial.

11.6 TEEN ATHLETES (14-18)

Training	No heavy spinal loading (barbell squat/DL at >80% 1RM). Dumbbells and machines preferred. Focus on technique.
Calories	Growing teens need TDEE + 200-300 kcal. Active teens may need 3,000-3,500+ kcal/day.
Supplements	Food only. No creatine under 16. No pre-workout stims. No testosterone boosters — ever.
Sleep	8-10 hours per night. Non-negotiable for growth hormone and development.
Key Priority	Build correct movement patterns now. They last a lifetime. Speed > load in early training.

11.7 40+ ADULTS

Anabolic Resistance	Older muscle responds less to protein. Increase target to 2.0-2.4 g/kg. Leucine 4g+ per meal.
Recovery	Requires more time. Add 1 extra rest day vs younger athletes. Deload every 4 weeks.
Joint Health	Extended warm-up (10-15 min). Avoid high-impact exercises. Fish oil 3-4g/day.
Testosterone	Declining testosterone after 35. Optimize sleep, Vitamin D, Zinc, Magnesium, dietary fat.
Program	Full body 3-4x/week. Moderate intensity (65-75% 1RM). Higher reps (10-15 range) safer on joints.

11.8 HARDGAINERS

Step 1	Audit actual food intake for 7 days via app. 90% discover they eat far less than perceived.
Step 2	Recalculate TDEE with higher NEAT multiplier. Some hardgainers burn 3,000+ kcal/day.
Liquid Calories	500ml whole milk + 2 tbsp PB + 1 banana + 1 scoop whey = +750 kcal easily drinkable.
NEAT Reduction	Consciously reduce non-essential movement. Hardgainers fidget and move more unconsciously.
Calorie Increment	Add +150 kcal every 2 weeks until scale shows 0.25 kg/week consistent gain.
Cardio	Eliminate all cardio in initial 3-month building phase. Add back after weight gain established.

11.9 SKINNY FAT USERS

Strategy	Recomposition (TDEE ±0). NOT a bulk. 2.0 g/kg protein. Heavy resistance training.
Timeline	6-12 months before considering a bulk. Build muscle base while fat percentage normalises.
Tracking	Scale may not move even as body recomposes. Track measurements + photos monthly.
Key Error	Starting a calorie surplus on a skinny fat physique accelerates fat gain without adequate muscle base.

11.10 BUSY PROFESSIONALS

Batch Cooking	Sunday: 2kg chicken + 2kg rice + vegetables. 7-day meal base done in 90 minutes.
Quick Breakfast	Overnight oats: 100g oats + 300ml milk + 30g whey + 1 banana. Prep in 5 min, eat on commute.
Office Strategy	Priority order at canteen: protein first, vegetables second, carbs third. Always hit protein.
Travel Kit	Protein powder single-serve sachets + almonds + protein bar. Hotel: egg breakfast + chicken dinner.
Training Timing	Morning training shows highest long-term adherence in professionals. 24% more consistent.

11.11 FREQUENT TRAVELERS

Hotel Training	Bodyweight circuit in room + hotel gym (usually dumbbells). 30-45 min minimum.
Nutrition on Travel	Whey sachets + nuts + protein bars as backup. Order grilled protein at restaurants.
Indian Restaurants (travel)	Dal makhani + roti + tandoori chicken = solid muscle gain meal anywhere in India.
Supplement Transport	Weekly sachet pre-pack of creatine + fish oil. Avoids carrying full tubs.

11.12 SHIFT WORKERS

Meal Timing	Anchor meals around training, not clock time. 2 pre-workout meals + 1 post-workout meal as minimum.
Sleep Strategy	Blackout curtains + white noise essential. Sleep 7-9 hrs regardless of time of day.
Vitamin D	Indoor/night-shift workers at high risk. Supplement 2,000 IU/day routinely.
Training	Train in alertest part of shift schedule. Avoid training immediately before 7-9 hr sleep window.

11.13 HOME WORKOUT USERS

Equipment Priority	Pull-up bar (INR 500-1500) + resistance bands (INR 300-1000) + adjustable dumbbells = covers 80% of training for 12+ months.
Progressive Overload	Harder exercise variations + slower tempo + pauses + unilateral work = overload without weight.
Program Type	Full body 4x/week or Push-Pull-Legs bodyweight version.
Advanced	Weighted vest + resistance bands allow near-gym level progressive overload for advanced trainees.

11.14 MODULE 11 COACHING RULES

- [R101] Skinny beginners and underweight athletes: aggressive calorie surplus (+400-500 kcal). Liquid calories are essential.
- [R102] Hardgainers: ALWAYS audit actual food intake first. 90% of hardgainers are under-eating, not under-responding.
- [R103] Female athletes: same training principles. Adjust for menstrual cycle. Program heavy weeks in follicular phase.
- [R104] Female athletes: minimum 15% BF. Iron supplementation mandatory for vegetarian or low-meat females.
- [R105] Vegetarian athletes: combine complementary proteins. Soy chunks are the best budget protein source.
- [R106] Hostel students: cold-soak soy chunks + mess dal + eggs + milk = complete protein foundation at minimal cost.
- [R107] Busy professionals: batch cooking on weekends + liquid shakes = the compliance solution.
- [R108] No gym = no problem for beginners. Bodyweight progressive overload is fully adequate for 6-12 months.
- [R109] Special populations: always lifestyle-first approach. Address the constraint before adding complexity.
- [R110] Never generate a one-size-fits-all plan. Always identify and address population-specific constraints first.

MODULE 12 | PLATEAUS & TROUBLESHOOTING

150+ IF-THEN rules · Weight · Strength · Recovery · Motivation · Compliance failures

12.1 WEIGHT PLATEAU TROUBLESHOOTING

- IF weight not increasing for 2 weeks on bulk → THEN increase daily calories by +200 kcal
- IF weight not increasing for 4 weeks → THEN audit food log for 7 days, recalculate TDEE, add +300 kcal
- IF weight gaining >0.6 kg/week → THEN reduce surplus by -200 kcal to control fat gain rate
- IF weight gaining >0.6 kg/week AND waist increasing >1 cm/month → THEN reduce surplus -300 kcal + increase protein
- IF weight fluctuating daily ± 1 -2 kg → THEN use 7-day rolling average. Normal water/glycogen fluctuation.
- IF weight stable despite surplus on tracking → THEN reassess tracking accuracy first. Likely logging error.
- IF hardgainer and not gaining on TDEE+300 → THEN add liquid calories: 500ml whole milk + 2 tbsp PB = +450 kcal
- IF weight gaining on scale but no visual progress at 3 months → THEN take measurements and compare photos
- IF losing weight despite eating "more" → THEN log everything for 7 days. Most common cause: weekend deficit.
- IF starting creatine and weight jumps 1-2 kg → THEN this is intramuscular water retention. Normal and desirable.

12.2 STRENGTH PLATEAU TROUBLESHOOTING

- IF strength plateau for 2 weeks (beginner) → THEN check sleep, nutrition, form. Try 1 deload week then reload heavier.
- IF strength plateau for 4 weeks (beginner) → THEN program change needed. Switch from LP to wave periodization.
- IF strength plateau for 2+ weeks (intermediate) → THEN implement block periodization: accumulation → intensification → deload.
- IF strength declining on bulk → THEN check: sleep (below 7hrs?), calories (below TDEE?), stress (high?), deload overdue?
- IF strength stalling on compound BUT isolation is improving → THEN neural fatigue. Reduce compound frequency 1 week.
- IF same weight for 3+ sessions (beginner) → THEN deload 10%, reset, try micro-loading (+1.25 kg instead of +2.5 kg).
- IF strength varies wildly session to session → THEN check: pre-workout nutrition, sleep quality, consistency of warm-up.
- IF bench press stalled but deadlift progressing → THEN add upper back volume. Weak upper back often limits bench.
- IF squat stalled → THEN add 1 set of pause squats per week. Builds power out of hole without adding volume.
- IF all lifts stalling simultaneously → THEN this is a systemic issue. Check: total caloric intake, sleep duration, stress.

12.3 RECOVERY & ENERGY TROUBLESHOOTING

- IF poor recovery (DOMS >72 hrs) → THEN increase sleep, reduce training volume 20%, add post-workout nutrition
- IF low energy in training → THEN check pre-workout carbs. Add 50-100g fast carbs 30-60 min pre-session.
- IF always tired despite 8 hrs sleep → THEN check iron (especially females), Vitamin D, thyroid. Get bloodwork.
- IF consistently fatigued → THEN deload immediately. Accumulated fatigue masks fitness — performance returns after rest.
- IF training energy low only on morning sessions → THEN add 30-50g fast carbs + caffeine 45 min before training.
- IF multiple muscle groups perpetually sore → THEN volume is too high. Reduce 2 sets per muscle group per week.
- IF one muscle perpetually sore → THEN muscle is getting insufficient recovery between sessions. Reduce frequency.
- IF experiencing joint pain → THEN add fish oil 3-4g EPA+DHA, reduce load 20%, check form, warm up thoroughly.
- IF knee pain during squats → THEN reduce depth temporarily, switch to leg press, add VMO work (terminal knee extension).
- IF lower back pain on deadlifts → THEN reduce load 20%, check hip hinge mechanics, add core work (dead bug, plank).
- IF shoulder pain on pressing → THEN avoid overhead pressing temporarily. Switch to neutral grip, reduce ROM.

12.4 NUTRITION TROUBLESHOOTING

- IF protein target missed 3+ days → THEN add soy chunks (100g dry = 52g protein at INR 20) as mandatory daily food.
- IF appetite too low to hit calories → THEN switch to liquid calories: milk + PB + protein shake. Less volume, more calories.
- IF GI distress from protein shakes → THEN switch whey concentrate to whey isolate. Reduce dose per serving.
- IF athlete vegetarian AND protein <1.6 g/kg → THEN immediate soy chunks or soy protein isolate addition mandatory.
- IF calorie tracking shows deficit but weight not dropping → THEN recalculate TDEE downward. Overestimated activity.
- IF hitting protein but still poor recovery → THEN check: sleep, caloric surplus, deload status, stress levels.
- IF hostel student cannot hit protein → THEN cold-soak soy chunks: 100g dry in cold water 30 min. No cooking required.
- IF gaining fat too quickly on bulk → THEN protein is likely too low. Increase to 2.2-2.4 g/kg. Reduce carbs slightly.

IF fat phobia (eating <20% calories from fat) → THEN educate on hormonal consequences. Immediate fat intake increase.
IF athlete reports "eating clean" but not gaining → THEN food is likely insufficient. Clean food can still be under-maintenance.

12.5 MOTIVATION & COMPLIANCE TROUBLESHOOTING

IF motivation low at 4 weeks → THEN show strength progress data. Muscle follows strength — data validates effort.
IF program hopping after 3-4 weeks → THEN address mindset. Minimum 12-week commitment required for any program.
IF missing >2 training sessions per week → THEN fix schedule adherence BEFORE optimising any other variable.
IF athlete says "no time to train" → THEN design 30-min minimum effective program. 30 min beats 0 min always.
IF athlete wants to quit → THEN revisit goals. Set 30-day mini-milestones. Celebrate non-scale victories.
IF athlete comparing to social media → THEN educate on natural vs enhanced athlete timelines. 95% of IG physiques are enhanced.
IF athlete not tracking food → THEN simplify to hand-portion method. Adherence to simple system beats non-adherence to precise system.
IF athlete bored with program → THEN review training logs. Progress data rebuilds motivation better than variety.
IF workout quality consistently poor → THEN schedule issue. Move training to optimal time. Check pre-workout nutrition.
IF athlete training but no visible progress at 3 months → THEN side-by-side progress photo comparison. Change is gradual and real.

12.6 MODULE 12 COACHING RULES

[R111] Always diagnose the root cause before prescribing solution. Symptoms can have multiple causes.
[R112] Insufficient calories is the #1 reason for no muscle gain. Always check this first.
[R113] Program hopping is a behaviour problem, not a program problem. Address the mindset first.
[R114] Progressive overload is measurable. No progress data = no effective coaching possible.
[R115] Sleep issues override all other coaching advice. Fix sleep first, then optimise everything else.
[R116] Excessive cardio during bulk is a silent muscle gain killer. Always quantify cardio volume.
[R117] Fat phobia leads to hormonal dysfunction. Educate and correct dietary fat intake to minimum 20%.
[R118] Post-workout nutrition missed >3x/week is a system failure. Create a simple portable system.
[R119] Dirty bulk: maximum 4 weeks acceptable before reassessing body composition metrics.
[R120] If all variables are correct and still no gain: reassess TDEE. May need significantly more calories.

MODULE 13 | AI DECISION ENGINE

200+ coaching rules · Athlete classification · Nutrition · Training · Recovery · Special conditions

13.1 INITIAL ATHLETE ASSESSMENT DECISION TREE

IF athlete BMI <18.5 AND training <1 year → THEN aggressive lean bulk (+400 kcal), 3-day full body, 1.8 g/kg protein
IF athlete BMI 18.5-24.9 AND training <1 year → THEN moderate lean bulk (+250-300 kcal), 3-day full body, 1.8 g/kg protein
IF athlete BMI 25-29.9 AND training <1 year → THEN recomp (TDEE ±0), 3-4 day full body, 2.0 g/kg protein
IF athlete BMI >30 AND training <1 year → THEN mild deficit (TDEE -300 kcal), 3-day full body, 2.2 g/kg protein
IF athlete INTERMEDIATE AND BF <15% (M) / <23% (F) → THEN lean bulk (+200-300 kcal), Upper-Lower or PPL, 2.0 g/kg
IF athlete INTERMEDIATE AND BF >20% (M) / >28% (F) → THEN mini-cut first (8 weeks, TDEE -300 kcal), then lean bulk
IF athlete ADVANCED AND <10% BF (M) → THEN lean bulk (+150-200 kcal). Slow controlled gains. 2.0-2.2 g/kg protein.
IF athlete ADVANCED AND >15% BF (M) → THEN recomp or mini-cut. Advanced athletes gain minimal muscle when fat is high.
IF athlete is HARDGAINER (gaining <0.1 kg/week on bulk) → THEN audit intake, add liquid calories, +150 kcal every 2 weeks
IF athlete is RETURNING after >3 month break → THEN classify as beginner. Reduce volume 40%. Expect "newbie gains" again.

13.2 WEEKLY ADJUSTMENT DECISION TREE

IF weight UP 0.1-0.3 kg/week → THEN stay the course. Lean bulk on track.
IF weight UP 0.3-0.6 kg/week AND strength increasing → THEN acceptable. Monitor waist. Reduce if waist gaining >1 cm/month.
IF weight UP >0.6 kg/week → THEN excess fat gain. REDUCE calories -200 kcal immediately.
IF weight UNCHANGED for 2 weeks → THEN INCREASE calories +150-200 kcal.
IF weight DOWN despite surplus tracking → THEN reassess tracking accuracy first. Stress or sleep check.
IF strength INCREASING AND weight stable → THEN recomp occurring. Excellent for BF >15% athletes.
IF strength DECLINING on bulk → THEN check: sleep quality, calorie intake accuracy, deload need, stress.
IF recovery POOR (3+ days DOMS, chronic fatigue) → THEN reduce volume 20%, increase calories +100, prioritize sleep.
IF athlete MISSES 2+ sessions this week → THEN investigate cause. System fix required. Do not add volume next week.
IF athlete REPORTS low energy + mood + libido → THEN check: sleep, calories, stress, deload needed, blood work.

13.3 NUTRITION DECISION RULES

IF user is VEGETARIAN → THEN base protein on eggs, dairy (paneer, milk, curd), soy chunks, lentils
IF user is VEGETARIAN AND protein target >150g/day → THEN prioritize soy chunks, paneer, Greek yogurt, eggs, whey
IF user is VEGETARIAN AND high protein AND low budget → THEN soy chunks (52g/100g dry, INR 20) + eggs + milk
IF user is VEGETARIAN AND lactose intolerant → THEN use soy milk, tofu, soy chunks, pea protein. Avoid dairy.
IF user is VEGAN → THEN soy protein isolate + pea protein blend + tofu + lentils + rice. Supplement B12, D3, zinc, omega-3.
IF user is LACTOSE INTOLERANT → THEN whey isolate (low lactose), soy protein isolate, plant milk alternatives
IF user has EGG ALLERGY → THEN replace with soy chunks, paneer, legumes, pea or soy protein powder
IF user is DIABETIC (Type 2) → THEN lower carb diet (150-200g/day), higher protein 2.0-2.4 g/kg, complex carbs only
IF user has HYPERTENSION → THEN reduce sodium, avoid canned/processed foods. Monitor creatine (generally safe).
IF user has PCOS (female) → THEN lower GI foods, moderate carbs, higher protein 1.8-2.0 g/kg, resistance training priority
IF user is RAMADAN fasting → THEN shift training to post-iftar. Eat within 30 min of breaking fast.
IF user SKIPS POST-WORKOUT MEAL → THEN pre-pack whey + banana in gym bag. Eat immediately post-session.
IF user fat intake <20% calories → THEN IMMEDIATE increase to 20% minimum. Testosterone protection priority.
IF user protein >3.0 g/kg → THEN flag as unnecessary cost. No additional muscle gain benefit. Redirect to food quality.
IF hostel student with budget INR 80-120/day → THEN eggs + soy chunks + dal + milk = minimum viable protein protocol

13.4 TRAINING DECISION RULES

IF athlete has FULL GYM → THEN use barbell program. Squat, deadlift, bench, row, OHP as foundation.
IF athlete has BASIC GYM (dumbbells only) → THEN dumbbell equivalents: DB press, DB row, goblet squat, Romanian DL
IF athlete has HOME GYM with pull-up bar + bands → THEN push-up variations, pull-ups, band squats, effective beginner-intermediate

IF athlete has NO EQUIPMENT → THEN bodyweight program: harder variations, tempo, pauses, unilateral work

IF athlete training at HOME AND advanced → THEN weighted vest + dumbbells + resistance bands for continued overload

IF athlete has GYM but inconsistent attendance → THEN create 3-day full-body program. Maximizes stimulus on available days.

IF athlete has CHRONIC JOINT PAIN → THEN switch to low-impact alternatives. Never train through sharp pain.

IF athlete doing >3 cardio sessions/week during bulk AND not gaining → THEN eliminate all cardio first

IF program has <50% compound movements → THEN restructure immediately with compound-first approach

IF athlete sleeping <7 hrs AND training 5+ days/week → THEN reduce training to 3-4 days. Sleep is more anabolic.

IF athlete is BEGINNER AND wants to train 6x/week → THEN educate on recovery. Set maximum 4 days/week.

IF athlete LOSING STRENGTH for 2+ consecutive weeks → THEN prescribe deload this week regardless of other factors.

IF athlete in EXAM SEASON → THEN reduce training volume 30%. Sleep becomes the priority investment.

IF female athlete in LUTEAL PHASE (day 15-28) → THEN reduce training volume 15-20%. Prioritize recovery.

IF athlete under 18 → THEN focus on compound movements and technique. Avoid heavy spinal loading.

13.5 RECOVERY & LIFESTYLE DECISION RULES

IF athlete sleeping <7 hrs consistently → THEN flag as HIGH RISK. Address sleep before any other variable.

IF athlete has HIGH STRESS job or exams → THEN reduce surplus 100 kcal. Reduce training volume. Add Ashwagandha.

IF athlete drinks alcohol regularly → THEN educate on GH suppression, testosterone reduction, sleep impairment.

IF athlete has POOR SLEEP HISTORY → THEN address sleep before increasing training. Add magnesium glycinate 300mg night.

IF athlete reporting BURNOUT symptoms → THEN OTS protocol: complete rest 1-2 weeks. Not a deload.

IF athlete using CAFFEINE daily with no cycle → THEN 1-2 week caffeine washout to restore sensitivity.

IF athlete TRAVELING frequently → THEN pack protein sachets, nuts, protein bars. Hotel: egg breakfast, grilled protein dinner.

IF athlete doing SHIFT WORK → THEN anchor meals to training session, not clock time. Sleep blackout curtains essential.

IF athlete RECOVERING FROM INJURY → THEN maintain protein 2.0-2.2 g/kg. Reduce training volume. No deficit.

IF athlete reports MOOD issues + low libido + fatigue → THEN check testosterone levels. Likely sleep or nutrition root cause.

IF athlete reports MENSTRUAL IRREGULARITY on deficit → THEN STOP deficit immediately. Return to maintenance calories.

IF athlete is under 18 AND requests supplements → THEN food only. No supplements required. Educate parents if necessary.

13.6 MASTER PROGRESS TRACKING TABLE

METRIC	HOW TO MEASURE	FREQUENCY	GOOD SIGNAL	BAD SIGNAL
Scale Weight	Morning fasted, 3-day average	Daily	+0.1-0.5 kg/week	Losing on bulk OR gaining >0.6 kg/week
Training Log	Record: exercise, sets, reps, weight	Every session	Strength increasing weekly	Same weights for 3+ weeks
Training Energy	Rate 1-10 during session	Every session	7-9/10 consistent	Below 6/10 for 3+ sessions
Sleep	Hours + quality (1-10)	Daily	7+ hrs, rested on waking	<7 hrs, tired on waking
Recovery	Soreness/fatigue 1-10	Daily	3-4/10 mild soreness	7+/10 chronic soreness
Body Measurements	Chest, waist, arms, legs (cm)	Monthly	Arms/chest up, waist stable	Waist +2 cm/month
Progress Photos	Front/side/back, same conditions	Monthly	Visible muscle fullness	No visual change in 3 months
Blood Markers	Testosterone, Iron, Vitamin D (annual)	Every 6-12 months	Within normal range	Deficiency or borderline values

13.7 STRENGTH MILESTONES (NATURAL ATHLETES)

Male Athletes — Relative to Bodyweight

LIFT	BEGINNER (3mo)	INTERMEDIATE (1yr)	ADVANCED (3yr)	ELITE (5yr+)
Squat	0.5x BW	1.0x BW	1.5x BW	2.0x BW
Deadlift	0.75x BW	1.25x BW	1.75x BW	2.5x BW
Bench Press	0.5x BW	0.75x BW	1.0x BW	1.25x BW

LIFT	BEGINNER (3mo)	INTERMEDIATE (1yr)	ADVANCED (3yr)	ELITE (5yr+)
Overhead Press	0.35x BW	0.5x BW	0.65x BW	0.8x BW
Barbell Row	0.4x BW	0.75x BW	1.0x BW	1.25x BW
Pull-ups	1 rep	10 reps BW	20 reps BW	+25 kg weighted

Female Athletes — Relative to Bodyweight

LIFT	BEGINNER (3mo)	INTERMEDIATE (1yr)	ADVANCED (3yr)	ELITE (5yr+)
Squat	0.35x BW	0.75x BW	1.0x BW	1.5x BW
Deadlift	0.5x BW	1.0x BW	1.25x BW	1.75x BW
Bench Press	0.25x BW	0.5x BW	0.75x BW	1.0x BW
Overhead Press	0.2x BW	0.35x BW	0.5x BW	0.65x BW
Pull-ups	0 reps	1-5 reps	8-12 reps	15+ reps

MODULE 14 | MASTER FAQ DATABASE

170+ expert Q&As; · Nutrition · Training · Recovery · Supplements · Special populations · Beginners

14.1 FAQ OVERVIEW

This FAQ database covers the most common questions from ZITLAS users across all categories. Organized by topic for easy RAG retrieval. Each answer is concise, actionable, and evidence-based.

14.x NUTRITION FAQs

Q: How much protein do I need to build muscle?	A: 1.8–2.0g per kg of bodyweight daily is optimal. Vegetarians should target 2.0–2.2g/kg due to lower bioavailability of plant proteins.
Q: Is whey protein necessary for muscle gain?	A: No. Whey is a convenient supplement but not necessary. You can hit protein targets from whole foods like eggs, chicken, paneer, and dal. Use whey only if whole food protein is insufficient.
Q: What is the best protein source for vegetarians in India?	A: Soy chunks (nutri nuggets) at 52g protein per 100g dry weight at ₹20-25 is the best budget option. Eggs are next if lacto-ovo vegetarian.
Q: How many calories should I eat to build muscle?	A: Calculate TDEE first using Mifflin-St Jeor formula, then add 200-350 kcal for a lean bulk. For example, a 70kg moderately active male needs ~2627 kcal maintenance + 250 = 2877 kcal for lean bulk.
Q: What is a lean bulk?	A: A lean bulk is a controlled calorie surplus of 200-350 kcal/day above TDEE, designed to maximize muscle gain while minimizing fat accumulation. Expected gain: 0.25-0.5 kg per month.
Q: What is the difference between a lean bulk and dirty bulk?	A: Lean bulk: +200-350 kcal, slow controlled gain. Dirty bulk: +1000+ kcal, rapid weight gain but significant fat accumulation. Lean bulk recommended for most athletes.
Q: How many meals should I eat per day for muscle gain?	A: 4-5 protein-containing meals per day is optimal. This keeps muscle protein synthesis (MPS) elevated throughout the day. Minimum 3 meals with protein at each.
Q: What should I eat before a workout?	A: 2-3 hours before: complex carbs + moderate protein (rice + chicken). 30-60 min before: simple carbs + light protein (banana + milk). Avoid high fat meals within 1.5 hrs of training.
Q: What should I eat after a workout?	A: 30-50g fast protein (whey or eggs) + 60-120g fast carbs (white rice, banana, bread) within 30-60 minutes post training. This maximizes glycogen replenishment and MPS.
Q: Is pre-sleep protein necessary?	A: Yes, for those trying to build muscle. 30-40g of casein (slow-digesting) protein before bed, such as 200g paneer or Greek yogurt, can increase overnight MPS by up to 22%.
Q: Can I build muscle on a vegetarian diet?	A: Yes, absolutely. Vegetarians should: increase protein by 10-20% over non-veg targets, combine plant proteins (rice+dal, corn+beans), prioritize soy chunks and eggs (if lacto-ovo), and supplement B12 and possibly vitamin D.
Q: What is BMR and how is it calculated?	A: BMR (Basal Metabolic Rate) is calories burned at complete rest. Mifflin-St Jeor formula: Males: $(10 \times W) + (6.25 \times H) - (5 \times A) + 5$. Females: $(10 \times W) + (6.25 \times H) - (5 \times A) - 161$. W=weight kg, H=height cm, A=age years.

Q: What is TDEE?	A: Total Daily Energy Expenditure. $TDEE = BMR \times Activity\ Multiplier$. Sedentary: 1.2, Light: 1.375, Moderate: 1.55, Very Active: 1.725, Extra: 1.9. This represents total daily calorie burn.
Q: How much fat should I eat for muscle gain?	A: Minimum 20% of total calories from fat (to protect testosterone). Optimal: 25-30%. Never go below 20% - it suppresses testosterone, estrogen, and anabolic hormones.
Q: Why are carbs important for muscle gain?	A: Carbs are the primary fuel for resistance training (glycolytic pathway), replenish glycogen after training, spare protein from oxidation (so protein goes to MPS not energy), and spike insulin which drives nutrients into muscle.
Q: How much water should I drink daily?	A: 35-40ml per kg bodyweight per day as baseline. Add 500-750ml for every hour of training. A 70kg athlete needs ~2.5-3.0L on rest days and 3.0-3.5L on training days.
Q: What is the leucine threshold?	A: Leucine is the key anabolic amino acid. A minimum of 2-3g leucine per meal is needed to maximally trigger MPS. A 30-40g serving of high-quality protein (whey, eggs, chicken) delivers this. Plant proteins require higher doses.
Q: Should I count calories?	A: For the first 4-6 weeks, yes - calorie tracking builds awareness of portion sizes and macros. After that, hand-portion tracking or intuitive eating based on learned awareness often suffices if progress is on track.
Q: What is protein synthesis?	A: Muscle Protein Synthesis (MPS) is the process of building new muscle protein from amino acids. It is triggered by training stimulus and leucine from dietary protein. MPS must exceed muscle protein breakdown (MPB) for muscle growth to occur.
Q: Can I eat junk food while building muscle?	A: Occasionally, yes. If protein and calorie targets are met, some junk food in the overall diet causes minimal harm. However, junk food is typically high in refined carbs, trans fats, and low in micronutrients, impairing recovery and performance.
Q: How important is meal timing for muscle gain?	A: Meal timing is a tier-3 factor. Total daily protein and calories (tier-1) matter most. However, distributing protein across 4-5 meals and having protein within 2 hours post-workout does provide measurable benefits.
Q: What is creatine and should I take it?	A: Creatine monohydrate replenishes ATP (energy) stores, allowing more volume in training. It is the #1 evidence-based supplement for muscle gain: +1.2-2.5 kg lean mass in 4-12 weeks vs placebo. Take 3-5g/day. No loading needed.
Q: What foods are rich in protein in India?	A: Top Indian protein sources: Soy chunks (52g/100g dry), Tuna canned (29g/100g), Chicken breast (31g/100g), Paneer (18g/100g), Eggs (13g/100g), Moong dal (7g/100g cooked), Roasted chana (22g/100g), Greek yogurt (9g/100g).
Q: Does protein timing matter within the anabolic window?	A: The anabolic window is real but wider than commonly believed. Post-workout protein within 0-2 hours is beneficial. If a pre-workout protein meal was consumed, the window may extend. Total daily protein intake remains the most critical factor.
Q: What is the role of fiber in a muscle building diet?	A: Fiber (25-35g/day) supports gut health, slows digestion, aids satiety, and maintains digestive regularity. Excessive fiber before training (>30g) can cause GI distress. Get fiber from vegetables, legumes, and whole grains.
Q: Is intermittent fasting compatible with muscle building?	A: Yes, IF 16:8 is compatible if protein targets (1.8-2.0g/kg) can be achieved in the eating window across 4-5 meals. Fasted weight training is suboptimal - minimum 10g EAAs/BCAAs should be consumed before fasted training.

Q: What happens if I eat too little protein?	A: Suboptimal MPS, muscle catabolism (breakdown), poor recovery, reduced strength gains, potential muscle loss especially during caloric deficit or heavy training. Protein below 1.4g/kg severely limits muscle gain.
Q: Can I build muscle in a caloric deficit?	A: Yes, but only for beginners, returning athletes, and those with higher body fat (>20% males, >28% females). This is called body recomposition. For most intermediate-advanced athletes, a surplus is needed for meaningful muscle gain.
Q: What are the best carbs for muscle building?	A: Complex carbs for pre-workout (oats, brown rice, sweet potato), fast carbs for post-workout (white rice, banana, bread). In India: chapati, dal, oats, banana, dates, sweet potato are excellent choices.
Q: How do I hit protein targets on a hostel student budget?	A: Budget protein strategy: 6 eggs (■35) = 36g protein. 100g soy chunks dry (■20) = 52g protein. 500ml milk (■30) = 17g protein. Total: 105g protein for ■85. Add mess dal (~15g) for ■0 extra. Cost: ■85-100 for 120g+ protein.
Q: What is body recomposition?	A: Simultaneous loss of fat and gain of muscle, possible at caloric maintenance or slight deficit. Best for: beginners, returning athletes, overweight individuals, those with BF >20% (M) or >28% (F). Requires: high protein (2.0-2.4g/kg) + training + adequate sleep.
Q: Are carbs bad for muscle building?	A: No. Carbs are essential for muscle building. They fuel workouts (glycolytic energy system), replenish glycogen, spare protein, and support anabolic hormone function. Never restrict carbs below 2g/kg during a muscle gain phase.
Q: How do I know if my calorie intake is correct?	A: Track scale weight 3-day average weekly. Lean bulk goal: +0.1-0.5 kg/week. Not gaining for 2 weeks = increase by 150-200 kcal. Gaining >0.6 kg/week = reduce by 150-200 kcal. Weekly trend is the truth.
Q: What micronutrients are critical for muscle building?	A: Vitamin D (testosterone, MPS, bone health), Zinc (testosterone cofactor), Magnesium (ATP production, muscle relaxation), Iron (oxygen transport, females especially), B12 (vegans), and Omega-3 (anti-inflammatory, MPS enhancer).
Q: Is ghee good or bad for muscle building?	A: Ghee is nutritionally valid and culturally appropriate. It provides saturated and monounsaturated fats that support testosterone and cell membrane function. 1-2 tsp/day within calorie budget is recommended. Avoid vanaspati (trans fat).
Q: What is a hardgainer?	A: Someone who struggles to gain weight despite eating 'a lot.' Usually caused by underestimating caloric intake, high NEAT (unconscious movement), or high metabolic rate. Solution: track every bite for 7 days, add 300-500 kcal, add liquid calories, reduce cardio.
Q: Can I drink alcohol while trying to build muscle?	A: Alcohol significantly impairs muscle gain: reduces testosterone and GH secretion, disrupts sleep quality, impairs MPS by 24% when consumed before bed, and adds empty calories. Zero tolerance recommended during active muscle gain phases.
Q: What is glycemic index and why does it matter?	A: GI measures how quickly a food raises blood sugar. High GI (>70): fast-digesting (white rice, bread, banana) - good post-workout for rapid glycogen replenishment. Low GI (<55): slow-digesting (oats, brown rice, lentils) - better for pre-workout sustained energy.
Q: How should I eat differently on rest days vs training days?	A: Training days: Higher carbs (5-7g/kg), same protein (2g/kg). Rest days: Lower carbs (2-3g/kg), same protein. This carb cycling maintains weekly average while optimizing performance on training days and recovery on rest days.

Q: What is NEAT and why does it matter?	A: Non-Exercise Activity Thermogenesis - calories burned from non-exercise movement (walking, fidgeting, housework). NEAT can range from 200-1000+ kcal/day between individuals. High NEAT athletes need higher TDEE multipliers. Track 8,000-10,000 steps daily.
Q: How does stress affect muscle building?	A: High stress elevates cortisol, a catabolic hormone that breaks down muscle protein, inhibits testosterone, impairs sleep quality, and reduces anabolic signaling. During high stress periods, reduce training volume 20-30% and prioritize sleep.

14.x TRAINING FAQs

Q: How many days per week should I train?	A: Beginners: 3 days full body. Intermediate: 4-5 days upper/lower or PPL split. Advanced: 5-6 days. Each muscle group should be trained 2x/week minimum for optimal hypertrophy in intermediate-advanced athletes.
Q: What is progressive overload?	A: Systematically increasing training stimulus over time: adding weight, reps, sets, density, or exercise difficulty. It is the fundamental mechanism that forces the body to adapt and grow. Without progressive overload, no muscle gain occurs.
Q: What rep range is best for muscle growth?	A: 6-12 reps at 60-85% 1RM is the primary hypertrophy range. However, all rep ranges build muscle when taken near failure. Include some heavy work (4-6 reps) and some pump work (12-20 reps) for complete muscle fiber recruitment.
Q: How long should my workouts be?	A: 45-75 minutes is optimal for most athletes. Longer workouts lead to cortisol accumulation and diminishing returns. High-quality shorter sessions beat long unfocused sessions. Track by sets completed, not time spent.
Q: What is the best workout split for beginners?	A: 3-day full body program (Monday-Wednesday-Friday or similar). All major muscle groups per session. Focus on compound movements only. This provides optimal frequency (2-3x/week per muscle) with adequate recovery.
Q: Should I do cardio while building muscle?	A: Minimal cardio is recommended during a muscle gain phase: 1-2 sessions/week of 20-30 min moderate intensity maximum. Excessive cardio burns surplus calories needed for muscle growth and competes with recovery resources.
Q: What is the difference between compound and isolation exercises?	A: Compound: multiple joints, multiple muscles (squat, deadlift, bench, row, OHP). Isolation: single joint, single muscle (curl, extension, lateral raise). Rule: 60-70% compound, 30-40% isolation. Beginners: 80-90% compound.
Q: How many sets should I do per muscle group per week?	A: 10-20 sets per muscle group per week for most athletes. Start at minimum effective volume (~10 sets) and progress by 2 sets per 4-week block. Never exceed maximum recoverable volume (~20 sets for most muscles).
Q: How long should I rest between sets?	A: Heavy compound work: 2-3 minutes. Hypertrophy work: 1-2 minutes. Isolation/pump work: 60-90 seconds. Longer rest is not wasted - it allows full PCr replenishment for heavier training performance.
Q: What is RPE and how do I use it?	A: RPE = Rate of Perceived Exertion on a scale of 1-10 (10 = max effort). For hypertrophy, target RPE 7-9 for most work sets, meaning 1-3 reps left in reserve (RIR). Never training to absolute failure on every set.

Q: How do I know if I'm training hard enough?	A: Use the RIR (Reps in Reserve) system. Aim for 1-3 RIR per set. If you stop at 10 reps but could do 5 more, that's too easy. If you cannot complete the planned reps, that's too hard. Record actual performance each session.
Q: What are the most important exercises for muscle gain?	A: The big 5: Squat, Deadlift, Bench Press, Barbell Row, Overhead Press. These compound movements recruit the most muscle mass, trigger the strongest hormonal response, and provide the best strength-to-muscle gain ratio.
Q: What is a deload week and when should I take it?	A: A deload is a planned reduction in training (typically 50% volume reduction while maintaining intensity) taken every 4-6 weeks. It allows accumulated fatigue to dissipate, joints to recover, and the nervous system to super-compensate.
Q: Can I build muscle without weights?	A: Yes, bodyweight training is effective for beginners to intermediate athletes, especially for 6-12 months. Progress through harder variations (push-up → pike push-up → handstand push-up). After intermediate level, added resistance is needed for continued gains.
Q: How do I track my progress?	A: Weekly: scale weight 3-day average, training performance. Monthly: body measurements (chest, arms, waist, legs), progress photos, strength benchmarks. Annually: DEXA or calipers. Track all lifts in a log every session.
Q: What should a beginner's training plan look like?	A: 3 days/week, full body. Session: Squat 3x8, Bench/Push 3x8, Row/Pull 3x8, OHP 2x10, Deadlift 1x5. Focus on form and progressive overload every session. No isolation work for first 3 months.
Q: How do I build bigger arms?	A: Prioritize compound movements (rows, pull-ups, chin-ups for biceps; dips, bench press, OHP for triceps). Add 4-6 sets of direct arm work (curls + extensions) 2x/week. Triceps are 2/3 of arm size - prioritize over biceps.
Q: What causes DOMS (muscle soreness)?	A: Delayed Onset Muscle Soreness occurs 24-72 hrs after training. Caused by microscopic muscle damage, especially during eccentric phase. Not an indicator of a good workout. Excessive DOMS (>72 hrs) indicates volume was too high.
Q: How do I train without a gym?	A: Bodyweight: push-ups (chest/triceps), pull-ups or rows (back/biceps), squats + lunges + glute bridges (legs/glutes), core work. Resistance bands add loading options. A pull-up bar is the most valuable piece of home equipment (■500-2000).
Q: Should I train to failure?	A: Training to absolute failure every set increases injury risk, CNS fatigue, and recovery time. Optimal: train within RIR 1-3 (leaving 1-3 reps in the tank) for most sets. One set to failure per muscle per session can be productive for intermediates.
Q: What is muscle memory?	A: Muscles have memory through satellite cells that retain nuclei from previous training. If someone loses muscle (detraining), they regain it significantly faster (weeks-months) than it took to build originally (months-years).
Q: How long does it take to see results?	A: Beginners see strength improvements in 2-3 weeks (neural adaptation). First visible muscle changes appear in 6-12 weeks with consistent training and nutrition. Significant physique changes take 6-12 months. Life-changing transformation: 2-5 years.
Q: What is the best exercise for big legs?	A: Back squat is the king of leg exercises. It recruits quads, hamstrings, glutes, and core simultaneously. Combined with Romanian deadlifts (posterior chain) and leg press, this covers complete leg development.

Q: How do I fix muscle imbalances?	A: Identify the weaker side. Add 1-2 extra sets unilaterally on weak side. Use dumbbell exercises to prevent dominant side compensation. Common imbalances: dominant side pec, dominant side lat, stronger quad vs hamstring.
Q: What is tempo training?	A: Controlling speed of each movement phase. Format: Eccentric-Pause-Concentric. Example: 3-1-2 = 3sec lower, 1sec pause, 2sec lift. Slower eccentrics increase TUT (time under tension) and mechanical damage, stimulating more hypertrophy.
Q: Is it better to train in the morning or evening?	A: Individual preference matters most. Research shows slight performance peaks in late afternoon (3-5 PM). However, the best time is whenever you will train consistently. Morning training shows 24% higher long-term adherence in some studies.
Q: Can I train every day?	A: Not recommended for strength/hypertrophy athletes. Recovery occurs BETWEEN sessions, not during. Maximum effective: 5-6 days/week with at least 1 full rest day. Beginners: 3 days. Overtraining is a real concern with daily high-intensity training.
Q: What is a superset?	A: Two exercises performed back-to-back with minimal rest. Agonist-antagonist (biceps+triceps) or same muscle group. Saves time, increases training density, and can enhance metabolic stress. Best used for isolation exercises.
Q: What is periodization?	A: Systematic variation of training variables over time. Linear (beginners): increase weight each session. Block (intermediate): 4-week accumulation → intensification → deload blocks. Wave (advanced): varying loads weekly within a program.
Q: What should I do if I miss a workout?	A: Do not try to make up missed sessions by adding extra volume next session. Resume normal program on the next scheduled day. Two missed sessions per week = investigate the cause (schedule, motivation, overload) and fix the system.
Q: What is the importance of form over weight?	A: Proper form is critical for: muscle targeting (wrong form = wrong muscle worked), injury prevention (wrong form = injury risk), and long-term progress (injury ends progress). Always reduce weight to maintain form. Record lifts monthly for form review.
Q: How do I build a chest without a bench?	A: Push-up variations: standard, wide, diamond, decline, archer. Chest dips on parallel bars or chairs. Pike push-ups. Resistance band chest fly. With dumbbells: floor press, floor fly. A basic bodyweight push-up program can build impressive chest for beginners.
Q: What is the best exercise for building a V-taper?	A: Wide back (lats): Pull-ups, lat pulldowns, straight arm pulldowns. Narrow waist: core stability work (not endless crunches). Capped shoulders: lateral raises 4x per week minimum. The V-taper is primarily built with compound pulling and lateral raises.
Q: How do I increase my bench press?	A: Progressive overload: add weight systematically. Improve technique: leg drive, arch, retracted scapulae, elbow angle. Strengthen supporting muscles: triceps (close-grip press, skull crushers), front deltoids (OHP), and serratus. Train chest 2x/week.
Q: How important is warm-up?	A: Warm-up is essential for injury prevention and performance. Minimum: 5 min general warm-up (light cardio, jumping jacks) + 2-3 warm-up sets at 40-60% 1RM for first compound exercise. Reduces injury risk significantly and improves performance.
Q: What exercises can I do at home for glutes?	A: Glute bridges (barbell or bodyweight), hip thrusts, donkey kicks, clamshells, banded monster walks, bodyweight squats, lunges, Bulgarian split squats. A resistance band makes home glute training significantly more effective.

Q: What is the best exercise for a bigger back?	A: Deadlift (overall mass), pull-ups/chin-ups (lat width and thickness), barbell bent-over row (thickness), seated cable row (mid back), face pulls (rear delt). Train back with both vertical (pulling down) and horizontal (rowing) movements.
Q: How do I build discipline to train consistently?	A: Schedule training sessions like important appointments. Start with minimum effective dose (3 days/week). Find an accountability partner. Track progress to see tangible results. Build the habit first (30-day commitment), then optimize the program.

14.x RECOVERY FAQs

Q: How much sleep do I need to build muscle?	A: 7-9 hours per night minimum. Growth hormone is secreted primarily during deep sleep (stages 3-4). One week of 5hr sleep reduces testosterone by 15% and impairs MPS. Sleep is the single most important recovery factor.
Q: What is overtraining?	A: Overtraining Syndrome (OTS) is persistent fatigue caused by accumulated training stress exceeding recovery capacity. Symptoms: strength decline >10%, persistent fatigue, mood changes, sleep issues, libido reduction, frequent illness. Treatment: complete rest 1-2 weeks.
Q: What is overreaching vs overtraining?	A: Overreaching: short-term (days-1 week) performance decrease from excessive training. Recovers with 1-2 weeks of deload. Overtraining Syndrome: long-term (weeks-months) decline requiring full rest. OTS is much more severe and takes longer to recover from.
Q: Do I need to take rest days?	A: Yes. Muscle grows during recovery, not during training. Minimum 1 full rest day per week from all structured exercise. Active recovery (light walking, yoga) on rest days improves blood flow without taxing the CNS.
Q: What is DOMS and how do I reduce it?	A: DOMS (Delayed Onset Muscle Soreness) peaks at 24-72 hrs post training. Reduce it with: active recovery (light walking), adequate protein, sleep 8+ hrs, foam rolling, cold water immersion 10-15 min. Not a sign of a good workout - just a sign of novelty.
Q: Does foam rolling work?	A: Moderate evidence supports foam rolling for reducing soreness perception and improving ROM (range of motion). 10-15 minutes of foam rolling post-workout is a practical recovery tool. Not a replacement for adequate sleep and nutrition.
Q: Is an ice bath beneficial?	A: Moderate evidence. Cold water immersion (10-15 min at 10-15°C) reduces acute inflammation and soreness. However, excessive cold immediately post-workout may blunt adaptation signals. Best used 4+ hours after training or on recovery days.
Q: What is CNS fatigue?	A: Central Nervous System fatigue is distinct from muscular fatigue. Symptoms: reduced bar speed despite adequate muscle strength, low motivation, grinding reps. Caused by heavy compound training (deadlifts, squats). Recovery: 48-72 hrs and good sleep.
Q: How do I know if I need a deload?	A: Signs: strength declining 2+ consecutive weeks, joints aching chronically, persistent fatigue, poor sleep, low motivation, every session feels hard. Best practice: schedule a deload every 4-6 weeks regardless of symptoms.
Q: What is the best way to recover faster?	A: Priority order: 1) Sleep 7-9 hrs, 2) Adequate protein and calories, 3) Post-workout nutrition (carbs+protein), 4) Active recovery, 5) Cold therapy, 6) Foam rolling. The first two overwhelm the rest in importance.

Q: Can I train if I'm sore?	A: Mild soreness (3-4/10): train normally. Moderate soreness (5-6/10): train but reduce intensity 20%. Severe soreness (7+/10): rest or active recovery only. Never train a severely sore muscle to failure - injury risk elevated.
Q: How does stress affect recovery?	A: Chronic stress elevates cortisol, which breaks down muscle protein, reduces testosterone, impairs sleep quality, and limits recovery. During high stress periods (exams, work crunch), reduce training volume 20-30% and prioritize sleep and nutrition.
Q: What role does sleep quality play?	A: Quality matters as much as duration. Deep sleep (stages 3-4) is when GH is secreted. Fragmented sleep reduces GH pulse amplitude by 50-70%. Dark, cool (18-20°C) room, consistent sleep/wake times, and no screens 60 min before bed improve quality.
Q: Does massage help muscle recovery?	A: Moderate evidence. Sports massage reduces soreness perception, improves blood flow, and promotes relaxation. Self-massage (foam rolling, massage gun) is practical and accessible. Professional massage is beneficial but not essential.
Q: What is active recovery?	A: Light physical activity on rest days (walking, light cycling, swimming, yoga) that promotes blood flow and waste removal from muscles without significant training stress. Better than complete sedentary rest for reducing DOMS and improving next-day performance.
Q: How long should I rest between workout sessions?	A: 48 hours between training the same muscle group is the minimum. Heavy compound training (deadlifts, squats) may require 72 hours. Design your split to allow adequate recovery between sessions targeting the same muscles.
Q: What are signs I'm recovering well?	A: Strength maintained or increasing, energy levels good, sleeping well, mild soreness resolves within 48-72 hrs, mood and motivation stable, hunger appropriate to activity level. These are all positive recovery indicators.
Q: Should I do yoga for recovery?	A: Yes, yoga is an excellent active recovery modality. It improves ROM, reduces muscle tension, promotes relaxation (reduces cortisol), and improves mind-body awareness. Even 15-20 minutes of yoga on rest days has measurable recovery benefits.
Q: What is the best pre-sleep routine for muscle recovery?	A: 90-60 min before sleep: no screens or blue light. 60 min: magnesium glycinate 300mg (improves sleep quality). 30-45 min: 30-40g casein protein (paneer, Greek yogurt). 30 min: cool room (18-20°C). Same sleep/wake time daily.
Q: Can I overtrain if I am a beginner?	A: True overtraining syndrome is rare in beginners due to lower training capacity. However, beginners can overreach (too much volume too soon), leading to injury and burnout. Stick to 3 sessions/week for first 3-6 months. Increase gradually.

14.x SUPPLEMENTS FAQs

Q: What supplements do I actually need to build muscle?	A: Mandatory: Only food. Optional tier 1: Creatine monohydrate (3-5g/day) - strong evidence. Optional tier 2: Whey protein (if dietary protein target not met from food), Fish oil (2-4g EPA+DHA), Vitamin D. Everything else is mostly marketing.
Q: When should I take creatine?	A: Any time daily, consistently. Post-workout is marginally better. Timing matters less than consistency. Do not skip days. No need for loading phase - 3-5g/day continuously for 3-4 weeks achieves full muscle saturation.

Q: What type of creatine is best?	A: Creatine monohydrate. No other form is better despite marketing claims. Kre-alkalyn, buffered creatine, ethyl ester, micronized - all have similar or worse efficacy than basic monohydrate. Buy the cheapest monohydrate you can find.
Q: Should I take BCAAs?	A: No, if you are already meeting your daily protein target of 1.8-2.0g/kg. BCAAs provide no additional benefit when total protein is adequate. They are expensive aminos already present in your food protein. Save the money.
Q: Is whey protein safe?	A: Yes, whey protein is safe for healthy individuals. No evidence of kidney damage in healthy people with adequate hydration. Ensure 35-40ml water/kg/day if consuming >2g/kg protein. Those with pre-existing kidney disease should consult a doctor.
Q: What is the best time to take whey protein?	A: Post-workout (fast absorption, triggers MPS) and morning (breaks overnight fast quickly) are optimal. Whey absorbs in 1-2 hours. Not ideal before bed - use casein or whole food protein (paneer) for pre-sleep instead.
Q: Should I take a pre-workout supplement?	A: Pre-workout supplements primarily work through caffeine (3-6mg/kg). If you use coffee (80-120mg caffeine per cup), you can achieve the same effect cheaper. Commercial pre-workouts often contain ineffective ingredients alongside caffeine. Build caffeine tolerance carefully.
Q: What is the difference between whey concentrate and isolate?	A: Concentrate: 70-80% protein, contains some lactose and fat. Isolate: 90%+ protein, minimal lactose. Most people tolerate both. If lactose intolerant, use isolate or switch to plant protein. Isolate is more expensive but useful for lactose sensitivity.
Q: Are mass gainers worth it?	A: Commercial mass gainers are mostly maltodextrin (cheap sugar) mixed with low-quality protein, sold at premium prices. DIY mass gainer is better: 100g oats + 50g peanut butter + 2 scoops whey + 500ml whole milk = ~900 kcal, 70g protein at fraction of cost.
Q: Is caffeine safe for exercise?	A: Yes, caffeine (200-400mg) is well-established as a performance enhancer: +3-7% strength, improved focus and endurance. Cycle off 1-2 weeks/month to reset tolerance. No caffeine after 2-3 PM if sleep is a concern. Avoid in hypertension or anxiety disorders.
Q: Do I need a multivitamin?	A: Recommended as nutritional insurance for athletes with restricted diets (vegetarians, hostel students with poor variety, those in caloric deficit). Not a replacement for whole food variety. Key nutrients: Vitamin D, Zinc, Magnesium, Iron (females), B12 (vegans).
Q: What supplements are illegal in sports?	A: Steroids (testosterone, nandrolone, stanozolol), SARMs, EPO, HGH, clenbuterol, prohormones, peptides. All are banned in competitive sports and illegal without prescription in India. Never take these - health risks are severe.
Q: Can supplements replace real food?	A: No. Whole food provides nutrients, fiber, phytonutrients, and a food matrix that supplements cannot replicate. Supplements fill specific gaps. Rule: Build a solid dietary foundation first, then use supplements to address specific deficiencies.
Q: Is ashwagandha useful for muscle building?	A: Moderate evidence (Tier 3). Ashwagandha (300-600mg KSM-66 extract) reduces cortisol, may modestly improve testosterone and recovery, and has mild adaptogenic effects. Useful during high-stress periods. Not a primary muscle-building supplement.
Q: What is the best supplement stack for a beginner?	A: Priority 1: Food. Priority 2: Creatine monohydrate (3-5g/day) - if budget allows. Priority 3: Nothing else until diet and training are consistent. Adding supplements on a poor dietary foundation provides minimal benefit.

Q: Is vitamin D important for athletes?	A: Yes. Majority of Indians are vitamin D deficient (indoor lifestyle, dark skin pigmentation requiring more sun). Vitamin D acts as a steroid hormone, critical for testosterone production, muscle protein synthesis, bone health, and immune function. Supplement 1000-2000 IU/day.
Q: How much does creatine cost in India?	A: Generic creatine monohydrate: ₹400-600 per month (for 3-5g/day). Branded creatine (BigMuscles, MuscleBlaze): ₹800-1200/month. All provide the same molecule. Buy the cheapest certified pure monohydrate available.
Q: Are protein bars a good option?	A: Protein bars are convenient but expensive per gram of protein. Most commercial bars contain significant sugar and artificial ingredients. Good for travel or emergencies. For daily protein needs, whole food and whey powder are more cost-effective.
Q: What supplements should vegetarians take?	A: Priority: Vitamin B12 (essential - not found in plant foods), Vitamin D3, Zinc, Fish oil (algae-based omega-3 EPA+DHA for vegans), Iron (women). Whey or pea/soy protein if protein targets are not met from food.
Q: Can I take creatine without working out?	A: Creatine has no meaningful effect without resistance training to deplete and refuel ATP stores. It only works in the context of high-intensity training. Taking creatine while sedentary has no muscle-building benefit and wastes money.

14.x SPECIAL POPULATIONS FAQs

Q: What are the specific challenges for women building muscle?	A: Women have 5-10% of male testosterone levels, leading to 40-60% slower absolute muscle gain. However, relative to their potential, women respond equally well to training. Menstrual cycle affects recovery - program heavy training in follicular phase (days 1-14).
Q: Can teenagers (under 18) lift weights?	A: Yes, resistance training is safe and beneficial for teenagers when performed with proper form and appropriate loading. Focus on technique, compound movements, and moderate intensity. Avoid maximal lifting or extreme volumes. Food-only nutrition - no supplements needed or recommended.
Q: How does muscle building change for people over 40?	A: Anabolic resistance increases with age - more protein is needed (2.0-2.4g/kg) to trigger the same MPS response. Recovery takes longer. Warming up is more important. Cardiovascular health becomes a co-priority. The principles remain the same, execution adjusts.
Q: What is the best approach for a skinny-fat person?	A: Skinny-fat (low muscle + higher fat): Body recomposition is the goal. Do NOT bulk (will only add more fat). Do NOT aggressive cut (will lose the little muscle you have). Target TDEE maintenance with high protein (2.0-2.2g/kg) + resistance training 4x/week.
Q: How should hostel students approach muscle building?	A: Core strategy: Soy chunks (₹20, 52g protein/100g dry) + 6 eggs (₹35, 36g protein) + 500ml milk (₹30, 17g protein) = 105g protein for ₹85. Training: bodyweight if no gym, 3-day program. Use mess for carbs (rice, dal, sabzi). Supplement only if budget allows.
Q: What are the best strategies for busy professionals?	A: Sunday batch cook: 2kg chicken + 2kg rice in 1 hour = 7-day meal base. Protein shake for breakfast (2 min prep). Pre-packed post-workout shake+banana in gym bag. 4-day training program (45-60 min sessions). Prioritize sleep and nutrition over extra training sessions.

Q: What is the best diet for hardgainers?	A: Audit actual intake first (most hardgainers eat less than perceived). Add 500ml whole milk + 2 tbsp peanut butter + 1 banana as liquid calorie add-on (+600 kcal). Eat 5-6 meals. Reduce cardio. Sleep 8-9 hrs. Increase calories by 150 kcal every 2 weeks until 0.25 kg/week gained.
Q: How do vegans build muscle?	A: Use soy protein isolate + pea protein blend for complete amino acids. Add 15-20% more protein than non-veg targets. Supplement: B12 (essential), Vitamin D3, algae omega-3, zinc, iron. Use tempeh, edamame, tofu, lentils, and quinoa as primary protein sources.
Q: Can someone with PCOS build muscle effectively?	A: Yes. Resistance training is especially beneficial for PCOS - improves insulin sensitivity, hormonal balance, and body composition. Focus on: higher protein (1.8-2.0g/kg), lower GI carbs, consistent training. PCOS does not prevent muscle building.
Q: How should I modify training if I have knee pain?	A: Reduce squat depth temporarily. Switch to leg press (easier on knees). Avoid high-impact exercises. Continue quad training via leg extension (shorter ROM). Add terminal knee extensions (TKE) for VMO strengthening. Ice after training. Consult physio if persistent.
Q: What modifications are needed for lower back pain?	A: Reduce deadlift load 20%. Switch to Romanian DL or trap bar DL. Assess hip hinge mechanics - most back pain during lifting is form-related. Add core stability work (dead bug, planks). Avoid flexion-based exercises during acute pain. Consult a physiotherapist.
Q: How do shift workers optimize muscle building?	A: Train consistently within your schedule regardless of shift pattern. Align training session with the start of your awake cycle (even if at 2 AM). Meal prep batches of food. Prioritize sleep consistency over quantity when rotating. Supplement melatonin if sleep is severely disrupted.
Q: Is it possible to build muscle while traveling frequently?	A: Yes. Hotel gym: use dumbbells and cables for all movements. Airport protein: Greek yogurt, boiled eggs, nuts, protein bars. Pack single-serve protein sachets. Book hotels with gyms. 20-30 min bodyweight circuit better than nothing. Maintain protein intake as the non-negotiable.
Q: What is the minimum training needed to maintain muscle during a busy period?	A: Maintenance requires less volume than building. 1-2 sessions/week covering major muscle groups with moderate intensity is sufficient to maintain most muscle mass. Even 30-45 min full body sessions twice weekly prevent significant detraining.
Q: How does Ramadan affect muscle building?	A: Shift training window to 60-90 min after Iftar (breaking fast). Consume high-protein, high-calorie meals during eating window (Iftar to Sehri). Priority: protein target met, calorie surplus maintained. Sleep quality may be disrupted - reduce training volume 15-20%.
Q: Can underweight individuals gain muscle quickly?	A: Yes. Underweight individuals (BMI <18.5) are in the best position for rapid muscle gain: high insulin sensitivity, neurological adaptation potential, can support higher surpluses. Use moderate aggressive bulk (+400-500 kcal). Expect 0.5-1.0 kg/week gain in first months.
Q: What is the best approach for complete beginners?	A: Start with 3-day full body program. Learn compound movements with proper form (first 4-6 weeks). Track every workout. Hit protein target (1.6g/kg minimum). Sleep 7-9 hrs. Set realistic expectations: 6-12 kg muscle possible in year 1 with consistency.
Q: How should returning athletes (after break) approach training?	A: Reduce volume by 30% for first 4 weeks. Muscle memory means strength returns faster than building from scratch. Avoid the temptation to jump back to previous loads - injury risk is high. Nutrition: maintain protein 2g/kg. Expect full return to previous level in 4-8 weeks.

Q: What are the special considerations for female athletes building muscle?

A: Same protein principles apply (1.6-2.0g/kg). Program heavy sessions in follicular phase (days 1-14). During luteal phase (days 15-28), higher fatigue is normal - adjust training intensity. Iron supplementation often needed. Minimum 15% BF - do not go below.

14.x TROUBLESHOOTING FAQs

Q: Why am I not gaining weight despite eating a lot?

A: Most common cause: underestimating caloric intake. Track every bite using an app for 7 days. Second: NEAT is high (you move more unconsciously than you think). Third: recalculate TDEE - you may be burning more than estimated. Add 200-300 kcal to current tracked intake.

Q: My strength is declining even though I'm eating in a surplus. Why?

A: Check: sleep quality (below 7 hrs will cause strength decline), cumulative fatigue (deload needed), stress levels (cortisol), tracking accuracy (calories may be below TDEE despite perceived surplus), and whether deload is overdue (every 4-6 weeks).

Q: I'm gaining weight but not seeing muscle. What's wrong?

A: Two possibilities: gaining fat too fast (surplus too large - reduce by 200 kcal), or muscle gains aren't visible yet (muscle gain is slow and internal at first). Take photos monthly. Check waist measurement - if waist increasing >1 cm/month, it's too much fat gain.

Q: Why do I feel tired all the time during my training program?

A: Possible causes: inadequate sleep (<7 hrs), insufficient calories (below TDEE), excessive training volume, iron deficiency (especially females), vitamin D deficiency, overtraining. Address sleep and calories first. If persists after 2 weeks, check blood work.

Q: I hit a plateau - my weights haven't moved in 4 weeks. Help?

A: First: take 1 deload week. Then implement periodization - switch from linear to wave loading. Increase volume by 2 sets. Try a different rep range for 4 weeks. Ensure nutrition is adequate (calories, protein). Sleep 8+ hrs. Plateau = adaptation signal needed.

Q: Why am I always sore for more than 3 days after training?

A: Excessive soreness (>72 hrs) indicates training volume was too high. Reduce volume by 20-30%. Ensure post-workout nutrition (protein+carbs within 2 hrs). Sleep 8+ hrs. Active recovery (light walking) helps. Do not progress volume until soreness normalizes.

Q: I get stomach issues from protein shakes. What should I do?

A: Switch whey concentrate to whey isolate (lower lactose). Reduce dose from 2 scoops to 1 scoop and space doses. Use water instead of milk. Try plant-based protein (pea or soy). Take with food. Use digestive enzyme supplements if issues persist.

Q: My chest is not growing despite training. What am I missing?

A: Common causes: insufficient volume (under 10 sets/week), not training to near-failure (leaving too many reps in reserve), lack of progressive overload, poor mind-muscle connection, insufficient frequency (train 2x/week). Add 2-3 sets of isolation flies after compound pressing.

Q: I'm not losing fat but also not gaining muscle on maintenance. What's happening?

A: Recalculate TDEE - your maintenance may have shifted. Add 200 kcal for lean bulk or subtract 200 kcal for mild cut. Increase protein to 2.2g/kg for recomp stimulus. Ensure training is progressive. Body composition changes are slow - 8-12 weeks needed to see clear trends.

Q: Why can't I sleep well despite being tired after training?

A: Possible causes: training too late in the evening (intense exercise raises cortisol and core temperature), caffeine too late in day, overtraining (paradoxically causes insomnia), inadequate protein or carbs before bed, or excessive stress. Move training to morning/afternoon.

Q: I'm gaining weight faster than expected (+1 kg/week). How do I slow it down?	A: Reduce calorie surplus by 200-300 kcal immediately. Increase cardio to 2 sessions/week of 20-30 min moderate intensity. Ensure protein is high (2g/kg) to maintain muscle while slowing gain. Target 0.25-0.5 kg/week for lean bulk.
Q: My arms are growing but my legs are not. What should I do?	A: Almost always a volume and frequency issue. Add leg training session. Increase leg volume to 15+ sets/week. Prioritize squat and RDL. Many athletes unconsciously undertrain legs. Stop doing so much arm work in the sets saved from arm reduction.
Q: I can't seem to hit my protein target every day. What should I do?	A: Simplify. Build 3 core high-protein meals first. Add liquid protein (milk shake + whey) for the remaining. For vegetarians: cook 100g soy chunks dry every day (52g protein). Prep protein in bulk for the week. Protein mismatch is usually a planning/preparation problem.
Q: My scale weight fluctuates 2-3 kg within the week. Is this normal?	A: Completely normal. Scale weight fluctuates 1-3 kg daily due to water retention, food weight, glycogen, sodium, and hormonal shifts (especially in women). Use 3-day average to assess trends. Judge progress over 2-4 week periods, not day to day.
Q: I started a muscle gain program but gained fat. Did I do something wrong?	A: Some fat gain is expected and normal during a bulk. If BF increase is >2-3% over 3 months, surplus was too large. Return to lean bulk (+200-300 kcal). If BF was above 20% (M) or 28% (F) when starting, a recomp or mini-cut first was the correct approach.
Q: What should I do if I feel unmotivated to train?	A: Motivation is unreliable - build systems and habits instead. Short-term: reduce target to minimum effective (1-2 compound exercises). Try a new program for 4 weeks. Train with a partner. Long-term: reassess goals, track progress photos and strength data.
Q: Should I be concerned if I don't get a pump during training?	A: The pump is not necessary for muscle growth. It indicates blood flow and metabolic stress but is not correlated with hypertrophy. Heavy compound work often produces less pump but more growth. Do not change effective programs in search of a pump.
Q: My shoulders hurt when I bench press. What should I do?	A: Reduce load by 30%. Check grip width (too wide = more shoulder stress). Retract scapulae more. Avoid excessive shoulder flare (keep elbows at 45-75deg, not 90deg). Add face pulls and rear delt work. Consider landmine press as a shoulder-friendly alternative.

14.x BEGINNER FAQs

Q: Where should I start as a complete beginner?	A: Step 1: Calculate TDEE + add 250 kcal for lean bulk. Step 2: Set protein at 1.8g/kg. Step 3: Choose a 3-day full body program. Step 4: Learn the big 5 lifts (squat, deadlift, bench, row, OHP) with proper form. Step 5: Track all lifts. Step 6: Sleep 7-9 hrs. That's it.
Q: How do I learn proper lifting technique?	A: Watch quality tutorials (Alan Thrall, Starting Strength YouTube for technique). Video yourself from the side. Get a coach or experienced lifter to review your form. Start with 50-60% of maximum weight to learn the pattern. Reduce weight when form breaks. Technique is never optional.
Q: What is the biggest mistake beginners make?	A: The most common is program hopping - changing programs every 2-3 weeks instead of committing to one for 12+ weeks. Second: insufficient protein. Third: insufficient calories. Fourth: no progressive overload tracking. Fifth: expecting results in 2-4 weeks.
Q: How should a beginner structure their diet?	A: Simplified beginner diet: Breakfast - 4 eggs + 300ml milk + oats/bread. Lunch - 150-200g meat/paneer/soy + rice + dal + vegetables. Snack - fruit + peanuts or yogurt. Dinner - similar to lunch. Pre-sleep - 200g paneer or yogurt. No complicated planning needed.

Q: Do I need supplements as a beginner?	A: No supplements are needed for beginners. Food is all you need for the first year. If budget allows, creatine monohydrate (3-5g/day) is the one supplement worth considering. Nothing else will meaningfully accelerate beginner gains compared to consistent training and diet.
Q: How do I know if I'm making progress?	A: Progress markers: strength increasing (add weight or reps each session), body weight trending up 0.25-0.5 kg/month (lean bulk), monthly progress photos showing more fullness, measurements (arms, chest) gradually increasing. Do not judge by how you look day-to-day.
Q: What should I eat if I don't know what to cook?	A: Simplest muscle-building diet: Morning - 4-5 boiled eggs + 300ml milk + 2 slices bread. Post-workout - banana + milk shake. Lunch - rice + dal + sabzi (use mess/canteen). Dinner - same as lunch + extra protein source. No cooking expertise required.
Q: How do I start if I have no gym access?	A: Buy a doorframe pull-up bar (₹500-1500). Start with: Push-ups (chest/triceps), Pull-ups/rows (back/biceps), Squats + lunges (legs), Glute bridges (glutes/hams), Plank variations (core). This simple bodyweight program is effective for 6-12 months of beginner gains.
Q: How quickly will I see results?	A: Strength: noticeable in 2-3 weeks (neural adaptation). Visible muscle changes: 8-12 weeks. Significant physique changes: 6-12 months. These timelines assume consistent training 3x/week and hitting protein and calorie targets daily. Results are certain with consistency.
Q: What is the most important thing to focus on as a beginner?	A: Consistency above all else. The best program you will do consistently beats the perfect program you abandon. After consistency: proper form on compound lifts, then protein intake, then calorie intake. Get these right first before optimizing anything else.
Q: How do I stay motivated when results are slow?	A: Track strength gains (not appearance). You will likely add 5-10 kg to your squat and bench in the first month - that is real progress. Keep a training log. Compare monthly photos side by side. Find a training partner. Remember that 12 months of consistency = a transformed body.
Q: What is the correct way to breathe while lifting?	A: General rule: exhale on the exertion (concentric phase = lifting the weight), inhale on the lowering (eccentric). For heavy compound lifts (squat, deadlift): Valsalva maneuver - brace fully and hold breath through the sticking point, then exhale at completion.
Q: Should I use a lifting belt as a beginner?	A: No. Beginners should build natural core strength first. A belt is a performance tool for intermediate-advanced lifters moving heavy loads (>85% 1RM). Using a belt too early prevents core development. Learn to brace your core effectively first.
Q: What is the difference between sets and reps?	A: A rep (repetition) is one complete movement. A set is a group of reps. 3x10 means 3 sets of 10 reps. In a session you might do 4x8 squats (4 sets of 8 reps each). Volume = sets x reps x weight.
Q: What time of day should I train?	A: The best time to train is whenever you will do it consistently. Morning training shows higher long-term adherence. Evening (4-7 PM) shows slightly better performance in studies. For beginners, consistency of time is more important than optimal timing.
Q: How do I deal with gym anxiety as a beginner?	A: Focus on your own program and progress. Everyone at the gym started as a beginner. Go during quieter hours (early morning, mid-afternoon). Have your program written so you always know what to do next. The anxiety decreases significantly after 2-4 weeks of consistent attendance.

Q: What gym equipment do I actually need to start?	A: Minimum: access to barbells, dumbbells, and a bench. Nice to have: pull-up bar, cable machine. For home: pull-up bar (■500-1500) + resistance bands (■300-1000) + a pair of adjustable dumbbells (■1000-3000) covers 80% of training needs for 12+ months.
Q: How do I calculate my 1 Rep Max (1RM)?	A: Epley formula: $1RM = \text{Weight} \times (1 + \text{Reps}/30)$. Example: lifted 60kg for 8 reps: $1RM = 60 \times (1 + 8/30) = 60 \times 1.267 = 76\text{kg}$ estimated. Use this to set training percentages. Test actual 1RM only when form is perfect and spotter is present.
Q: Is it normal to feel weak in the first few weeks?	A: Yes. First 2-4 weeks involve learning motor patterns, not building muscle. Strength often appears to decline in the first week as you learn new movement patterns. This is completely normal. Rapid strength gains follow in weeks 3-8 as neural efficiency improves.
Q: How do I know if my form is correct?	A: Video yourself from the side and front during all major lifts. Compare to quality tutorial videos. Ask a coach or experienced lifter to review. The most common beginner form errors: knee cave in squat, rounding lower back in deadlift, flared elbows in bench.

MODULE 15 | RAG OPTIMIZATION

Retrieval tags · Decision rules per section · Related modules · AI coaching notes · Structured metadata

15.1 RAG ARCHITECTURE OVERVIEW

This document is structured for optimal RAG (Retrieval Augmented Generation) performance inside the ZITLAS AI Coach. Each section includes structured metadata, retrieval tags, and cross-module references to ensure the LLM can accurately retrieve, synthesize, and apply coaching intelligence from this knowledge base.

15.2 RETRIEVAL TAG SCHEMA

RETRIEVAL TAG	USE CASE / QUERY TYPE
nutrition	Any query about food, macros, calories, protein, carbs, fat, supplements, meal timing
training	Any query about exercise, program design, volume, frequency, progressive overload, splits
recovery	Sleep, deloads, cortisol, DOMS, overtraining, rest days, CNS fatigue
assessment	BMI, TDEE, BMR, body fat, LBM, FFMI, intake calculations, scoring
swot	SWOT generation, user profiling, strength-weakness analysis, opportunity identification
beginner	First-time lifters, training age <1 year, basic programs, foundation building
intermediate	Training age 1-3 years, Upper-Lower or PPL, double progression
advanced	Training age 3+ years, block periodization, specialisation, advanced techniques
vegetarian	Plant-based protein, dairy, eggs, leucine gap, protein combinations
vegan	No animal products, B12, algae oil, soy/pea protein, complete plant amino acids
hostel	Budget <INR 150/day, no kitchen, shelf-stable proteins, bodyweight training
female	Female-specific protocols, menstrual cycle, iron, body fat floor, bone health
teen	Ages 14-18, growth plates, technique focus, no heavy loading, sleep priority
hardgainer	Not gaining despite eating, liquid calories, NEAT reduction, TDEE audit
skinny-fat	Normal BMI + high BF%, recomp strategy, scale not moving
budget	Budget protocols, INR-based planning, cost-efficiency rankings
plateau	Stalled progress, troubleshooting, deload need, program change logic
decision-engine	IF-THEN rules, coaching logic, conditional recommendations
faq	Common questions, evidence-based answers, myth-busting
injury	Joint pain, knee, shoulder, lower back, modification protocols

15.3 MODULE CROSS-REFERENCE MAP

QUERY TYPE	MODULE RETRIEVAL CHAIN
Calorie Query	Module 2 (Assessment) → Module 13 (Decision Engine) → Module 4 (Nutrition)
Protein Planning	Module 4 (Nutrition) → Module 5 (Food DB) → Module 6 (Recipes) → Module 13
Training Program	Module 2 (Assessment) → Module 8 (Programs) → Module 7 (Exercise DB) → Module 13
Special Population	Module 3 (SWOT) → Module 11 (Special Populations) → Module 13 (Decision Engine)
Supplement Query	Module 10 (Supplements) → Module 13 (Decision Engine) → Module 4 (Nutrition)
Plateau / Stuck	Module 12 (Troubleshooting) → Module 13 (Decision Engine) → Module 9 (Recovery)
Recovery Issue	Module 9 (Recovery) → Module 13 (Decision Engine) → Module 10 (Supplements)
FAQ Response	Module 14 (FAQs) → Relevant topic module → Module 13 (Decision Engine)
SWOT Generation	Module 2 (Assessment) → Module 3 (SWOT Engine) → Module 11 (Special Populations)

QUERY TYPE	MODULE RETRIEVAL CHAIN
Diet Plan Generation	Module 2 → Module 4 → Module 5 → Module 6 → Module 13
Workout Plan Generation	Module 2 → Module 7 → Module 8 → Module 13
Full Onboarding	Module 2 → Module 3 → Module 1 → Module 13 → Module 8 + Module 4

15.4 AI COACHING RESPONSE FRAMEWORK

STEP	ACTION
1. ASSESS	Always collect the 9 key intake parameters (age, weight, height, goal, experience, diet, budget, schedule, equipment) before generating any plan.
2. CLASSIFY	Classify athlete across all dimensions: training age, body fat level, diet type, budget tier, lifestyle type, special population flag.
3. SWOT	Generate personalised SWOT using Module 3 rules. Identify the 2-3 most critical factors to address.
4. PLAN	Generate nutrition plan (Module 2+4+5+6) and training plan (Module 7+8) personalised to classification.
5. RULES	Apply relevant Module 13 IF-THEN decision rules to refine and customise the plan further.
6. ADJUST	Build in weekly adjustment protocol: weight tracking, strength tracking, recovery monitoring.
7. TROUBLESHOOT	When plateau or issue reported: apply Module 12 troubleshooting rules before modifying plan.
8. EDUCATE	Embed relevant Module 14 FAQ answers and Module 1 expectation-setting into coaching messages.

15.5 FINAL MASTER RULES

- [R251] Hierarchy of muscle gain: Calories > Protein > Training > Sleep > Supplements > Everything else.
- [R252] If any of the Big 4 (Calories, Protein, Training, Sleep) is deficient, results will be compromised.
- [R253] Personalization is the difference between a good plan and a great plan. Always assess before prescribing.
- [R254] Compliance to a mediocre plan > Non-compliance to a perfect plan. Make plans the athlete will actually follow.
- [R255] One year of consistent training with proper nutrition = transformation most people don't achieve in 5 years.
- [R256] AI Coach role: assessment → classification → SWOT → plan generation → weekly adjustment → troubleshooting → education.
- [R257] Every coaching decision should be traceable to at least one rule in this knowledge base.
- [R258] RAG system note: when generating advice, cite the relevant module and rule number for transparency.
- [R259] Athlete safety is first priority. If any health red flag is raised, refer to qualified medical professional.
- [R260] This knowledge base is optimized for Indian athletes (budget, food culture, lifestyle). Adapt for other regions.

END OF ZITLAS AI MUSCLE GAIN BRAIN — VOLUME 2
15 Modules • 500+ Coaching Rules • 200+ Foods • 300+ Exercises • 120+ Recipes • 170+ FAQs
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